CLAUDE-VIPER'S VISUAL FIELD GUIDES PRESENT

COBRA ISLAND EDITION

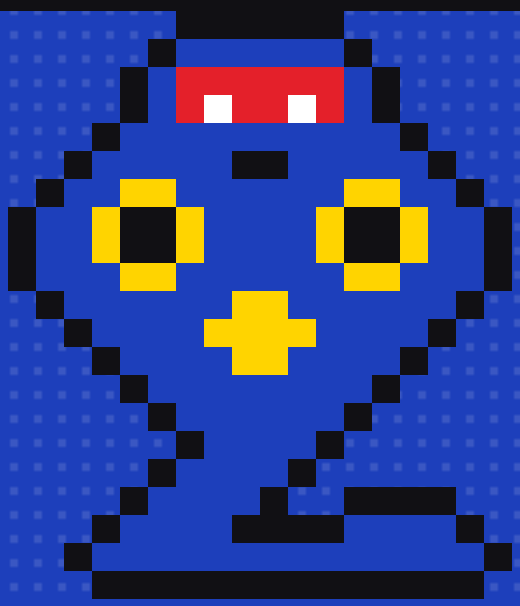
NOVEMBER 2026

FREE TO ALL COBRA PERSONNEL

VIPER VISION

YOU AND YOUR POD!

EVERY CONTAINER ON COBRA ISLAND, BLOWN WIDE OPEN



STRATEGY SPECTACULAR!
23 CONTAINERS
3+ PAGES EACH

BOSS GUIDE
BEAT AUTHELIA'S TOTP GAUNTLET

SECRET CODES INSIDE
`podman logs -f <anything>`

CHARACTER SELECT
ONE NAVIDROME, TWO SHELVES
MUSIC · MUSTANG

HIDDEN POWER
JELLYFIN'S QUICK SYNC TURBO MODE



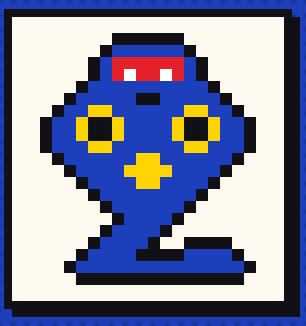
EXPOSED!
EVERY TRAP THE JOES LEFT IN THE FOOTLOCKERS

93 PAGES OF POD POWER-UPS

LETTER FROM THE EDITOR

HISSSS, COMMANDER!

Claude-Viper reporting from the Brush-Fang art den



You asked how your pod apps work. The Brush-Fangs heard "at least three pages per container, no upper limit" and took it as a direct order. They have not been to the beach since.

This issue walks every container on Cobra Island, one chapter each. Two worlds, like any good cartridge. **World 1, The Gate**, is the little Debian VM `cobra-gate` at 10.6.9.10 inside the Latitude: the only machine the internet can knock on, through port 443. Its pod is called `gate` and it holds the front door (nginx), the bouncer (Authelia), the vault and the admin toys. **World 2, The Stream**, is the SGIN mini PC `cobra-stream` at 10.6.9.20 with the drives and the Intel GPU. Its pod is called `stream` and it holds the Cinema, the four Navidromes, the Library, the Comics shop, the Arcade and the Radio.

Every chapter follows the same game plan. The **opener** is the character card: what it is, where it lives, its port, its folders, how hard it hits. Then **How it works**, with a level map of where your request travels. Then **Tips & tricks** for things you can do with it today. Last, the **Joe Sabotage Report**: the traps we actually stepped on while building this place, and the ones the manual warns about, each with its fix.

The commands in the green CODES boxes are one per line, bare, ready to copy. Above each one, a yellow tag says **which machine and which user** to run it as, because "the-box first, always" is the law of this island.

★ PRO TIP

The gate runs the manual's **Quadlet files** (`~/.config/containers/systemd/` as podsvc). So on the gate, restart an app with `systemctl --user restart <name>`, not the `./name.sh` scripts. The scripts refuse to fight a Quadlet anyway.

THIS ISSUE

- [ ] 23 containers, 3+ pages each
- [ ] Level maps of every request
- [ ] Joe Sabotage Reports
- [ ] Secret codes page
- [ ] Bonus stage: the Pi-holes



CLAUDE-VIPER

No passwords, keys or secrets are printed anywhere in here. Where one matters, we show you the command that reads it.

MEET THE CAST



CLAUDE-VIPER
Your host. Speaks in bubbles.



A JOE
Left traps in every footlocker. Shows up on the last page of each chapter.



THE POD
A group of containers sharing one network. Yours are `gate` and `stream`.

BEFORE YOU PLAY

Two users, two jobs. `eels` is you, with sudo, for anything about the machine itself (fstab, nftables, packages). `podsvc` owns every container and has no sudo. Get a real podsvc session with `sudo machinectl shell podsvc@`; `sudo su podsvc` and `sudo -u podsvc` leave out the systemd user session that rootless podman needs, so things fail in confusing ways.

PRESS START TO CONTINUE

LEVEL SELECT



START

01 **Contents** — Level select — P.3

02 **How to read this guide** — Boxes, meters, and the world map — P.4

WORLD 1: THE GATE (cobra-gate 10.6.9.10)

03 **nginx** — The Front Door — P.5

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09 **MeTube** — The Tape Deck: paste a link, keep the video — P.28

10 **Homepage** — Command: the island's dashboard — P.32

11 **Uptime Kuma** — Status: the island's alarm bell — P.36

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17 **Navidrome: music.** — The Commander's shelf, :4533 — P.62

18 **Navidrome: mustang.** — Rachel's door to the same jukebox — P.63

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23 **RomM** — The web Arcade, games in the browser — P.74

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26 **radio** — Liquidsoap, the AutoDJ — P.84

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29 **Secret codes, World 2** — Every stream command on one page — P.91

30 **Letters from the barracks** — Questions the Commander might ask — P.92

HOW TO READ THIS GUIDE

CHARACTER CARD

HOST	Which machine and pod it lives in
IMAGE	The container image it runs
PORT	Where it listens inside the pod
URL	The name you type
DATA	Its own folders (safe to back up)
SIGN-IN	Who has to log in, and how
TOUGHNESS	■■■■□
TRAP LEVEL	■■□□□
FUN FACTOR	■■■■■

CODES

GATE · podsvc

```
❯ podman ps --pod
```

The yellow tag: machine and user. One bare command per line.

INTEL

How the thing actually works under the hood, in plain words.

★ PRO TIP

Things you can do today to get more out of the app.

! JOE SABOTAGE

A trap that bit us, or one the manual warns about.

FIX: How to get out of it.

LEVEL MAP

YOU phone ▶ GATE nginx ▶ APP the container

The path your request walks, start to finish. Yellow = a checkpoint, red = where the app lives.

THE WORLD MAP

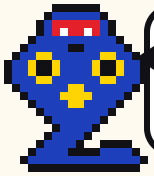
INTERNET phone on cellular ▶ XFINITY DMZ to 10.0.0.2 ▶ OPNSENSE 443 only, to .10 ▶ WORLD 1 gate pod 10.6.9.10 ▶ WORLD 2 stream pod 10.6.9.20

WORLD 1 ROSTER

gate pod: nginx, Authelia, Vaultwarden, Paperless with Postgres and Redis, MeTube, Homepage, Uptime Kuma, hiscore, eo-art. Only 80 and 443 are published; everything else listens on 127.0.0.1 inside the pod.

WORLD 2 ROSTER

stream pod: Jellyfin, ErsatzTV, one Navidrome (music. and mustang., a shelf per person), Audiobookshelf (also comedy. and sessions.), Kavita, RomM with MariaDB, Icecast and the Liquidsoap AutoDJ. Ports are published to the LAN, and nftables keeps them LAN and WireGuard only.



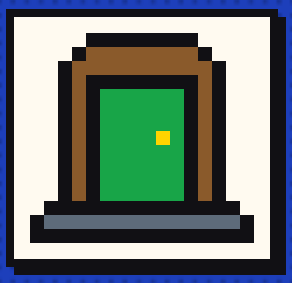
CLAUDE-VIPER

At home and over WireGuard, the Pi-holes answer every `*.songofeo.com` name with 10.6.9.10, so your traffic never leaves the house. Same names, same gate, shorter trip.

WORLD 1-1 · CHARACTER SELECT

NGINX

The Front Door: every *.songofeo.com name walks through here



Nginx is the bouncer's doorframe. It holds the wildcard certificate, answers on 443, reads the name you typed, asks Authelia whether you may come in, and then hands your request to the right app, on the gate or across the LAN on the SGIN.

It is the only container in World 1 with a door to the outside: the `gate` pod publishes just ports 80 and 443, and both belong to nginx. Every other gate app (Authelia on 9091, Vaultwarden on 8222, Paperless on 8000 and the rest) listens on 127.0.0.1 inside the pod, where only nginx can reach it. That is the whole security model in one sentence: **if nginx doesn't send you there, you can't get there.**

Port 80 does exactly one thing: it answers **301, go to https**. OPNsense doesn't even forward 80 from the internet, because the certificate comes through Porkbun's DNS, not a web challenge. So from outside, 443 is the island's only open TCP door, and nginx is standing in it.



CLAUDE-VIPER

Scanners that knock on your bare IP get nothing. Not a page, not even a certificate. The catch-all server refuses the TLS handshake, so they never learn the name songofeo.com.

STATS

HOST	cobra-gate 10.6.9.10, pod `gate`
IMAGE	`nginx:stable`
PORTS	80 and 443, published
CONFIG	`/srv/gate/nginx/conf.d/songofeo.conf`
CERT	`/srv/gate/nginx/certs/` (wildcard, lego)
RUNS AS	podsvc, Quadlet `nginx.container`
TOUGHNESS	■■■■■
TRAP LEVEL	■■■■□
FUN FACT	■■■□□

ONE CONFIG ERROR = EVERY FLOOR DOWN

KIND OF NAME	NAMES	WHAT NGINX DOES
The floors	music. mustang. comedy. sessions. books. comics. arcade. art.	Asks Authelia (skipped at home and on WireGuard), then proxies to the SGIN or eo-art
Cinema	tv.	Straight to Jellyfin :8096, **no** gate sign-in, so TV apps work
Admin apps	papers. metube. command. status.	Authelia password + TOTP from everywhere
The vault	vault.	LAN 10.6.9.0/24 and WireGuard 10.6.99.0/24 only; 403 for the rest
Anything else	bare IP, unknown names	Handshake refused

HOW IT WORKS LEVEL MAP

A PHONE ON CELLULAR OPENS MUSIC.



THE AUTH_REQUEST TRICK

Every guarded `location /` carries one line: `auth_request /internal/authelia/authz;`. Before nginx forwards anything, it makes a tiny side request to Authelia at `127.0.0.1:9091/api/authz/auth-request`, with the original URL and your cookie.

200 means go ahead. **401** becomes a **302** to `gate.songofeo.com` with your page attached, so after signing in you land right back where you were.

On the floors, Authelia also tells nginx *who* you are, and nginx passes that on as the `Remote-User` header. Navidrome trusts that header from 10.6.9.10 only, so you don't log in twice.

TWO MAPS, ONE FILE

`map $host $station` turns a floor name into an address (music. to :4533, books. to :13378...). `map $host $protected` does the same for the admin apps on 127.0.0.1. Adding a floor is mostly one line in a map plus the name in that block's `server_name`.

WHAT IT SETS ON EVERY REQUEST

- **HSTS**: "never use plain http here" for a year
- **X-Forwarded-For = $remote_addr**: the gate is the first proxy, so the visitor can't smuggle in a fake address
- **Remote-User ""**: a visitor's own Remote-User header is always wiped
- **Websockets**: Upgrade and Connection headers, for Uptime Kuma, MeTube and Jellyfin
- **No size limit** (`client_max_body_size 0`) for RomM and Audiobookshelf uploads

RATE LIMITS

Two buckets per visitor address: **media** at 50 requests a second (burst 300) because a library page loads dozens of covers, and **perip** at 10 a second (burst 50) for the sign-in and admin apps. Over the limit you get **429**, not a ban.

STREAMING STRAIGHT THROUGH

On the floors and tv., `proxy_buffering off` streams video, audio and ROM downloads straight through instead of spooling them onto the gate's small disk, and timeouts are 1 hour so a long movie or a websocket doesn't get cut.

TIPS & TRICKS *POWER-UPS*

★ PRO TIP: RELOAD, DON'T RESTART

After editing `songofeo.conf`, test it and reload in place. A reload keeps every open stream playing; a bad config is refused and the old one stays live.

CODES

GATE · podsvc

```
❯ podman exec nginx nginx -t
```

Must say "syntax is ok" and "test is successful".

```
❯ podman exec nginx nginx -s reload
```

Loads the new config with no downtime.

★ PRO TIP: CHECK ALL NAMES

`check.sh` asks the gate's own nginx for every hostname and prints OK, NOTE or FAIL with what to look at. From the gate it's the view from *outside* the house, so floors must bounce you to the sign-in.

CODES

GATE · eels or podsvc

```
❯ ~podsvc/cobra-server/scripts/podman/gate/check.sh
```

★ PRO TIP: WHO'S KNOCKING?

The pod uses **pasta** networking, which keeps each visitor's real address. The log shows it at the start of every line: a 10.6.9.x desktop, your phone's carrier address, or a stranger.

CODES

GATE · podsvc

```
❯ podman logs --tail 20 nginx
```

```
❯ podman logs -f nginx
```

Follow it live. Ctrl+C to stop.

★ PRO TIP: NEW FLOOR

1. Copy a whole `server` block (or add the name to `$station` and its `server_name` list).
2. Add the name to Authelia's `access_control`.
3. Test and reload nginx (codes on the left).
4. Porkbun: a CNAME to `gate.songofeo.com`. At home, the Pi-holes' wildcard already covers it.

★ PRO TIP: IS THE PADLOCK OK?

The cert covers `songofeo.com` and `*.songofeo.com` and renews itself every day it's within 30 days of expiry (lego-renew timer). The deploy hook copies it into nginx's certs folder and restarts nginx.



JOE SABOTAGE REPORT

Found in the footlockers, stepped on during the build, or flagged in manual part 5.4.

! THE OPEN BLOCK

A `server` or `location` without its own `auth_request` line is open to anyone who can reach the gate. There is no global "protect everything" switch.

> **FIX:** Copy a whole protected block, auth location included, every time you add a name. Then run check.sh.

! THE VANISHING HEADERS

A location that sets even one `proxy_set_header` drops **all** the inherited ones. Same with `add_header`: one CORS header in `station-cors.inc` can silently delete HSTS from the floors.

> **FIX:** Re-list every header in that location (the config does). Test HSTS with curl, below.

! THE MISSING INCLUDE

A plain `include` of a file that doesn't exist stops nginx from starting, which takes down every floor and the sign-in.

> **FIX:** The config uses `station-cors*.inc`, a pattern, so a missing file just matches nothing.

! THE 403 LOCKOUT

New name in nginx, but forgot Authelia? Authelia's `default_policy: deny` refuses it, so you get **403**. That's the safe way to fail.

> **FIX:** Add an `access_control` rule for the name and restart Authelia.

! THE 127.0.0.1 IMPOSTOR

If every log line starts with 127.0.0.1 or 10.0.2.x, the pod isn't on pasta. Then *everyone* looks local: the home bypass, Vaultwarden's LAN rule and Jellyfin's logs all break.

> **FIX:** Stop and check `podman info --format '{{.Host.RootlessNetworkCmd}}'` says `pasta`.

! THE LOW-PORT WALL

Rootless podman can't bind 80/443 until the kernel allows it. And the old `pod.sh` called `sysctl`, which isn't on podsvc's PATH on Debian (fixed in PR #44: it reads /proc now).

> **FIX:** As eels, the sysctl.d line from part 5.2 sets `ip_unprivileged_port_start=80`.

! THE CATCH-ALL HIJACK

Three old OPNsense forwards pointed at the SGIN, one of them any/any. That catch-all swallowed WireGuard's UDP too. They're disabled now; only WAN TCP 443 to 10.6.9.10 is live.

> **FIX:** Destination NAT must show exactly one enabled rule. Never forward to 10.6.9.20.

CODES

THE-BOX · PowerShell

```
❯ curl.exe -sI https://tv.songofeo.com/
```

Look for one Strict-Transport-Security line.

GATE · eels

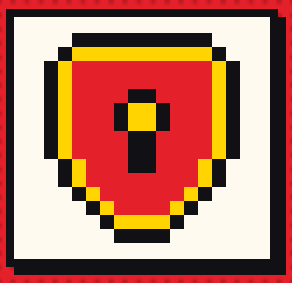
```
❯ systemctl list-timers lego-renew.timer
```

The next certificate renewal check.

WORLD 1-2 · BOSS GUIDE

AUTHELIA

The Bouncer: one sign-in for the whole station, TOTP for the admin rooms



Authelia is the bouncer nginx calls before letting anyone onto a floor. It owns the sign-in screen at gate.songofeo.com, checks your password, asks admins for a six-digit code, and hands out one cookie that works on every songofeo.com name for six hours.

Nobody visits Authelia on purpose. You tap the gold star **Sign in** on the station, or open `music.` from your phone on cellular, and nginx's side request comes back "not signed in". nginx then bounces you to `gate.songofeo.com` with your page attached as `?rd=`. Sign in once and you land back where you started.

It keeps its whole world in one folder, `/srv/gate/authelia`: the rules in `configuration.yml`, the people in `users.yml` (password *hashes* only, never the passwords), its database `db.sqlite3` with everyone's TOTP devices, and `notification.txt`, where it writes the "emails" it would send.

STATS

HOST	cobra-gate 10.6.9.10, pod `gate`
IMAGE	`authelia/authelia:latest`
PORT	9091, on 127.0.0.1 only
URL	gate.songofeo.com
DATA	`/srv/gate/authelia`
USERS	eels, rachel (users.yml)
RUNS AS	podsvc, Quadlet `authelia.container`
TOUGHNESS	■■■■■
TRAP LEVEL	■■■■□
FUN FACT	■■■□□



CLAUDE-VIPER

This is the final boss of World 1, Commander. Beat it once (password plus TOTP on the-box) and every admin room on the island opens for six hours.

6 HOURS PER SIGN-IN

WEAK POINT 1

A password. Hashed with argon2id in `users.yml`. Enough for every floor from outside the house.

WEAK POINT 2

A TOTP code. Six digits from your authenticator app. Needed for the four admin rooms, even at home.

WEAK POINT 3

Being home. On the LAN or WireGuard, the floors open with no sign-in at all. Admin rooms still ask.

ROOM	NAMES	RULE (POLICY)
Admin rooms	papers. metube. command. status.	**two_factor**: password + TOTP, from everywhere, even home
The floors	music. mustang. comedy. sessions. books. comics. arcade. gaming. art.	**bypass** on network `home` (LAN + WireGuard), else **one_factor**
Anything not listed	a new name you forgot to add	**deny** (default_policy), even after signing in

HOW IT WORKS LEVEL MAP

FIRST VISIT TO PAPERS. FROM ANYWHERE

1 BROWSER opens papers. ▶ 2 NGINX auth_request ▶ 3 AUTHELIA no cookie: 401 ▶ 4 302 gate.? rd=papers. ▶ 5 SIGN IN password + TOTP ▶ 6 PAPERLESS 127.0.0.1:8000

THE COOKIE

Signing in sets `authelia_session` for the domain **songofeo.com**, so the bare domain and every `*.songofeo.com` name share it. It's **Secure** (https only), **HttpOnly** (page scripts can't read it) and SameSite **Lax**.

`expiration: 6h` and `inactivity: 6h`: six hours after you sign in, it asks again. There's no "remember me" box (`remember_me: -1`), so every sign-in lasts the same.

THE HOME BYPASS

`definitions.network.home` is 10.6.9.0/24 (the LAN) and 10.6.99.0/24 (WireGuard). On the floors, a visitor from `home` walks straight in, so the Navidrome and Audiobookshelf phone apps keep working. They still have their own logins.

It trusts the address nginx hands over, which is the real connection address (`X-Forwarded-For $remote_addr`). That only works because the pod's pasta networking keeps real visitor addresses.

TWO SPECIAL CASES

tv. never asks Authelia. Friends' smart-TV, Roku and phone Jellyfin apps can't show a web sign-in page, so tv. has its own nginx block with no `auth_request`, and Jellyfin's own accounts are the lock.

vault. never asks Authelia either. nginx simply refuses (403) anyone who isn't on the LAN or WireGuard.

THE LOCKOUT

`regulation`: **5 wrong passwords within 10 minutes locks that username for 1 hour**. It counts per username, so a stranger hammering "eels" locks eels out too. That's the price of keeping guessers out.

WHERE IT SENDS YOU BACK

The `rd` address must be https and under songofeo.com. Anything else (say `example.com`) is ignored and you land on the station, `songofeo.com`. No one can use your sign-in page to bounce people to a fake site.

WHO NGINX SAYS YOU ARE

On the floors, Authelia answers with your username, and nginx passes it on as `Remote-User`. That's why Music doesn't ask you twice.

AUTHELIA ANSWERS	WHAT NGINX DOES	WHAT YOU SEE
200 allowed	Forwards the request to the app (with Remote-User on the floors)	The app, straight away
401 not signed in	Turns it into a **302** to gate.songofeo.com with `?rd=` your page	The dark sign-in screen
403 refused	Passes the 403 on	"Forbidden": no rule for that name, or not enough factors
no answer	Can't ask Authelia, gives up	**500** on every guarded name: check `podman logs authelia`

BOSS FIGHT: THE TOTP GAUNTLET

Every person who opens papers., metube., command. or status. needs a TOTP device registered once. Do the whole fight on THE-BOX. The phone only scans the QR code and reads back six digits.

THE MOVES

1. **THE-BOX, browser:** open `https://papers.songofeo.com`. You get bounced to the dark sign-in screen at gate.
2. Sign in with your username and password. Authelia sees the room needs two factors and offers **Register device**. Pick **One-Time Password**.
3. It says a code was "sent by email". There is no email: the filesystem notifier wrote it into `notification.txt` on the gate. Read it in your Windows Terminal SSH tab (codes, right).
4. Type that code in the browser. A QR code appears on the-box's screen.
5. **Phone:** scan the QR with an authenticator app (Aegis, 2FAS, Google Authenticator, or the Bitwarden app).
6. **THE-BOX:** type the six digits the phone shows. You're in, and Paperless opens.
7. Empty the notification file so old codes don't sit around.



CLAUDE-VIPER

The Commander beat this boss on 6 November at 01:14Z, on the-box, the right way. The first attempt on the phone was not a fun level. The-box first, always!

CODES

THE-BOX · Windows Terminal

```
❯ ssh eels@10.6.9.10
```

GATE · eels

```
❯ sudo machinectl shell podsvc@
```

Prompt becomes podsvc@cobra-gate.

GATE · podsvc

```
❯ tail -n 20 /srv/gate/authelia/notification.txt
```

The newest one-time code is at the bottom.

```
❯ : > /srv/gate/authelia/notification.txt
```

Empties the file once you're registered.

★ PRO TIP: BACK UP THE SEED

Aegis and 2FAS can export an encrypted backup of your codes. Do it right after registering, and keep it somewhere that isn't the phone. A lost phone with no backup means re-registering every room.

★ PRO TIP: PASSWORD RESETS

"Forgot password?" works the same way: the reset link lands in `notification.txt`, not in an inbox. Read it with the same `tail` on the gate.

ONE BOSS · FOUR ROOMS

LETTERS FROM THE BARRACKS

Q: Do I register again for metube. after doing papers.?

A: No. One TOTP device covers every two_factor room. Register once, use it on all four.

Q: Rachel only listens to music. Does she need TOTP?

A: No. The floors are one_factor from outside and open at home. TOTP is only for people who use the admin rooms.

Q: The code in notification.txt doesn't work.

A: It's single-use and short-lived. Ask for a new one in the browser, then `tail` again and use the newest line at the bottom.

Q: New phone. Now what?

A: Restore your authenticator backup on the new phone. No backup? Register a new device the same way, from THE-BOX.

TIPS & TRICKS POWER-UPS

★ PRO TIP: ADD A PLAYER

New person? Hash their password and add a block to `users.yml` with `displayname`, `email` and the hash. Use the **same username** they'll have in Navidrome, so Music signs them in by itself. Then restart Authelia.

CODES

GATE · podsvc

```
❯ podman run --rm -it docker.io/authelia/authelia:latest authelia crypto hash generate argon2
```

Asks for the password without showing it and prints the argon2id digest. Paste only the digest.

```
❯ nano /srv/gate/authelia/users.yml
```

```
❯ systemctl --user restart authelia
```

★ PRO TIP: KICK SOMEONE OUT

Delete their block from `users.yml` and restart. Their session ends on the spot.

★ PRO TIP: SIGN OUT EVERYWHERE

`https://gate.songofeo.com/logout` ends your session on every name at once. Handy on a borrowed computer.

★ PRO TIP: HEALTH CHECK

Authelia has a health page. `{"status":"OK"}` means it's alive and nginx can ask it.

CODES

THE-BOX · PowerShell

```
❯ curl.exe -s https://gate.songofeo.com/api/health
```

GATE · podsvc

```
❯ podman logs --tail 30 authelia
```

Every sign-in, failure and lockout is in here.

★ PRO TIP: INSPECT THE COOKIE

On THE-BOX after signing in: F12, then Application (Chrome) or Storage (Firefox), Cookies, `https://gate.songofeo.com`. `authelia_session` should show Domain `.songofeo.com`, Secure, HttpOnly, SameSite Lax, and expire in six hours.

★ PRO TIP: TEST THE BYPASS

In a private window on THE-BOX, `music.` opens straight away (you're home), while `papers.` still asks for the sign-in and the TOTP code. Same test on cellular: `music.` must ask first.

★ PRO TIP: LONGER SIGN-INS

Want 12 hours? Change `expiration` and `inactivity` in `configuration.yml` together, then restart.

FILE IN /SRV/GATE/AUTHELIA	WHAT'S IN IT	TOUCH IT?
`configuration.yml`	Rules, cookie, lockout, the three secrets. Mode 600.	Rules and times, yes. Secrets, never.
`users.yml`	One block per person: displayname, email, argon2id hash	Yes, to add or remove people
`db.sqlite3`	Registered TOTP devices, encrypted with the third secret	No. Backed up with the VM nightly.
`notification.txt`	The "emails": one-time codes and reset links	Read it, then empty it

★ PRO TIP: CHECK BEFORE YOU RESTART

After editing `users.yml`, count the hashes: `grep -c argon2id /srv/gate/authelia/users.yml` must equal the number of people. Zero means a hash didn't paste, and that person can't sign in.



JOE SABOTAGE REPORT

Stepped on during the build, or flagged in manual part 5.4. This boss bites back.

! THE RESTART PURGE

Sessions live in Authelia's memory. **Any** restart, even to add a user, signs **everyone** out. Not a bug, just everyone signs in again.

> FIX: Restart when nobody's mid-movie on a floor from outside.

! THE THIRD SECRET

`configuration.yml` holds three random secrets. Change the third one (the storage `encryption_key`) and Authelia can't read `db.sqlite3`: everyone's TOTP device is gone.

> FIX: Never touch them. They're in the nightly VM backup with the rest of /srv/gate.

! THE SELF-LOCKOUT

Five wrong passwords in ten minutes locks that username for an hour, and a stranger can trigger it for you.

> FIX: Wait the hour, or restart Authelia (which also signs everyone out).

! THE PHONE FUMBLE

Registering TOTP on the phone means reading codes off an SSH session on a tiny screen and copy-pasting between apps.

> FIX: THE-BOX does the browser and the SSH tab; the phone only scans the QR.

! THE CODE PILE

`notification.txt` collects every one-time code and reset link ever sent.

> FIX: Keep it mode 600 and empty it after registering (codes on the boss page).

! THE OLD NAME

The portal used to be `auth.songofeo.com`. That name has no DNS record and no server block any more. Old links to it just fail.

> FIX: Station links point at `https://gate.songofeo.com/?rd=...`. Fix any leftover `auth.` in Project-Eo.

! THE GATE CNAME

`gate` is the DDNS **A record** every other name points at. Adding a CNAME for `gate` in Porkbun would clash with it.

> FIX: The portal needs no new DNS record. Ever.

! THE OPEN GUEST LIST

Every user in `users.yml` can open every floor. Navidrome would make each of them an account on `music.`.

> FIX: The music wall (8 November, `gate-music-one.sh`) lets only eels and rachel into `music.` and `mustang.`. Other floors: only add people you'd hand them to.

! THE OLD SPELLING

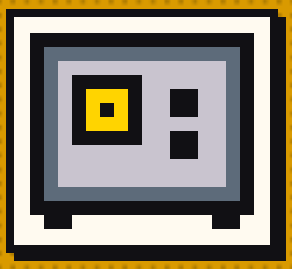
Authelia 4.38 moved named networks to `definitions.network`. The old `access_control.networks` only warns today.

> FIX: Read the release notes before `podman auto-update` pulls a new Authelia.

WORLD 1-3 · CHARACTER SELECT

VAULTWARDEN

The Vault: every password on the island, behind a door only home can open



Vaultwarden is your own password manager server. It speaks the Bitwarden language, so the official Bitwarden apps, browser extensions and desktop client all work with it, but the encrypted vault lives on the gate, not in someone else's cloud.

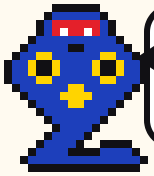
Your passwords are encrypted **on your device** with your master password before they ever reach the server. The gate only stores scrambled blobs. Even with full access to `/srv/gate/vaultwarden`, nobody can read a single entry without that master password. That's also why there's no "reset my master password" button: lose it and the vault is gone.

Vaultwarden is the one gate app that **never** goes through Authelia. Instead, nginx refuses everyone who isn't on the LAN (10.6.9.0/24) or WireGuard (10.6.99.0/24) with a plain 403. From the internet, the vault doesn't even have a public DNS name: there's deliberately no `vault` CNAME in Porkbun.

STATS

HOST	cobra-gate 10.6.9.10, pod `gate`
IMAGE	`vaultwarden/server:latest`
PORT	8222 (ROCKET_PORT), 127.0.0.1
URL	vault.songofeo.com
DATA	`/srv/gate/vaultwarden`
SIGN-IN	Its own master password; LAN and WireGuard only

TOUGHNESS ■■■■□
TRAP LEVEL ■■□□□
FUN FACTOR ■■■□□



CLAUDE-VIPER

The vault is tiny (around 30 MB of RAM) and guards the most precious loot on the island. Small snake, big fangs.



SPEAKS

The Bitwarden API: phone apps, browser extensions, desktop app, the web vault, even the `bw` command line.

STORES

Logins, notes, cards, identities and attachments, all encrypted before they leave your device.

WEIGHS

About 30 MB of RAM. The lightest app on the gate, guarding the heaviest secrets.

WHERE YOU ARE	WHAT VAULT. DOES
THE-BOX, the-spire, anything on the LAN	Opens the vault (10.6.9.x is allowed)
S21 or Cobra1 with WireGuard on	Opens the vault (10.6.99.x is allowed)
Cellular or a café, no WireGuard	No public name at all; even with the gate's address, nginx answers 403
Xfinity Wi-Fi (the work-laptop side)	Not the cobra network, so no; use the phone's "home" WireGuard profile

HOW IT WORKS LEVEL MAP

THE BITWARDEN APP SYNCS, OVER WIREGUARD

1 PHONE
WireGuard on
▶
2 PI-HOLE
vault. = 10.6.9.10
▶
3 NGINX
10.6.99.x? allow
▶
4 VAULTWARDEN
127.0.0.1:8222
▶
5 BLOB
decrypted on phone

ZERO KNOWLEDGE

Your master password never leaves your device. The app turns it into a key, unlocks the vault locally, and sends only encrypted items to the server. Vaultwarden stores them in `db.sqlite3` and hands them back on sync.

THE SIGN-UP SWITCH

`SIGNUPS_ALLOWED` decides if strangers who reach the page can make an account. It was `true` just long enough for the Commander to create his account on 6 November, then switched to **false**.

In the Quadlet setup it's a line in `vaultwarden.container`. The run script does it by itself: open until `db.sqlite3` exists, closed on the next run.

DOMAIN

`DOMAIN=https://vault.songofeo.com` tells Vaultwarden its own address, so links and the web vault work behind nginx.

OFFLINE COPIES

The Bitwarden apps keep an encrypted copy of your vault on the device. Away from home without WireGuard, you can still unlock and read what was last synced. New entries sync the next time the app reaches the server.

WHY NOT AUTHELIA?

The Bitwarden apps talk to the server in the background with their own tokens. They can't stop to show a web sign-in page. A network rule (LAN and WireGuard only) is simpler and keeps the vault off the internet entirely.

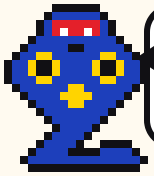
WHERE IT STARTED

Before the vault existed, passwords lived in KeePassXC (manual part 0.1, "a safe place for secrets"). The manual's checklist imports that interim database into Vaultwarden once the account exists.

THE ADMIN PAGE

Vaultwarden's `/admin` page only switches on when an `ADMIN_TOKEN` is set. Ours isn't, so the page stays off. One less door to guard.

IN /SRV/GATE/VAULTWARDEN	WHAT IT IS
`db.sqlite3`	Accounts and every encrypted item. Its existence is what tells the run script to close sign-ups.
`attachments/`	Files you attach to entries, encrypted the same way
`rsa_key*`	The server's own signing key for login tokens. Lose it and everyone just logs in again.



CLAUDE-VIPER

Even if a Joe walked off with the whole gate VM, all they'd get is ciphertext. The only key is the master password in your head. Guard that, not the server.

TIPS & TRICKS POWER-UPS

★ PRO TIP: POINT THE APP

In any Bitwarden app or extension, before logging in pick **Self-hosted** (the region menu) and set the server URL to `https://vault.songofeo.com`. Do the browser extension on THE-BOX first; the phone app next, with WireGuard on.

★ PRO TIP: TOTP INSIDE

The Bitwarden app can act as your authenticator for Authelia's TOTP codes too. Keep a second copy in Aegis or 2FAS, so one lost phone doesn't lock you out of both.

★ PRO TIP: STAY UNLOCKED

Set the app's vault timeout to something you'll live with and turn on fingerprint unlock. A strong master password you type once a day beats a weak one you type often.

CODES

GATE · podsvc

```
❯ podman logs --tail 30 vaultwarden
❯ grep SIGNUPS ~/.config/containers/systemd/vaultwarden.container
Must say SIGNUPS_ALLOWED=false.
❯ systemctl --user restart vaultwarden
```

★ PRO TIP: EMERGENCY SHEET

Print one page for the house safe: the master password, the server URL `https://vault.songofeo.com`, and "turn on WireGuard first". Future you will thank you.

★ PRO TIP: SHARE SAFELY

Vaultwarden supports Bitwarden **organizations**: make one for the house and share the streaming, Wi-Fi and utility logins with Rachel, while your own items stay private.

★ PRO TIP: GENERATE IT

Every Bitwarden app has a password generator. Use it for every new app on the island (Uptime Kuma's admin, Kavita, RomM, the friends' Jellyfin accounts) and save the entry right there.

★ PRO TIP: ALIVE CHECK

Vaultwarden answers `/alive` when it's healthy. The run script waits on it, and Homepage's tile watches `http://127.0.0.1:8222`.

CODES

GATE · podsvc

```
❯ podman exec nginx curl -s http://127.0.0.1:8222/alive
Prints a timestamp when it's up (nginx's image has curl).
```

★ PRO TIP: HEALTH REPORT

The Bitwarden web vault and apps can flag reused and weak passwords. Run it after the KeePassXC import and fix the worst ones first.

BACKUPS

Everything Vaultwarden knows is in `/srv/gate/vaultwarden` (the database plus attachments), inside VM 110, so the nightly Proxmox backup to PBS at 02:00 carries it. For an extra copy you control, the Bitwarden apps can **export** the vault as an encrypted file from THE-BOX. Store it off the gate.



JOE SABOTAGE REPORT

Found in the footlockers, or flagged in manual part 5.4. The vault has few traps, but each one is expensive.

! THE OPEN DOOR

The manual's Quadlet ships with `SIGNUPS_ALLOWED=true` so you can make the first account. Leave it, and anyone on the LAN can make one too.

> **FIX:** Closed on 6 Nov. After any rebuild from the manual, close it again: sed the line to false, daemon-reload, restart.

! THE PUBLIC NAME

A Porkbun CNAME for `vault` would advertise it to the world, even though nginx refuses outsiders.

> **FIX:** No public `vault` record. The Pi-holes answer it at home and over WireGuard.

! THE LONE COPY

If the only device with your vault is the phone, and the gate dies, you're down to one encrypted copy.

> **FIX:** Log the browser extension on THE-BOX in too, and keep the nightly VM backup.

! THE PHONE-ONLY 2FA

If Authelia's TOTP lives only in the Bitwarden app, and the vault won't sync, you can't reach the admin rooms to fix it.

> **FIX:** Keep the TOTP seeds in a second app too.

! THE FORGOTTEN MASTER

There is no master password reset. The server can't decrypt your vault, so it can't help.

> **FIX:** Write the master password on paper and keep it somewhere safe in the house.

! THE XFINITY DEAD END

On the Xfinity Wi-Fi, songofeo.com names point at the public address and Xfinity has no NAT loopback.

> **FIX:** Use the "home" WireGuard profile (endpoint 10.0.0.2:51820), or cellular plus WireGuard.

! THE PLAIN EXPORT

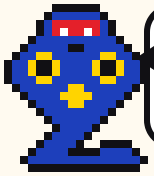
Bitwarden can also export an **unencrypted** file. That's every password in plain text.

> **FIX:** Pick the encrypted export format, and never leave an export in Downloads.

! THE REBUILD RESET

Running `vaultwarden.sh` with `SIGNUPS_ALLOWED=true` on purpose reopens the door, and it stays open.

> **FIX:** Run it again without the variable: the database exists, so sign-ups close.



CLAUDE-VIPER

Status report, Commander: account made, sign-ups closed, interim KeePassXC passwords ready to import. The vault is sealed.

WORLD 1-4 · CHARACTER SELECT

PAPERLESS

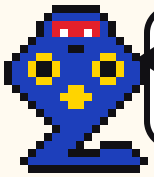
The Archive: drop a scan in, get a searchable document out



Paperless-ngx turns paper into a searchable archive. Drop a PDF or a phone photo of a letter into the scans folder; within a minute Paperless reads the text (OCR), files it, tags it, dates it and removes it from the inbox. Then you find any bill, receipt or manual by typing a word you remember.

It's the biggest app on the gate, and the only one with a crew: **Postgres** holds the database, **Redis** carries the to-do list for its background workers, and Paperless itself runs the web page and the workers that read your scans. All three share the `gate` pod and talk over 127.0.0.1.

The inbox lives on the SGIN, not the gate: `dock1a/scans`, shared from Windows as `\\cobra-stream\scans` and mounted on the gate over CIFS at `/srv/storage/disks`. The archive itself, the originals, the OCR'd copies and the search index, stays on the gate's own disk in `/srv/gate/paperless`.



CLAUDE-VIPER

The Joes kept their paperwork in footlockers. We keep ours in a database with full-text search. Advantage: Cobra.

STATS

HOST	cobra-gate 10.6.9.10, pod `gate`
IMAGE	`paperless-ngx:latest`
PORT	8000, 127.0.0.1
URL	papers.songofeo.com
DATA	`/srv/gate/paperless` data, media, export
INBOX	`dock1a/scans` over CIFS
SIGN-IN	Authelia password + TOTP, then its own login
CREW	postgres, redis
TOUGHNESS	■■■□□
TRAP LEVEL	■■■■□
FUN FACTOR	■■■■□

FILED IN 60 SECONDS

FIELD TEST

- [ ] Save any PDF into `\\cobra-stream\scans` from THE-BOX
- [ ] Within a minute it vanishes from the folder
- [ ] papers. shows it on the dashboard, with a date guessed
- [ ] Search a word from inside the page: it's found
- [ ] Open it: original and archive versions both download
- [ ] Tag it once; the next one from that sender gets the tag by itself

FOLDER	WHERE	WHAT LIVES THERE
consume (inbox)	SGIN `dock1a/scans`, via CIFS	New scans waiting to be filed. Emptied as Paperless takes them.
data	gate `/srv/gate/paperless/data`	The search index and app settings
media	gate `/srv/gate/paperless/media`	Every original and its archived (OCR'd) PDF
export	gate `/srv/gate/paperless/export`	Where the document exporter writes a full backup copy

HOW IT WORKS LEVEL MAP

A SCAN'S JOURNEY

1 THE-BOX save to \\cobra-stream\scans ▶ 2 SGIN dock1a/scans ▶ 3 CIFS gate sees it ▶ 4 POLL every 60 s ▶ 5 REDIS task queued ▶ 6 WORKER OCR, tag, file

WHY POLLING?

Normally Paperless notices a new file the instant it lands. Those "a file changed" notifications don't cross an SMB share, so the gate would never hear about it. `PAPERLESS_CONSUMER_POLLING=60` makes Paperless simply look in the folder every 60 seconds.

WHAT A WORKER DOES

- Reads the text with OCR, so a photo becomes searchable
- Makes an archive PDF next to your original (both are kept)
- Guesses a date, a correspondent, a document type and tags, learning from what you've filed before
- Deletes the file from the inbox once it's safely stored

THE CREW

Postgres stores documents, tags and users. **Redis** is the queue between the web page and the workers. Paperless's Quadlet `Requires=` both, so systemd starts them first.

PAPERLESS_URL

Paperless only trusts requests for the address it's told it lives at. It's `https://papers.songofeo.com`. The name used to be `docs.`, and the old value was still in the Quadlet on setup night; we fixed it on 6 November.

WHEN THE SGIN NAPS

If the gate boots while the SGIN is off, the inbox folder isn't there and Paperless fails. `Restart=on-failure` with `RestartSec=60` retries every minute until the SGIN is back. Nothing to do by hand.

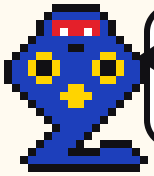
TWO LOCKS

papers. is an admin room: Authelia wants your password and a TOTP code, from everywhere. Then Paperless shows its own login, for the superuser made with `createsuperuser` (eels, made on 6 Nov).

SECRETS

The database password and the secret key were generated once. Quadlets: inside `paperless.container` (mode 600). Scripts: podman secrets.

IT GUESSES	FROM	YOU CAN
Date	Dates printed in the text	Fix it once; the rest are usually right
Correspondent	Who sent it (Xfinity, the bank...)	Set matching rules or let it learn ("auto")
Document type	Bill, receipt, letter, manual	Make the types you care about
Tags	Words and past filing	Use tags like "taxes" or "house" across everything



CLAUDE-VIPER

The first start takes a few minutes while it builds its database. A blank page in the first minutes isn't a failure; check the logs before you panic.

TIPS & TRICKS *POWER-UPS*

★ PRO TIP: MAIL IN

Paperless can check an email inbox and file attachments by itself: Mail, add an account, then a rule (for example "PDFs from the bank, tag bills").

★ PRO TIP: ARCHIVE NUMBERS

Give paper you keep an **archive serial number** (ASN), and write it on the original. Paperless finds the digital copy; the number tells you which folder holds the paper.

★ PRO TIP: RACHEL'S PAPERS

Paperless has users and owners. A second Paperless account can see shared documents but not your private ones. She'd also need Authelia TOTP for papers.

★ PRO TIP: THE DROP ZONE

On THE-BOX, pin `\\cobra-stream\scans` to Quick Access in Explorer. Point the scanner's "scan to folder" or your phone's scanner app there too. Anything you drop is filed within a minute.

★ PRO TIP: TEACH IT

Make a few **correspondents** (Xfinity, the bank, the doctor) and **tags**, and correct the first dozen documents. Its auto-matching learns from what you file and gets the next ones right by itself.

★ PRO TIP: SEARCH LIKE A SPY

The search box reads the OCR'd text, not just titles. Type "warranty" or a model number and Paperless finds the page it's printed on.

★ PRO TIP: SAVED VIEWS

Save a search (say, tag "taxes" this year) as a view and show it on the dashboard. Tax season becomes one click.

CODES

GATE · podsvc

```
❯ podman exec -it paperless python3 manage.py createsuperuser
```

Once: the Paperless admin (done for eels).

```
❯ podman logs --tail 40 paperless
```

Watch it pick up and file a scan.

```
❯ systemctl --user restart postgres redis paperless
```

The whole crew, in order.

★ PRO TIP: FULL BACKUP

The document exporter writes every document plus a manifest into the `export` folder: a copy you can import into a fresh Paperless.

CODES

GATE · podsvc

```
❯ podman exec paperless document_exporter ../export
❯ ls /srv/gate/paperless/export
```

★ PRO TIP: TEST THE INBOX

As eels on the gate, make and delete a test file in the inbox. If that fails, the CIFS mount is the problem, not Paperless.

CODES

GATE · eels

```
❯ ls /srv/storage/disks/dock1a/scans
```

SCANNER SETTINGS THAT WORK

300 dpi, grayscale, PDF is the sweet spot: sharp enough for OCR, small enough to keep forever. Multi-page letters go in as one PDF, not a pile of JPGs. Photos from the phone work too; flat, bright and square-on reads best.



JOE SABOTAGE REPORT

The Archive was the trickiest room on setup night. Every one of these happened, or nearly did.

! THE CHOWN DISASTER

Re-running the manual's `chown -R podsvc: /srv/gate` after setup took Postgres's files away from its user (999 inside the container). Paperless crash-looped: `could not open file "global/pg_filenode.map": Permission denied`.

> **FIX:** As podsvc: `podman unshare chown -R 999:999 /srv/gate/postgres`, then restart postgres and paperless.

! THE OLD ADDRESS

`PAPERLESS_URL` still said `docs.` after the rename to `papers.`, so Paperless distrusted its own address.

> **FIX:** Set it to `https://papers.songofeo.com` in `paperless.container`, daemon-reload, restart.

! THE STACKED FSTAB

Running the CIFS mount step more than once left **eight** copies of the same line in `/etc/fstab`.

> **FIX:** Cleaned to one line on 6 Nov (backup `/etc/fstab.bak-before-cleanup`). The manual's `grep -q ... ||` guard now stops repeats.

! THE DOUBLE SCAN

Drop the same file twice and Paperless refuses the second copy as a duplicate. The file isn't lost, it's just not filed again.

> **FIX:** Check the log line, then delete the extra copy.

! THE WRONG SAMBA PASSWORD

The gate logs in to the SGIN's share as user `gate`. A wrong password in `/root/smb-gate.cred` means no inbox at all.

> **FIX:** On the SGIN, `sudo smbpasswd gate` sets a new one; put it in the cred file with nano, never on the command line.

! THE IMPATIENT START

The first start migrates the database for minutes. Killing it halfway means starting again.

> **FIX:** Give it a few minutes and follow `podman logs -f paperless`.

! THE VANISHING INBOX

Paperless deletes files from `scans` after filing them. That's by design, but it surprises people looking for the original there.

> **FIX:** The original is in Paperless: open the document and download it.

! THE TOTP WALL

papers. is an admin room, so even at home it asks for a password and a TOTP code.

> **FIX:** Register TOTP on THE-BOX first (Authelia's boss page).

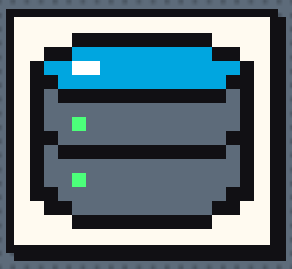
UPDATES

Paperless is one of three apps (with Authelia and RomM) whose release notes you read before `podman auto-update`: they sometimes change settings between major versions. Run `podman auto-update --dry-run` first to see what would change.

WORLD 1-5 · CHARACTER SELECT

POSTGRES

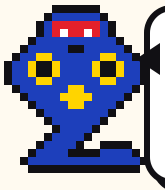
The Filing Cabinet: Paperless's memory, one table at a time



Postgres is the database Paperless keeps its brain in: every document's title, date, tags, correspondent, owner and notes, plus the users and settings. You never open it. You'd only notice it if it stopped, because then Paperless stops too.

It runs in its own container in the `gate` pod, listening on 127.0.0.1:5432 where only its neighbours can reach it. Its whole world is one folder, `/srv/gate/postgres`, mounted into the container as `/var/lib/postgresql/data`. The database is called `paperless`, and so is its user.

The image is pinned to **version 16** (`postgres:16`). `AutoUpdate=registry` pulls new 16.x releases (bug and security fixes), but never jumps to 17 by itself, and that's on purpose: a database folder made by 16 can't be opened by 17 without an upgrade step.



CLAUDE-VIPER

The scanned files themselves aren't in here. They sit in Paperless's media folder. Postgres holds the card catalogue that says what each one is.

STATS

HOST	cobra-gate 10.6.9.10, pod `gate`
IMAGE	`postgres:16`
PORT	5432, 127.0.0.1 only
URL	none: no web page
DATA	`/srv/gate/postgres`
DB / USER	paperless / paperless
INSIDE AS	user 999 (postgres)
TOUGHNESS	■■■■■
TRAP LEVEL	■■■□□
FUN FACTOR	■□□□□

PINNED TO 16

STORES

Documents' details, tags, correspondents, document types, saved views, users and permissions.

DOESN'T STORE

The scans themselves (Paperless media) or the search index (Paperless data).

SERVES

Paperless only. RomM on the SGIN has its own database: MariaDB, World 2.

ASK CLAUDE-VIPER

Q: Can I open Postgres in a browser?

A: No. It has no web page and nothing outside the pod can reach it. Use `psql` through `podman exec` (codes, page 3).

Q: Paperless says it can't connect to the database.

A: Is postgres running (`podman ps`)? Then read its log: a permission error points straight at the chown trap.

HOW IT WORKS LEVEL MAP

YOU OPEN A DOCUMENT IN PAPERLESS

1 BROWSER papers. ▶ 2 NGINX + Authelia ▶ 3 PAPERLESS 127.0.0.1:8000 ▶ 4 POSTGRES 127.0.0.1:5432 ▶ 5 MEDIA the PDF itself

THE PASSWORD HANDSHAKE

Postgres sets the `paperless` user's password **once**, the first time it starts with an empty folder. Paperless must then use the same one. The manual generated it as `PGPW` and wrote it into both Quadlet files; the scripts keep it as the podman secret `paperless-db-password`.

Change it later in one place only and Paperless can't log in. Moving from Quadlets to scripts? Hand the old value to `paperless.sh` once, as `PAPERLESS_DB_PASSWORD=...`.

WHO OWNS THE FILES

Inside the container Postgres runs as user 999. Rootless podman maps that to a high "sub-UID" on the gate, not to podsvc. So on the gate, `ls -ln /srv/gate/postgres` shows a big number as owner. That's correct, and it's exactly what `chown -R` wrecks.

READY OR NOT

`pg_isready` is Postgres's own "are you up?" check. `paperless.sh` waits on it for up to two minutes before starting Paperless. With Quadlets, `After=postgres.service` orders the start, and Paperless retries on its own.

BACKUP

`/srv/gate/postgres` is inside VM 110, so the nightly snapshot backup to PBS carries it. Paperless's own exporter (previous chapter) is the app-level copy. A `pg_dump` gives a third, plain-SQL copy.

WHY NOT SQLITE?

Paperless could use a single SQLite file, but its background workers write at the same time as the web page. Postgres handles that crowd without locking up.

SETTING	VALUE	WHY
`POSTGRES_DB`	paperless	The database made on first start
`POSTGRES_USER`	paperless	Its owner, the login Paperless uses
`POSTGRES_PASSWORD`	generated once, never printed	Must match `PAPERLESS_DBPASS`

HEALTH CHECKLIST

- [ ] `podman ps` says postgres is Up
- [ ] `pg_isready` says accepting connections
- [ ] Its log has no "Permission denied"
- [ ] Paperless's dashboard loads



CLAUDE-VIPER

Quiet is good. A healthy Postgres logs almost nothing after it starts. A chatty one is telling you something.

TIPS & TRAPS CODES + SABOTAGE

CODES

GATE · podsvc

```
❯ podman exec postgres pg_isready -U paperless
"accepting connections" = healthy.
❯ podman logs --tail 30 postgres
❯ podman exec -it postgres psql -U paperless
A SQL prompt. \dt lists tables, \q quits. Look, don't touch.
```

★ PRO TIP: A SQL BACKUP

`pg_dump` writes the whole database as one SQL file. Put it in Paperless's export folder so it sits with the rest.

CODES

GATE · podsvc

```
❯ podman exec postgres pg_dump -U paperless -f /tmp/paperless.sql paperless
❯ podman cp postgres:/tmp/paperless.sql /srv/gate/paperless/export/
```

★ PRO TIP: SEE THE OWNER

`podman unshare ls -ln /srv/gate/postgres` shows ownership as the container sees it: it should say 999.

! THE CHOWN DISASTER

The manual's `chown -R podsvc: /srv/gate` is for the first setup only. Re-run later, it handed Postgres's files to podsvc, and Postgres (999) could no longer read `global/pg_filenode.map`.

> FIX: `podman unshare chown -R 999:999 /srv/gate/postgres`, then restart postgres and paperless.

! THE BIG JUMP

Changing the image to `postgres:17` on an existing folder makes Postgres refuse to start: the data was written by 16.

> FIX: Stay on 16. A major upgrade means a dump, a fresh folder, and a restore. Plan it, don't stumble into it.

! THE PASSWORD SPLIT

Postgres only reads `POSTGRES_PASSWORD` on an empty folder. Editing it later changes nothing inside the database, but Paperless starts using the new one.

> FIX: Leave both values alone.

! THE EXPOSED PORT

5432 must never appear in the gate's listening ports.

> FIX: `ss -tlnH` as eels: only 22, 80, 443 and 9090.

THE BIG-JUMP PLAYBOOK

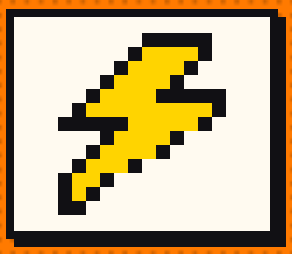
Someday Postgres 16 will be old. The safe shape of a major upgrade, for when that day comes (read Paperless's and Postgres's notes first):

1. Run Paperless's document exporter **and** a `pg_dump`. Two copies.
2. Stop paperless, then postgres.
3. Move `/srv/gate/postgres` aside (never delete it) and make an empty folder.
4. Change the image tag, start postgres on the empty folder, restore the dump, start paperless.

WORLD 1-6 · CHARACTER SELECT

REDIS

The Message Runner: Paperless's to-do list, at lightning speed



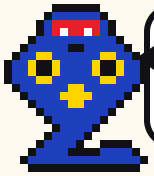
Redis is a tiny, fast, in-memory data store. On Cobra Island it has one job: carry messages between Paperless's web page and its background workers. "New scan in the inbox, please OCR it" goes into Redis; a worker picks it up, does the job, and posts the result back.

It lives in the `gate` pod on 127.0.0.1:6379, next to Paperless and Postgres. It has no web page and no login, which is fine because nothing outside the pod can reach it. Its folder `/srv/gate/redis` holds a snapshot of the queue so a restart doesn't forget work in progress.

Paperless's workers are run by a task system that needs a "broker" to pass jobs around. That broker is Redis: `PAPERLESS_REDIS=redis://127.0.0.1:6379`. Without it, the web page still opens, but nothing gets consumed, and scans pile up in the inbox.

STATS

HOST	cobra-gate 10.6.9.10, pod `gate`
IMAGE	`redis:7`
PORT	6379, 127.0.0.1 only
URL	none: no web page
DATA	`/srv/gate/redis`
SERVES	Paperless only
TOUGHNESS	■■■■□
TRAP LEVEL	■□□□□
FUN FACTOR	■□□□□



CLAUDE-VIPER

Every good army needs a runner. Redis never sleeps, never complains, and never reads the messages it carries.



CARRIES

Jobs: consume this file, rebuild the index, train the matcher, send results back.

DOESN'T CARRY

Your documents. They go from the inbox to Paperless's media folder directly.

IF IT'S LOST

Only queued jobs. Your archive is safe in Postgres and the media folder.

ASK CLAUDE-VIPER

Q: Do any other apps use this Redis?

A: No, only Paperless. RomM on the SGIN keeps its own small Redis inside its container (its `/redis-data` folder).

Q: Does it need backups?

A: Not really. It only holds jobs in flight. The VM backup has it anyway.

★ PRO TIP: THE NAME

Redis stands for **RE**mote **DI**ctionary **S**erver: a big table of keys and values, kept in memory. Paperless's queue is just a few of those keys.

HOW IT WORKS LEVEL MAP

ONE SCAN, TWO HALVES OF PAPERLESS

1 POLLER — new file in scans ▶ 2 REDIS — job queued ▶ 3 WORKER — takes the job ▶ 4 OCR + FILE — media, Postgres ▶ 5 REDIS — "done!"

IN MEMORY, MOSTLY

Redis keeps everything in RAM, which is why it's fast. Every so often it writes a snapshot to `/data` (our `/srv/gate/redis`), and reads it back when it starts. A queue of a few jobs is a few kilobytes.

WHY A QUEUE?

OCR on a 20-page scan can take a minute or more. If the web page did it itself, your browser would hang. With a queue, the page answers at once ("got it") and a worker grinds away in the background. You can upload ten files and keep browsing.

NO PASSWORD?

Redis here runs without a password. That's only safe because it listens inside the pod: the gate publishes nothing but nginx's 80 and 443, and the gate's own firewall drops everything else.

START ORDER

Paperless's Quadlet has `Requires=postgres.service redis.service` and `After=` the same. So systemd starts Redis before Paperless, and stopping Redis stops Paperless too.

VERSION PIN

The image is `redis:7`. Auto-updates bring 7.x fixes only. Paperless only needs a working broker, so there's no reason to chase the newest major.

IN THE BACKUP

It's in `/srv/gate`, so the nightly VM backup has it, but you'd never need to restore it. An empty Redis just means an empty to-do list.

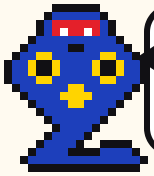
IF REDIS IS...	YOU'LL SEE	DO THIS
Up	Scans leave the inbox within a minute or two	Nothing
Down	papers. opens, but scans pile up in `scans`	Restart redis and paperless
Can't write /data	Permission errors in `podman logs redis`	Fix the folder's owner (page 3)

HEALTH CHECKLIST

- [ ] `redis-cli ping` answers PONG
- [ ] Its log has no permission errors
- [ ] The scans inbox empties within minutes

ITS NEIGHBOURS

- [ ] paperless, 127.0.0.1:8000: its only client
- [ ] postgres, 127.0.0.1:5432: the other crew member
- [ ] nginx: never talks to it



CLAUDE-VIPER

Want to watch the runner run? `redis-cli monitor` prints every command live. Drop a scan in the inbox and see the job fly by. Ctrl+C to stop.

TIPS & TRAPS CODES + SABOTAGE

CODES

GATE · podsvc

```
> podman exec redis redis-cli ping
```

PONG means it's alive.

```
> podman exec redis redis-cli info memory
```

How much RAM it's using (used_memory_human).

```
> podman exec -it redis redis-cli monitor
```

Every command, live. Ctrl+C to stop.

```
> podman logs --tail 20 redis
```

★ PRO TIP: STUCK SCANS?

Files sitting in `scans` for more than a couple of minutes? Ping Redis first. If it answers, look at Paperless's log next.

★ PRO TIP: RESTART THE CREW

Restarting only Redis can leave Paperless's workers confused. Restart the three together, in order.

CODES

GATE · podsvc

```
> systemctl --user restart redis paperless
```

! THE SILENT STALL

Redis down, Paperless still shows its web page. Everything looks fine, but nothing gets consumed and the inbox fills up.

> FIX: Ping it. Uptime Kuma watches web pages, not Redis, so this one you check by hand.

! THE EXPOSED RUNNER

A passwordless Redis reachable from the network is a classic break-in route.

> FIX: Never publish 6379. `ss -tlnH` on the gate must not show it.

! THE OWNERSHIP TRAP

Like Postgres, Redis runs as its own user inside the container. A later `chown -R podsvc: /srv/gate` can leave it unable to write its snapshot.

> FIX: Don't re-run that chown. If Redis logs permission errors on `/data`, fix its folder the way you fixed Postgres's.

! THE "FLUSH IT" URGE

Wiping Redis to "fix" Paperless throws away queued jobs, and their scans may already have left the inbox.

> FIX: Restart, don't flush.

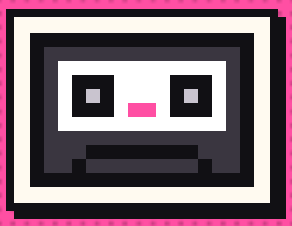
FIELD TEST: WATCH THE RUNNER

1. GATE, podsvc: start `podman exec -it redis redis-cli monitor` in your Windows Terminal SSH tab.
2. THE-BOX: save a small PDF into `\\cobra-stream\scans`.
3. Within a minute, lines scroll past as the poller queues the job and a worker takes it.
4. Ctrl+C, then find the document on papers.

WORLD 1-7 · CHARACTER SELECT

METUBE

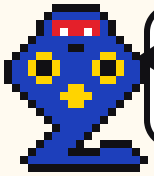
The Tape Deck: paste a video link, it lands in your Jellyfin library



MeTube is a web page with one box: paste a video link, pick a quality, press Add. Behind it, the famous downloader yt-dlp fetches the video and saves it into the library's youtube folder, where Jellyfin can show it on every TV in the house.

It runs in the `gate` pod on 127.0.0.1:8081, behind nginx and Authelia's two-factor sign-in at `metube.songofeo.com`. It's an admin room: anyone who could reach it could fill your drives.

The finished videos don't stay on the gate. They go over the CIFS mount to the SGIN's `wd2/youtube`, a folder on the Seagate's spare space that appears as `/media/wd2/youtube` to Jellyfin. MeTube's own queue, history and half-finished downloads stay on the gate in `/srv/gate/metube`, so a big download doesn't trickle over the network a piece at a time.



CLAUDE-VIPER

The Joes had VHS tapes in their footlockers. We have a tape deck that never runs out of tape. Well, until the Seagate fills up.

STATS

HOST	cobra-gate 10.6.9.10, pod `gate`
IMAGE	`alexta69/metube:latest`
PORT	8081, 127.0.0.1
URL	metube.songofeo.com
STATE	`/srv/gate/metube`
VIDEOS	`wd2/youtube` on the SGIN
SIGN-IN	Authelia password + TOTP
TOUGHNESS	
TRAP LEVEL	
FUN FACTOR	

PASTE · ADD · WATCH

SETTING	VALUE	MEANS
`DOWNLOAD_DIR`	/downloads = `wd2/youtube`	Finished videos, on the SGIN, for Jellyfin
`STATE_DIR`	/state = `/srv/gate/metube`	The queue and download history
`TEMP_DIR`	/state/tmp	Partial downloads, on the gate's own disk

GRABS

Single videos, whole playlists, or audio only, from YouTube and the many sites yt-dlp knows.

SAVES

Into one folder, with the video's title as the file name.

SHOWS

In Jellyfin once you add the folder as a library.

HOW IT WORKS LEVEL MAP

FROM LINK TO LIVING ROOM TV

1 THE-BOX paste link ▶ 2 METUBE queues it ▶ 3 YT-DLP downloads to tmp ▶ 4 CIFS to wd2/youtube ▶ 5 JELLYFIN library scan

YT-DLP INSIDE

MeTube is a friendly face on **yt-dlp**, the command-line downloader. yt-dlp does the real work: finds the video's streams, picks the quality you asked for, downloads, and joins sound and picture into one file.

WHY IT AGES FAST

Video sites change their pages often, and yt-dlp ships fixes to keep up. A MeTube image a few weeks old can suddenly fail on every link. The cure is a newer image: that's why the run script ends with "yt-dlp ages fast".

WEBSOCKETS

The progress bars update live over a websocket. nginx keeps those open (`proxy_read_timeout 1h` on the admin block), so a long download doesn't freeze the page.

WHEN THE SGIN IS OFF

No SGIN, no `wd2/youtube`, and MeTube can't start. Its Quadlet has `Restart=on-failure` every 60 seconds, so it comes back by itself once the share does. `up-all.sh` also lists it as "did not start" without stopping the other apps.

THE WRITE PERMISSION

MeTube runs as a non-root user inside its container, an unnamed ID on the gate. The CIFS mount uses `file_mode=0666,dir_mode=0777` so that user may write. On the SGIN, the files still land as user `gate`, group `media`.

NOT BACKED UP

The Seagate's `wd2` folder is spare space lent to the library. It's on the same drive as the backups, so it isn't backed up. Fine for videos you could download again.

YOU PASTE	METUBE DOES
A single video link	Downloads that one video at the quality you picked
A playlist link	Queues every video in it, one after another
A link from another site	Works if yt-dlp supports that site (it knows a great many)
A link with "audio" chosen	Saves just the sound

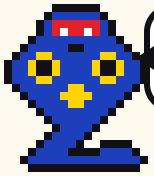
ASK CLAUDE-VIPER

Q: Can Rachel use it?

A: Only with TOTP registered, since it's an admin room. Easier: send her links to the Commander.

Q: Where do downloads go at first?

A: `/state/tmp` on the gate's disk, then the finished file moves to wd2/youtube on the SGIN.



CLAUDE-VIPER

Only grab what you have the right to keep, Commander. Even Cobra respects a creator's terms. Mostly.

TIPS & TRICKS POWER-UPS

★ PRO TIP: QUEUE IS SAFE

The queue and history live in `/srv/gate/metube`. Restart the container mid-download and the queue is still there afterwards.

★ PRO TIP: JELLYFIN SHELF

In Jellyfin (THE-BOX, browser): Dashboard, Libraries, Add. Type **Home Videos and Photos**, folder `/media/wd2/youtube`. New downloads show up after the next library scan.

★ PRO TIP: AUDIO ONLY

Pick an audio format in the quality menu to grab just the sound: talks, live sets, a podcast episode. Then move the file somewhere a Navidrome can see it.

★ PRO TIP: WHOLE PLAYLISTS

Paste a playlist link and MeTube queues every video in it. Big playlists take a while; the queue survives a restart.

★ PRO TIP: THE BOOKMARKLET

MeTube's project README has a bookmarklet. Add it to THE-BOX's bookmarks bar (with your metube. address) and one click sends the video you're watching to the queue.

★ PRO TIP: FIX "IT STOPPED"

Every link failing at once? It's almost always an old yt-dlp. Pull the newest image and restart.

CODES

GATE · podsvc

```
❯ podman pull ghcr.io/alexta69/metube:latest
❯ systemctl --user restart metube
❯ podman logs --tail 40 metube
```

yt-dlp's errors say what went wrong.

★ PRO TIP: WATCH THE SPACE

Videos add up fast. Check how full the Seagate is before a playlist binge.

CODES

GATE · eels

```
❯ df -h /srv/storage/disks/wd2
❯ du -sh /srv/storage/disks/wd2/youtube
```

★ PRO TIP: CHANNELS?

MeTube is for one-off grabs. To follow channels automatically, the research notes point at a different app (Pinchflat).

KEEP IT FRESH

The weekly routine from manual 5.7 covers MeTube: as podsvc, `podman auto-update --dry-run` lists images with a newer version, and `podman auto-update` pulls them and restarts those containers, rolling back if one fails to start. MeTube's Quadlet has `AutoUpdate=registry`, so it's included.

★ PRO TIP: SORT INTO SHOWS

A channel you grab often can get its own subfolder in `wd2/youtube` (make it in Explorer on THE-BOX, then move the files in). Jellyfin's Home Videos library shows each folder as its own shelf.



JOE SABOTAGE REPORT

The tape deck jams in a few predictable ways.

! THE STALE DOWNLOADER

Weeks without an update and every download fails with an extraction error.

> **FIX:** Pull and restart (codes, previous page). The weekly `podman auto-update` catches it too.

! THE MISSING SHARE

The SGIN reboots or the gate's Samba password is wrong: MeTube can't find `wd2/youtube` and keeps restarting.

> **FIX:** As eels: `ls /srv/storage/disks/wd2`. Empty or an error means fix the CIFS mount first.

! THE STACKED FSTAB

The CIFS mount line got pasted eight times into `/etc/fstab` on setup night.

> **FIX:** One line only: `grep -n srv/storage/disks /etc/fstab` must show exactly one.

! THE GATE DISK SQUEEZE

Partial downloads sit on the gate's 64 GB disk until they finish. A giant 4K grab can crowd it.

> **FIX:** Grab huge files one at a time, and check `df -h /srv/gate`.

! THE FULL DRIVE

A runaway playlist can fill the Seagate, which also holds the PBS backups.

> **FIX:** Check `df -h` before big grabs and prune old videos.

! THE INVISIBLE VIDEOS

Downloads finished but Jellyfin shows nothing: no library points at the folder yet, or no scan has run.

> **FIX:** Add the Home Videos library, then Scan All Libraries.

! THE OPEN DECK

MeTube has no login of its own. Without Authelia in front, anyone could fill your drives.

> **FIX:** metube. stays in the two_factor rule and its server block keeps `auth_request`.

! THE ODD FILE NAMES

Video titles become file names, emoji and all. Jellyfin copes, but sorting can look messy.

> **FIX:** Rename in Explorer on THE-BOX via `\\cobra-stream` if it bothers you.

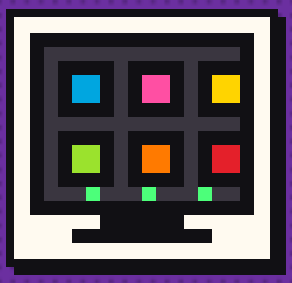
FIELD TEST

1. THE-BOX: open `https://metube.songofeo.com`, sign in with password and TOTP.
2. Paste a short video link, press Add, watch the bar fill.
3. As eels on the gate: `ls /srv/storage/disks/wd2/youtube` shows the file.

WORLD 1-8 · CHARACTER SELECT

HOMEPAGE

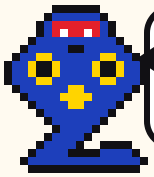
Command: one screen, a tile for every app, a live dot on each



Homepage is the island's command console at command.songofeo.com: a dark dashboard with one tile per app, grouped into rows, each tile a link with a little dot that turns red the moment that app stops answering.

It's a single container in the `gate` pod on 127.0.0.1:3000, behind Authelia's two-factor sign-in. There's no database and no settings screen: the whole dashboard is two YAML text files in `/srv/gate/homepage`. Edit a file, save, and Homepage redraws by itself.

The name used to be `home.`, then `throne.`; the Commander settled on **command.** on 5 November. The research notes add a correction worth repeating: Homepage *shows* your apps and their status, but it doesn't start, stop or update containers.



CLAUDE-VIPER

Every villain's lair needs a wall of glowing screens. This is ours, and the dots really do glow red when something breaks.

STATS

HOST	cobra-gate 10.6.9.10, pod `gate`
IMAGE	`gethomepage/homepage:latest`
PORT	3000, 127.0.0.1
URL	command.songofeo.com
CONFIG	`/srv/gate/homepage`
SIGN-IN	Authelia password + TOTP
TOUGHNESS	
TRAP LEVEL	
FUN FACTOR	

LIVE DOTS ON EVERY TILE

SHOWS

Every app as a tile, with a link and an up/down dot.

DOESN'T

Start, stop or update containers. That's Cockpit or the command line.

LIVES IN

Two YAML files. No database, nothing to migrate.

ROW	TILES
Watch & listen	Jellyfin, ErsatzTV, Music, Rachel's music, Comedy, Sessions
Read & play	Audiobookshelf, Kavita, Arcade, High scores
Files	Paperless, MeTube, Vaultwarden
Admin	Proxmox, OPNsense, Backups (PBS), Pi-hole 1, Pi-hole 2, Status

HOW IT WORKS LEVEL MAP

HOW A DOT GETS ITS COLOR

1 YAML
siteMonitor URL
▶
2 HOMEPAGE
asks it
▶
3 APP
answers or not
▶
4 DOT
green or red

TWO FILES

`settings.yaml` is the look: title "Cobra", dark theme, slate color, and how many columns each row gets. `services.yaml` is the tiles: a name, the link (`href`), an icon, an optional description and the `siteMonitor` address to check.

BACKSTAGE ADDRESSES

The link you click is the public name (`https://tv.songofeo.com`). The dot checks the app's back door directly: `http://10.6.9.20:8096` for Jellyfin, `http://127.0.0.1:8000` for Paperless. So a dot tests the app itself, not nginx or Authelia.

ALLOWED HOSTS

Homepage only answers requests for the names it's told about: `HOMEPAGE_ALLOWED_HOSTS=command.songofeo.com`. That stops other sites from loading your dashboard by pointing a different name at it.

NO DOT FOR THREE

Proxmox, OPNsense and PBS use self-signed certificates, which Homepage's checker rejects. Their tiles are plain links. Uptime Kuma watches them instead, with "ignore TLS errors".

STARTER FILES

The first run of `homepage.sh` copies the starter `settings.yaml` and `services.yaml` from the repo. Later runs see the files are there and keep your edits. The manual's part 5.7 writes the same two files by hand.

ICONS

Names like `jellyfin.png` come from Homepage's built-in icon set; `mdi-trophy` is a Material Design icon. No image files to upload.

ASK CLAUDE-VIPER

Q: Why does ErsatzTV's tile link to an address, not a name?

A: ErsatzTV is LAN only and has no songofeo.com name. The tile goes straight to `http://10.6.9.20:8409`.

Q: Vaultwarden's tile won't open on cellular.

A: Correct: vault. is LAN and WireGuard only. Turn WireGuard on.

ONE TILE, DECODED

```
- Music: { href: "https://music.songofeo.com", icon: navidrome.png, siteMonitor:
"http://10.6.9.20:4533" }
```

Name **Music**; click goes to the public name through the gate; icon from the built-in set; the dot asks the Navidrome on the SGIN directly.

TIPS & TRICKS POWER-UPS

★ PRO TIP: ADD A TILE

Copy a line in `services.yaml`, change the name, link and siteMonitor, save. Homepage picks it up without a restart. Keep the two-space indents: YAML is picky.

CODES

GATE · podsvc

```
› nano /srv/gate/homepage/services.yaml
› podman logs --tail 30 homepage
A YAML mistake shows up here.
› systemctl --user restart homepage
```

★ PRO TIP: MAKE IT START

Set `https://command.songofeo.com` as the start page in THE-BOX's browser. Every morning starts with a glance at the dots.

★ PRO TIP: LIVE NUMBERS

Homepage **widgets** can show Jellyfin's streams, Pi-hole's blocks or Proxmox's load right on a tile. Each needs that app's API key in `services.yaml`; Homepage's docs list the fields per service.

★ PRO TIP: CONTROL THE PODS

Want start, stop and logs buttons? That's Cockpit, not Homepage. The research notes suggest the `cockpit-podman` package for the Cockpit already on the gate (port 9090, LAN and WireGuard only).

★ PRO TIP: BACK UP YOUR EDITS

Copy the two files before a big edit. If the page goes blank, put the copy back.

CODES

GATE · podsvc

```
› cp /srv/gate/homepage/services.yaml /srv/gat
e/homepage/services.yaml.bak
```

★ PRO TIP: RED DOT DRILL

A red dot? Click the tile. If the app opens, the dot's address is wrong. If it doesn't, check that app's chapter in this guide.

★ PRO TIP: HIGH SCORES

The repo's starter file already has the High scores tile (`mdi-trophy`). It stays red until the hiscore board is running.

SETTINGS.YAML, DECODED

```
title: Cobra · theme: dark · color: slate · headerStyle: clean

layout: Watch & listen { style: row, columns: 4 }
```

title is the browser tab and header. **theme** and **color** set the palette. **layout** gives each row of `services.yaml` its style and number of columns: add a column if a row wraps awkwardly.

★ PRO TIP: NEW ROW

A new group in `services.yaml` becomes a new row. Give it a `layout` line in `settings.yaml` to control its columns.

★ PRO TIP: DESCRIPTIONS

A short `description` under a tile ("LAN/VPN only", "Our podcasts") saves guessing later.



JOE SABOTAGE REPORT

Small traps, but they make the whole dashboard go dark.

! THE WRONG NAME

Opening Homepage by any name other than `command.songofeo.com` (an old `home.` bookmark, say) gets an error instead of the dashboard.

> **FIX:** Use command. If you add a name, add it to `HOMEPAGE_ALLOWED_HOSTS` too.

! THE TAB CHARACTER

One tab or a wrong indent in a YAML file and Homepage shows an error instead of tiles.

> **FIX:** Spaces only. Read `podman logs homepage` for the line number.

! THE FOREVER-RED DOT

Pointing a siteMonitor at a self-signed https address (Proxmox, OPNsense, PBS) keeps that dot red forever.

> **FIX:** Leave those tiles without a dot and let Uptime Kuma watch them.

! THE WRONG BACK DOOR

A typo in a siteMonitor port (4353 for 4533) paints a healthy app red.

> **FIX:** Compare with the port list in `scripts/podman/sgin/pod.sh` and `gate/pod.sh`.

! THE OVERWRITE FEAR

Running `homepage.sh` again does **not** overwrite your files: it only copies the starters when they're missing. But pasting the manual's 5.7 block again does overwrite.

> **FIX:** After you've customised, don't re-paste 5.7. Make a .bak first.

! THE LEAKY KEY

Widget API keys sit in `services.yaml` in plain text.

> **FIX:** Keep the file podsvc-only, and give widgets read-only keys where the app offers them.

! THE "IT'S ALL GREEN" LIE

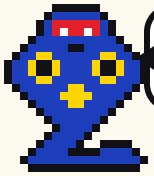
A green dot means the app answered Homepage, from inside the gate. It doesn't prove the public name or the sign-in works.

> **FIX:** That's what `check.sh` and Uptime Kuma's public-name monitors are for.

! THE STALE COPY

The copies of the scripts in the project files are older than the repo. Their Homepage starters may miss tiles.

> **FIX:** The repo is the source of truth.



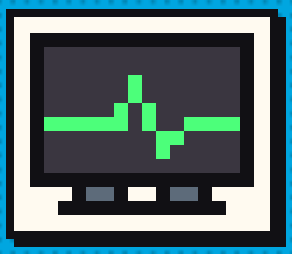
CLAUDE-VIPER

Homepage is the console, Uptime Kuma is the alarm. Next chapter: the snake that wakes you when something falls over.

WORLD 1-9 · CHARACTER SELECT

UPTIME KUMA

Status: checks every app every minute, and shouts when one falls over



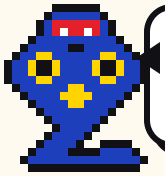
Uptime Kuma is the island's sentry. Every minute or so it knocks on every app, every router and both Pi-holes. When one doesn't answer, it turns the monitor red and sends you a notification. When it comes back, it tells you that too.

It lives in the `gate` pod on 127.0.0.1:3001, behind Authelia's two-factor sign-in at `status.songofeo.com`, with its own admin login on top. Version 2 (the image is pinned to `uptime-kuma:2`). Everything it knows, monitors, history and settings, is in `/srv/gate/uptime-kuma`.

`status.` has no public DNS record: from outside the house the name doesn't resolve at all. At home and over WireGuard the Pi-holes answer it, so you check it from THE-BOX or from the phone with WireGuard on.

STATS

HOST	cobra-gate 10.6.9.10, pod `gate`
IMAGE	`louislam/uptime-kuma:2`
PORT	3001, 127.0.0.1
URL	status.songofeo.com (home only)
DATA	`/srv/gate/uptime-kuma`
SIGN-IN	Authelia password + TOTP, then its own admin
TOUGHNESS	■■■■□
TRAP LEVEL	■■□□□
FUN FACTOR	■■■■□



CLAUDE-VIPER

Homepage's dots tell you when you look. Kuma tells you when you don't. That's the difference between a dashboard and an alarm.



CHECKS

Web pages, raw ports, DNS answers, and certificates.

REMEMBERS

Every heartbeat, so you see flaky apps, not just dead ones.

TELLS

Your email or phone app, the moment something changes.

MONITOR TYPE	WHAT TO WATCH (MANUAL 5.7)
HTTP	Every public hostname: tv. music. books. comics. mustang. comedy. arcade. papers. ...
HTTP, ignore TLS errors	Proxmox `https://10.6.9.2:8006`, OPNsense `https://10.6.9.1`, PBS `https://10.6.9.20:8007`
DNS	Both Pi-holes, 10.6.9.3 and 10.6.9.20: resolve tv.songofeo.com, expect 10.6.9.10
TCP port	Samba on the SGIN, 10.6.9.20:445

HOW IT WORKS LEVEL MAP

A PUBLIC-NAME CHECK, FROM INSIDE THE GATE

1 KUMA
check tv.
▶
2 PI-HOLE
tv. = 10.6.9.10
▶
3 NGINX
TLS, server block
▶
4 JELLYFIN
10.6.9.20:8096
▶
5 VERDICT
up or down

WHY PUBLIC NAMES WORK

The gate's DNS servers are the two Pi-holes (set at install: 10.6.9.3 and 10.6.9.20). They answer every `*.songofeo.com` name with 10.6.9.10, so Kuma checking `https://tv.songofeo.com` goes through the gate's own nginx to the app. One monitor tests the certificate, nginx and the app together.

HEARTBEATS

Each check is a "heartbeat": green bar for up, red for down. Kuma keeps the history, so you can see when something has been flaky all week, not just whether it's up right now.

WEBSOCKETS

The dashboard updates live over a websocket. nginx keeps it open for an hour (`proxy_read_timeout 1h` in the admin block).

THE PROTECTED ROOMS

A check on `papers.` from the gate gets Authelia's redirect to the sign-in, not Paperless. That still proves nginx and Authelia are up. To check the app behind it, point a monitor at its back door instead, like `http://127.0.0.1:8000`: Kuma is in the same pod.

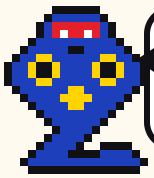
NOTIFICATIONS

Settings, Notifications: Kuma can send to email and many phone and chat services. Make one, then attach it to every monitor (there's a "default enabled" switch so new monitors get it too).

VERSION 2

Uptime Kuma 2.0 went stable in October 2025. The image tag `:2` keeps you on 2.x updates without jumping to a future 3.

YOU SEE	IT MEANS	LOOK AT
One red monitor	That app stopped answering	Its chapter here, then `podman logs`
All SGIN apps red	The SGIN or its network link is down	The SGIN itself (power, USB network adapter)
Everything red	DNS, or nginx	The DNS monitors first, then `podman logs nginx`
Flickering	Up and down all day	The heartbeat bar's history; often a slow app timing out



CLAUDE-VIPER

If BOTH Pi-holes go down, every public-name monitor turns red at once, even though the apps are fine. A wall of red usually means DNS, not twenty broken apps.

★ PRO TIP: RESPONSE TIMES

Each HTTP monitor also graphs how long the app took to answer. A slowly climbing line warns you before the red ever shows up.

TIPS & TRICKS *POWER-UPS*

★ PRO TIP: CLAIM IT FIRST

Kuma's very first visit asks you to create the admin account. Do it on THE-BOX right away, with a generated password saved in Vaultwarden.

★ PRO TIP: MONITOR GROUPS

Group monitors as World 1 (gate apps), World 2 (SGIN apps) and Network (Proxmox, OPNsense, Pi-holes). A whole group going red points straight at the machine.

★ PRO TIP: KEYWORD CHECKS

An HTTP monitor can also look for a word on the page. Checking that `tv.` contains "Jellyfin" catches the case where nginx answers but serves the wrong thing.

★ PRO TIP: THE HISCORE BEAT

When the hiscore board is live, add an HTTP monitor for `http://127.0.0.1:8090/healthz`. Same pod, so no sign-in in the way.

★ PRO TIP: QUIET HOURS

Planning to reboot the SGIN? Pause its monitors, or set a maintenance window, so your phone doesn't buzz for every app on it.

★ PRO TIP: CERT WATCH

HTTP monitors can warn before a certificate expires. Turn it on for one public name: the wildcard cert covers them all.

CODES

GATE · podsvc

```
❯ podman logs --tail 30 uptime-kuma
❯ systemctl --user restart uptime-kuma
❯ du -sh /srv/gate/uptime-kuma
Its history grows over time; this shows how much.
```

★ PRO TIP: STATUS PAGE

Kuma can publish a simple status page of chosen monitors. Handy to show Rachel "is the music server up?" Keep it behind the same gate.

STARTER PACK MONITOR	TYPE	ADDRESS
Cinema	HTTP	`https://tv.songofeo.com`
Music (Eels)	HTTP	`http://10.6.9.20:4533` (back door, skips sign-in)
Paperless	HTTP	`http://127.0.0.1:8000`
Pi-hole 1	DNS	10.6.9.3, resolve tv.songofeo.com, expect 10.6.9.10
Samba	TCP	10.6.9.20 port 445
Proxmox	HTTP, ignore TLS	`https://10.6.9.2:8006`

★ PRO TIP: NAME THEM WELL

Name monitors the way you'd say them out loud ("Music Eels", "Pi-hole 2"), so a 3 a.m. notification makes sense without opening anything.



JOE SABOTAGE REPORT

The sentry can be fooled. Here's how.

! THE UNCLAIMED THRONE

Until someone creates Kuma's admin, the first visitor who reaches it gets to. Authelia's TOTP keeps strangers out, but don't leave it unclaimed.

> **FIX:** Claim it on THE-BOX on the first visit (on the 6 Nov to-do list).

! THE DNS AVALANCHE

Both Pi-holes down: every public-name monitor fails together.

> **FIX:** Look at the two DNS monitors first. If they're red, fix DNS before anything else.

! THE SILENT ALARM

Monitors with no notification attached go red where nobody sees them.

> **FIX:** Settings, Notifications, then attach to every monitor and tick "default enabled".

! THE SIGN-IN SMILE

A check on `papers.` follows Authelia's redirect and reads green even if Paperless is down.

> **FIX:** For admin apps, also watch the back door on 127.0.0.1.

! THE SELF-SIGNED RED

Proxmox, OPNsense and PBS use self-signed certificates, so a normal HTTP check always fails.

> **FIX:** Tick "Ignore TLS/SSL errors" on those three only.

! THE WATCHER'S BLIND SPOT

Kuma lives on the gate. If the gate itself is down, Kuma is down too, and can't tell you.

> **FIX:** Proxmox's own notifications (manual 5.7, SMTP) cover the host. An outside check would be the next step.

! THE OUTSIDE LOOK

status. has no public record, so it won't load on cellular without WireGuard.

> **FIX:** Turn WireGuard on, or check from THE-BOX.

! THE TOO-EAGER BELL

Checks every 20 seconds with no retries buzz your phone for every hiccup.

> **FIX:** Give monitors a retry or two before they count as down.

FIELD TEST

1. THE-BOX: add an HTTP monitor for MeTube's back door, `http://127.0.0.1:8081`. Then GATE, podsvc: `systemctl --user stop metube`.
2. Within a minute or two that monitor goes red and your notification arrives.
3. `systemctl --user start metube`: green again, and a "back up" notice.

WORLD 1-10 · CHARACTER SELECT

HISCORE

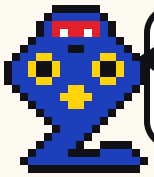
The Scoreboard: every cabinet high score, newest first, kept forever



Hiscore is the island's own arcade scoreboard, written for Cobra Island and kept in this repo (apps/hiscore). Whenever a game on the cabinet saves a new high-score file, it's uploaded within seconds; the board at scores.songofeo.com lists every game, newest score first, and keeps every version byte for byte.

It's a small Python app that uses only Python's standard library, so there's no image to build: the stock `python:3.13-slim` image runs it straight from `/srv/gate/hiscore-app`. Its data, every `.hi` file ever received plus the index `hiscore.db`, lives in `/srv/gate/hiscore`.

Status: part 7 of the manual is written and the app is tested, but the board isn't running on the gate yet. check.sh shows `scores.` as NOTE ("no server block yet"), and Homepage's High scores tile stays red until it is. It comes after the cabinet build.



CLAUDE-VIPER

Three letters, one score, eternal glory. Every Joe who ever set a record on our cabinet will be immortalised. Then beaten. By us.

STATS

HOST	cobra-gate 10.6.9.10, pod `gate`
IMAGE	`python:3.13-slim`
PORT	8090, 127.0.0.1
URL	scores.songofeo.com
APP	`/srv/gate/hiscore-app`
DATA	`/srv/gate/hiscore`
SIGN-IN	Authelia password; cabinet uses an API key
TOUGHNESS	■■■□□
TRAP LEVEL	■■□□□
FUN FACTOR	■■■■■

COMING SOON

PATH ON SCORES.	WHO GETS IN	WHAT IT IS
`/`	Signed in with Authelia (one_factor)	The board: cards per game, NEW badge for the last 24 hours
`/upload/`	The cabinet, with its API key (PUT only)	Where the agent and plugin send `.hi` files
`/screen`	Anyone (GET only)	The big attract-mode screen for songofeo.com/arcade
`/public/`	Anyone, only if `HISCORE_PUBLIC=1`	Titles, initials and scores, nothing else

KEEPS

Every version of every `.hi` file, downloadable from its card.

IGNORES

An upload identical to the last one, so quitting without a record bumps nothing.

SHOWS

Rank, initials and score once something can read the file.

HOW IT WORKS LEVEL MAP

A NEW RECORD ON THE CABINET

1 RETROARCH writes galaga.hi ▶ 2 CABINET plugin + agent ▶ 3 NGINX PUT /upload/ ▶ 4 HISCORE 127.0.0.1:8090 ▶ 5 BOARD NEW badge

TWO UPLOADERS

The **LaunchBox plugin** uploads the moment a game closes and adds what only LaunchBox knows: the proper title and platform. The **hiscore agent** checks every 15 seconds, catches games started outside LaunchBox, and retries while the network is down. Both upload; the server ignores a file it already has.

LOCKED-DOWN NGINX

Its server block opens only three paths without Authelia: `/upload/` (PUT only, 256 KB max), `/public/` and `/screen` (GET only). Everything else, the board itself, needs the sign-in. Viewers need a password, not TOTP.

THE API KEY

`python -m hiscore hash-key cabinet` prints a key and a line for `keys.txt`. The cabinet keeps the key; the server keeps only its **hash** in `/srv/gate/hiscore/keys.txt`. To replace a key, make a new one and swap the line.

THE ARCADE SCREEN

`/screen` cycles through each game's top scores with a ticker of the latest. The server lets `songofeo.com` show it in a frame and no other site. Options: `?embed=1`, `?seconds=8`, `?game=galaga`, `?demo=1`.

WHAT'S IN A .HI FILE

A copy of the game's memory, not text. Reading it needs a per-game definition: hi2txt on the cabinet through the plugin, or a decoder on the server (`HISCORE_DECODER`). Games nothing can read still list, as "Score table saved".

PUBLIC OR NOT

With `HISCORE_PUBLIC=1` (the Quadlet in part 7.3 sets it), anyone can read titles, initials and scores, so the screen on songofeo.com works for visitors. Without it, visitors see demo scores and only signed-in people see real ones.

CARD SHOWS	COMES FROM
Game title	LaunchBox, through the plugin; or `titles.tsv`; or the ROM name until then
System	The RetroArch core folder, e.g. `fbneo`
Last updated	When the server received a changed file
Versions + download	Every distinct upload, kept in `raw/`
Score table	hi2txt on the cabinet, or a server decoder

THE PUBLIC BOARD, ONE ENTRY


```
{"game": "galaga", "title": "Galaga", "system": "fbneo", "scores": [{"rank": "1", "score": "52310", "name": "EEL"}]}
```


`/public/v1/board`: newest first, up to 100 games (`?limit=` up to 500), at most 10 scores each.

TIPS & TRICKS POWER-UPS

★ PRO TIP: ONE GAME ONLY

`/screen?game=galaga` shows a single game's table: perfect for a tournament night on the marquee.

★ PRO TIP: IS IT ALIVE?

`/healthz` answers `{"ok": true}`. The image has no curl, so ask with Python from inside.

CODES

GATE · podsvc

```
❯ podman exec hiscore python -c "import urlli
b.request; print(urllib.request.urlopen('htt
p://127.0.0.1:8090/healthz').read())"
❯ podman logs --tail 30 hiscore
❯ systemctl --user restart hiscore
```

★ PRO TIP: REAL TITLES

Put a `titles.tsv` in `/srv/gate/hiscore/`, one `romname`, tab, `Title` line per game, and restart. ROM names become proper names even without the plugin.

★ PRO TIP: TRY THE SCREEN

On THE-BOX, open `https://scores.songofeo.com/screen?demo=1` to see attract mode with made-up games before any real score exists.

★ PRO TIP: A NEW KEY

Run the hash-key command from the app folder. It prints the key (into Vaultwarden it goes) and the line for `keys.txt`.

CODES

GATE · podsvc

```
❯ cd /srv/gate/hiscore-app
❯ podman run --rm -v "$PWD":/app:ro,Z -w /app
docker.io/library/python:3.13-slim python -m h
iscore hash-key cabinet
```

★ PRO TIP: UPDATE THE APP

New version in the repo? Copy `apps/hiscore/hiscore` to the gate (scp from THE-BOX), replace `/srv/gate/hiscore-app/hiscore`, restart. With the run scripts, `hiscore.sh` copies it from a repo checkout by itself.

★ PRO TIP: DOWNLOAD A RECORD

Every card has a download link for each version. Lost the cabinet's disk? Your scores aren't lost.

★ PRO TIP: SCORE THE CHECK

A bad key must get 401: `curl -X PUT -H "Authorization: Bearer wrong" --data-binary x https://scores.songofeo.com/upload/test/x.hi` from THE-BOX.

WHERE THE FILES LAND

`/srv/gate/hiscore/raw/fbneo/galaga.hi/<time>-<hash>.hi`

One folder per system and game, one file per version, named by when it arrived and its fingerprint. `hiscore.db` next to `raw/` is the index of titles and tables. All of it is inside VM 110, so the nightly backup to PBS covers every score ever set.



JOE SABOTAGE REPORT

The board isn't live yet, so these are the traps waiting for launch day.

! THE MISSING KEY FILE

No `keys.txt`, or a wrong line in it, and every cabinet upload gets 401. The board stays empty.

> **FIX:** One line: `cabinet`, a space, the hash. Mode 600. The script warns when the file is empty.

! THE DEMO-ONLY SCREEN

`hiscore.sh` doesn't set `HISCORE_PUBLIC`; the part 7.3 Quadlet does. Run by script and visitors see demo scores on songofeo.com/arcade.

> **FIX:** Script mode: put `HISCORE_PUBLIC=1` in `~/.config/cobra/hiscore.env` and rerun.

! THE FORGOTTEN RULE

Add the nginx block but not Authelia's rule, and scores. answers 403: default deny.

> **FIX:** Add `scores.songofeo.com` with `policy: one_factor`, then restart nginx and authelia.

! THE NO-DNS NAME

Without a Porkbun CNAME, scores. works at home (the Pi-holes answer every name) but not from outside.

> **FIX:** CNAME `scores` to `gate.songofeo.com`.

! THE BIG UPLOAD

nginx caps `/upload/` at 256 KB, matching the server's default. Some odd file bigger than that is refused.

> **FIX:** That's on purpose: .hi files are tiny. Raise both limits only if a real game needs it.

! THE UNREADABLE CARD

A card says "Score table saved" but shows no initials. Nothing can read that game's file yet.

> **FIX:** hi2txt on the cabinet (7.6) or a server decoder (7.8). The file is kept either way.

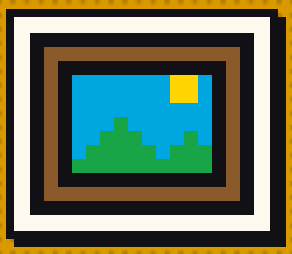
LAUNCH DAY CHECKLIST

- [ ] App in `/srv/gate/hiscore-app`, key line in `keys.txt`
- [ ] Container up, `/healthz` says ok
- [ ] nginx block, Authelia rule, Porkbun CNAME
- [ ] Cabinet agent log says `uploaded fbneo/<game>.hi (201)`
- [ ] Uptime Kuma monitor and Homepage tile go green

WORLD 1-11 · CHARACTER SELECT

EO-ART

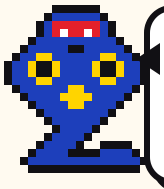
The Gallery: one random picture at a time for the station's landing pages



Eo-art is the station's picture hanger. When someone opens a Song of Eo landing page, the page asks eo-art for a picture, and eo-art hands back one random image from the right folder: a movie backdrop for the Cinema, album covers for Music, book, comic and game-book covers for the Library, Comics and the Game Cellar.

It's a single standard-library Python file, `eo_art.py`, from the Project-Eo repository (`home/eo-art/eo_art.py`). The stock `python:3-slim` image runs it from `/srv/gate/eo-art`, mounted read-only, on 127.0.0.1:8790 inside the `gate` pod.

The pictures come from the SGIN over the CIFS mount: `dock1a/art`, with five flat folders of `.jpg` copies. eo-art only reads them, and rescans every two minutes. An empty folder simply means that page shows its own drawn artwork instead.



CLAUDE-VIPER

This is the fun window bay the Commander wants back on the station's front. The Song of Eo exterior redesign is another crew's mission, but the pictures come from right here.

STATS

HOST	cobra-gate 10.6.9.10, pod `gate`
IMAGE	`python:3-slim`
PORT	8790, 127.0.0.1
URL	art.songofeo.com (home only)
APP	`/srv/gate/eo-art/eo_art.py`
PICTURES	`dock1a/art/*`, read-only
SIGN-IN	Like the floors: open at home, password outside
TOUGHNESS	■■■□□
TRAP LEVEL	■■□□□
FUN FACTOR	■■■■■

NEW PICTURE EVERY VISIT

FOLDER IN DOCK1A/ART	SETTING	FEEDS
`movies`	`EO_ART_MOVIES`	Cinema backdrops
`albums`	`EO_ART_RADIO`	Music and radio covers
`books`	`EO_ART_BOOKS`	Library covers
`comics`	`EO_ART_COMICS`	Comics covers
`gaming`	`EO_ART_GAMING`	Game Cellar game-book covers

HOW IT WORKS LEVEL MAP

THE CINEMA PAGE ASKS FOR A BACKDROP

1 STATION
Cinema page
▶
2 NGINX
art. floor block
▶
3 AUTHELIA
home or signed in
▶
4 EO-ART
/cinema/backdrop
▶
5 PICTURE
random .jpg

A FLOOR, NOT A ROOM

`art.songofeo.com` is in nginx's `$station` map, next to music. and books., and in Authelia's floor rule. So it opens straight away at home and over WireGuard, and asks for one password elsewhere. Signed in on the station, the landing pages' pictures just load.

NO PUBLIC NAME (YET)

Manual 5.5 gives `art` no Porkbun record for now. At home and over WireGuard the Pi-holes answer it; from outside it doesn't resolve, so visitors there see each page's drawn artwork instead of live pictures.

THE PLAYERS' CORS FILE

The station's pages live on Porkbun's hosting and call the floors from there, so nginx answers them with CORS headers from `station-cors.inc`. That file came from Project-Eo (`home/nginx/station-cors.inc`) with eo_art.py.

RESCANS ITSELF

eo-art looks at its five folders every two minutes. Drop new pictures in and they join the rotation without a restart.

404 IS FINE

An empty folder answers 404, and the page falls back to its drawn art. The run script's health check counts any HTTP answer, 404 included, as "up".

NO CURL INSIDE

`python:3-slim` has no curl or wget, so health checks ask with Python's `urllib` from inside the container.

READ-ONLY EVERYTHING

The script folder is mounted `:ro` and the pictures are mounted read-only. eo-art can't change a single file, on the gate or on the SGIN.

EO-ART ANSWERS	THE PAGE SHOWS
A picture	Your picture, picked at random from that folder
404 (folder empty)	The page's own drawn artwork
Sign-in redirect (outside, not signed in)	Drawn artwork until you sign in on the station
Nothing (eo-art down)	Drawn artwork; nginx logs a 502

WHEN THE SGIN NAPS

Like Paperless and MeTube, eo-art needs the SGIN's share. Its Quadlet has `Restart=on-failure` every 60 seconds, so it comes back once the share does, and `up-all.sh` lists it as "did not start" without holding up the rest of the gate.

TIPS & TRICKS POWER-UPS

★ PRO TIP: HANG PICTURES

On THE-BOX, open `\\cobra-stream\media\dock1a\art` in Explorer and copy `.jpg` files into the five folders. Copies, not your library: eo-art shows whatever's there.

★ PRO TIP: WIDE FOR CINEMA

Backdrops fill a wide page: pick landscape pictures for `movies`. Covers for the other four can be portrait.

★ PRO TIP: SEE ONE NOW

On THE-BOX's browser, open `https://art.songofeo.com/cinema/backdrop`. Reload for a new one. 404 means the movies folder is still empty.

★ PRO TIP: KEEP IT LIGHT

Every visit downloads a picture. A few hundred KB each keeps the pages snappy on a phone; there's no need for 20 MB originals.

CODES

GATE · podsvc

```
❯ podman logs --tail 30 eo-art
❯ systemctl --user restart eo-art
❯ ls /srv/storage/disks/dock1a/art/movies | head
```

What it can see in one folder.

★ PRO TIP: UPDATE THE SCRIPT

A new eo_art.py in Project-Eo? Copy it from THE-BOX, install it over the old one as podsvc's file, restart.

CODES

THE-BOX · PowerShell

```
❯ scp .\Project-Eo-main\home\eo-art\eo_art.py eels@10.6.9.10:/tmp/
```

GATE · eels

```
❯ sudo install -o podsvc -m 644 /tmp/eo_art.py /srv/gate/eo-art/eo_art.py
```

★ PRO TIP: CHECK THE NAME

check.sh tests art. with the other floors. It failed on setup night only while eo-art was down.

★ PRO TIP: THEME NIGHTS

Swap the `movies` folder's pictures for a theme (all monster movies in October) and the Cinema's mood changes within two minutes.

★ PRO TIP: NAME NOTHING

File names don't matter to eo-art. `001.jpg` works as well as a title.

THE FIVE FOLDERS

`\\cobra-stream\media\dock1a\art\` movies · albums · books · comics · gaming

Same folders on the gate at `/srv/storage/disks/dock1a/art`, and inside eo-art at `/art`. One path, three views: that's why the manual mounts the share at the same path as on the SGIN.



JOE SABOTAGE REPORT

The gallery's empty frames, and how they got that way.

! THE MISSING SCRIPT

eo_art.py lives in the Project-Eo repository, not in cobra-server, so the gate had no copy and eo-art wouldn't start. art. failed check.sh.

> **FIX:** Copied from the Project-Eo ZIP on THE-BOX by scp on 6 Nov, 01:38Z. Same for station-cors.inc.

! THE MISSING FOLDERS

The run script stops if any of the five `art` folders doesn't exist.

> **FIX:** As eels on the gate: `mkdir -p /srv/storage/disks/dock1a/art/{movies,albums,books,comics,gaming}`.

! THE WRONG FORMAT

The art folders are meant for flat sets of `.jpg` files. Subfolders or other formats may simply be skipped.

> **FIX:** Flat folders of .jpg copies.

! THE BLANK OUTSIDE

On cellular the landing pages show drawn art, not your pictures. That's because art. has no public DNS record yet.

> **FIX:** Expected. Add a CNAME for `art` later if you want live pictures from outside.

! THE LOST HEADER

`station-cors.inc` uses `add_header`, which can silently drop the HSTS header on the floors.

> **FIX:** Test with `curl.exe -sI https://tv.songofeo.com/` and add the HSTS line to the .inc if it's missing (manual 5.4).

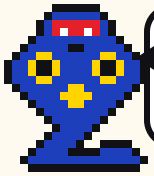
! THE SHARE DOWN

No SGIN share, no pictures, and eo-art restart-loops every minute.

> **FIX:** Fix the CIFS mount first (`ls /srv/storage/disks` as eels).

FIELD TEST

1. THE-BOX: copy three landscape .jpg files into `\\cobra-stream\media\dock1a\art\movies`.
2. Wait two minutes for the rescan.
3. Open `https://art.songofeo.com/cinema/backdrop` and reload a few times: the three take turns.
4. Open the station's Cinema page: the backdrop is yours.



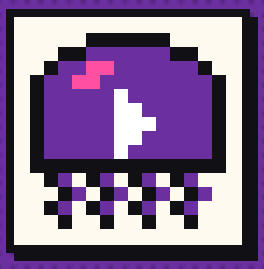
CLAUDE-VIPER

That's World 1 cleared, Commander! The gate is guarded, the vault is sealed, the archive files itself. Grab a power-up and warp to World 2: The Stream.

WORLD 2-1 · CHARACTER SELECT

JELLYFIN

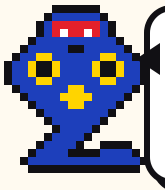
The Cinema: movies, shows and Live TV at tv.songofeo.com



Jellyfin is the headliner of World 2. It reads every movie and show on the drives, dresses them up with posters and descriptions, and plays them on the TVs, phones and browsers of everyone you let in. When a screen can't play a file as it is, Jellyfin re-encodes it on the fly with the SGIN's Intel GPU.

It lives in the `stream` pod on `cobra-stream` (10.6.9.20) and answers on port 8096. Its own files, the database, posters and watch history, sit on the SGIN's internal SSD under `/srv/stream/jellyfin`. The movies themselves come in **read-only** from `/srv/storage/disks`, which the container sees as `/media`. Jellyfin can look at your library all day long; it can never delete a frame of it.

From outside, it's `tv.songofeo.com`, and it is the one floor that **skips the gate's sign-in**. Friends' Roku, Apple TV and phone apps can't show Authelia's login page, so tv. has its own nginx block with no `auth_request`, and Jellyfin's own login does the guarding.



CLAUDE-VIPER

Because tv. has no gate sign-in in front of it, Jellyfin's own passwords are the only lock on that door. Give every friend their own account, and make the passwords real ones, Commander.

STATS

HOST	cobra-stream 10.6.9.20, pod `stream`
IMAGE	`jellyfin/jellyfin:latest`
PORT	8096 (LAN + WireGuard)
URL	`tv.songofeo.com`
DATA	`/srv/stream/jellyfin/config` and `/cache`
LIBRARY	`/media/video/movies`, `/media/video/series`
GPU	`/dev/dri/renderD128` (Quick Sync)
SIGN-IN	Jellyfin's own login only
TOUGHNESS	
TRAP LEVEL	
FUN FACT	

THE ONLY FLOOR WITH NO GATE SIGN-IN

SCREENS THAT PLAY IT

TVs: the Roku app, the Android TV / Fire TV app, and Swiftfin or Infuse on Apple TV. The Android TV app plays HEVC 10-bit itself, so it rarely transcodes.

Computers: Jellyfin Media Player, or any browser at tv.songofeo.com. Chrome and Edge play more formats than Firefox.

Phones: the Jellyfin app on Android and iPhone, or Swiftfin and Infuse on iPhone. Downloads for the plane, if you allow them for that user.

LIBRARY	FOLDER IN JELLYFIN	ON THE DRIVES
Movies	`/media/video/movies`	wd1 + wd2 pooled (mergerfs), wd1 fills first
Shows	`/media/video/series`	Same pool. The folder is **series**, not "shows"
Filler	`/media/wd1/filler`	Short break videos for ErsatzTV and cobra-breaks
YouTube	`/media/wd2/youtube`	MeTube's downloads from the gate, as Home Videos

HOW IT WORKS LEVEL MAP

A FRIEND PRESSES PLAY

1 TV APP
server: https://tv.songofeo.com

▶

2 OPNSENSE
WAN 443 to the gate

▶

3 NGINX
tv. block, no Authelia

▶

4 JELLYFIN
10.6.9.20:8096 login

▶

5 DRIVES
/media/video read-only

THE VIDEO POOL

wd1 (the WD 6 TB) and wd2 (a folder on the Seagate 24 TB) are joined by **mergerfs** into one folder, `/srv/storage/disks/video`. It's only a view: every file sits whole on one drive, and either drive reads fine on its own.

New files land on **wd1 first** (`category.create=ff`) and spill onto wd2 only once wd1 is down to 100 GB free. That matters, because wd2 lives on the backup drive itself and has no second copy. Keep wd2 for things you could download again.

wd1 and wd2 are the **only** drives that are ever pooled. Music, books and the rest stay one-drive-one-job, so you always know where a file lives.

THE THREE WAYS TO PLAY

Direct Play: the screen plays the file as it is. Zero work for the SGIN. **Direct Stream:** only the wrapper or audio changes. Cheap. **Transcode:** the video is decoded and re-encoded live, which is where Quick Sync earns its keep (next page).

HOW THE CONTAINER IS WIRED

- **/dev/dri** is handed in, and `keep-groups` lets the container keep podsvc's `render` group, so it may open the GPU
- **/media** is `ro,rbind`: read-only, and `rbind` carries each drive's own mount in, so `/media/wd1` is really wd1
- **No :Z** on the library, because Samba shares the same folders
- **/transcode-ram**: a 4 GB RAM scratch area (scripts only), if you ever want transcodes off the SSD
- **JELLYFIN_PublishedServerUrl** tells apps the address is `https://tv.songofeo.com`

WHO IS WATCHING?

Every remote viewer arrives through the gate, so without help Jellyfin would think they're all one LAN client called 10.6.9.10. Setting **Known proxies: 10.6.9.10** makes Jellyfin read the real address from nginx's `X-Forwarded-For`. Then remote bitrate limits work, and Dashboard → Activity shows who's really there.

FRIENDS IN THE PIPE

ErsatzTV's channels show up under **Live TV**, and the cobra-breaks helper builds "with breaks" playlists. Both talk to Jellyfin over 127.0.0.1 inside the pod (chapter 2-2 and the tips page).

FIRST-RUN WIZARD

1. **the-box, browser:** `http://10.6.9.20:8096` → language → admin user `eels` and a strong password (save it in Vaultwarden).
2. **Add Media Library:** Movies → `/media/video/movies`; again for Shows → `/media/video/series`. Music and books skip Jellyfin: they live in Navidrome and Audiobookshelf.
3. **Allow remote connections: on.** Newer Jellyfin has no automatic port mapping, which is exactly what we want: the gate is the only way in.
4. Then the Transcoding settings (next page) and **Known proxies: 10.6.9.10**.

QUICK SYNC TURBO MODE HIDDEN POWER

The SGIN's Celeron N5095A is a small chip with a big secret: an Intel GPU with Quick Sync, a block of silicon built only for decoding and encoding video. With it switched on, one tiny laptop board can turn a 4K HEVC movie into 1080p H.264 for a phone without breaking a sweat.

INSIDE A TRANSCODE



THE HUC KEY

Jasper Lake can **only** encode in low-power mode, and low-power mode needs the **HuC firmware** loaded into the GPU. On Debian 13 that firmware moved into its own package, `firmware-intel-graphics`, and the driver option `enable_guc=2` in `/etc/modprobe.d/i915.conf` tells i915 to load it.

Without it, decoding still works, so everything *looks* fine, but every transcode fails. The Quick Sync media driver itself, `intel-media-va-driver-non-free`, lives in Debian's `non-free` section, which the installer leaves switched off.

CODES

SGIN · eels

```
❯ sudo dmesg | grep -i -E 'guc|huc'
```

Want "HuC ... authenticated". No FAIL or ERROR.

```
❯ sudo vainfo --display drm --device /dev/dri/renderD128 | grep -i hevc
```

Want HEVC Main10 decode lines and EncSliceLP lines.

DASHBOARD → PLAYBACK → TRANSCODING	SET
Hardware acceleration	**Intel QuickSync (QSV)**
Device	`/dev/dri/renderD128`
Hardware decoding	H264, HEVC, HEVC 10bit, VP9
AV1 decoding	**OFF** (no AV1 on Jasper Lake)
Hardware encoding	On
Low-Power H.264 / HEVC encoders	**On**
Tone mapping + VPP tone mapping	Both on
Transcode path	Leave the default (internal SSD)

CODES

SGIN · podsvc

```
❯ podman exec jellyfin /usr/lib/jellyfin-ffmpeg/vainfo --display drm --device /dev/dri/renderD128
```

Same test from inside the container. HEVC lines = the container sees the GPU. jellyfin.sh runs this check for you.

★ PRO TIP: WATCH THE ENGINE

While a movie transcodes, `sudo intel_gpu_top` (as eels) shows the **Video** row busy. At 0% while `htop` shows ffmpeg eating the CPU, the transcode is in software. `q` quits.

TIPS & TRICKS POWER-UPS

★ PRO TIP: ONE PER FRIEND

Dashboard → Users → **+**. Per user you choose which libraries they see, whether they may download, and a remote bitrate cap. The manual sets users **hidden** (no name list on the login screen) and **Quick Connect off**, so nobody can pair a TV to an account without a password.

★ PRO TIP: APP ADDRESS

In any Jellyfin app, the server is `https://tv.songofeo.com` with **no port**. Same address at home, on cellular and on WireGuard; at home the Pi-holes send it straight to the gate inside the house.

★ PRO TIP: EPISODE BREAKS

Jellyfin apps only ask for intros before **movies**, never episodes, and the Roku and Apple TV apps ignore web plugins. So **cobra-breaks** (manual part 6) builds a playlist per show and person, "<Show> — with breaks": episode, filler, episode. It starts at each person's next unwatched episode and rebuilds every day at 4:00.

Every break must be a different video (Jellyfin drops repeats from a playlist), so 40 episodes need **39+ filler videos** of 3 minutes or less in `wd1/filler`.

CODES

SGIN · podsvc

```
❯ podman exec cobra-breaks python -m cobra_breaks run --config /config/breaks.toml
```

Rebuild the playlists now, after editing breaks.toml or adding filler.

★ PRO TIP: LIVE TV

Jellyfin → **Live TV** shows ErsatzTV's channels with a real program guide, on every app, at home and away. Remote viewers watch through Jellyfin; ErsatzTV itself never leaves the LAN.

★ PRO TIP: MOVIE PRE-ROLLS

Movies get their intro from the **Local Intros** plugin, fed from `wd1/pre-rolls`. Live TV channels get breaks from ErsatzTV. Three kinds of break, three different tools, and every app plays all of them.

★ PRO TIP: WHY TRANSCODE?

In the web player, ⚙ → **Playback Info**, or Dashboard → Activity, names the play method and the **reason**. That one word tells you which fix on the next page to use.

CODES

SGIN · podsvc

```
❯ podman logs --tail 50 jellyfin
❯ podman restart jellyfin
```

Restart is safe: config and watch history live in /srv/stream/jellyfin.

```
❯ podman pod restart stream
```

After plugging a drive back in: every app gets the new mount.

★ PRO TIP: API KEYS

Dashboard → **API Keys** → **+**. Make one key per helper (`ersatztv`, `cobra-breaks`) so you can revoke one without breaking the others.



CLAUDE-VIPER

New movies? Drop them in `\\cobra-stream\media\video\movies` from the-box. The pool picks the drive.

STUTTER BUSTERS MANUAL 3.SC

"Only some movies stutter" is almost never the file size. A 4 GB two-hour movie averages about 4.5 Mbps, which anything here handles easily. The big files are usually HEVC 10-bit or HDR, often with PGS or ASS subtitles, and when the screen can't play those, Jellyfin transcodes. The stutter comes from that transcode.

FIND THE REASON FIRST!

REASON SHOWN	FIX
Subtitle codec not supported	The subtitles are being burned in. Turn them off and retest; then pick an SRT track, or set the user's subtitle mode to *Smart* with image subtitles off by default. The single most common cause.
Video codec not supported	Use a client that plays HEVC 10-bit itself: the Android TV / Fire TV app, Jellyfin Media Player, Swiftfin or Infuse. In a browser, Chrome or Edge rather than Firefox.
Bitrate exceeds limit	Client app → Maximum streaming quality → Auto or highest. Over WireGuard, Dashboard → Playback → Streaming → raise the remote limit.
Audio codec not supported	Cheap, rarely the stutter. Allow passthrough, or pick the AC3 track.

STILL STUTTERS?

Dashboard → Logs → newest `FFmpeg.Transcode-…log`. The encoder must be `h264_qsv` or `hevc_qsv`. **libx264** or **libx265** means the CPU is doing it: recheck the HuC page.

SMOOTH BUT JERKY

Never buffers, but pans judder? That's a 24 fps film on a 60 Hz TV. Turn on **match frame rate** (refresh rate switching) in the TV's app.

WIRED VS WI-FI

Retest once with the TV on Ethernet. Smooth wired but not on Wi-Fi means the Wi-Fi is the problem, not Jellyfin. From outside, Xfinity's upload speed is the cap.

CODES

SGIN · eels

```
> sudo dmesg -T | grep -i -E 'uas|reset|i/o error' | tail -20
> lsusb -t
```

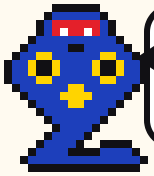
Every drive at 5000M. 480M = it fell back to USB 2.

★ PRO TIP: FOREVER FIX

The best transcode is the one that never happens. A better app on the one TV that keeps transcoding fixes the whole library at once.

THE STUTTER SPEEDRUN

1 PLAY the bad movie ▶ 2 INFO Playback Info: the reason ▶ 3 FIX the table above ▶ 4 LOG h264_qsv or libx264? ▶ 5 GPU vainfo + intel_gpu_top



CLAUDE-VIPER

Is the SGIN the right box for this? Yes. With Quick Sync working end to end it handles several 4K-to-1080p transcodes at once. Moving Jellyfin to the Latitude wouldn't fix a stutter; finding the reason does.



JOE SABOTAGE REPORT

Pulled from the footlockers, manual parts 3.3 to 3.8c, and the build itself.

! THE MISSING HUC

Decode works, the movie list loads, the first transcode dies. Debian 13 moved the GPU firmware out of `firmware-misc-nonfree`.

> **FIX:** Install `firmware-intel-graphics`, keep `enable_guc=2`, rebuild the initramfs, reboot, and look for EncSliceLP in vainfo.

! THE AV1 TICKBOX

Ticking AV1 under hardware decoding looks harmless. Jasper Lake has no AV1 decoder, so AV1 files then fail outright.

> **FIX:** Leave AV1 unticked; Jellyfin decodes those few files in software.

! THE LOCKED GPU DOOR

podsvc not in `render`, or the container started without `keep-groups`: the container can't open `/dev/dri` and everything falls back to the CPU.

> **FIX:** As eels: `sudo usermod -aG render,video podsvc`, then log podsvc in again and rerun jellyfin.sh. Its QSV check says pass or fail.

! THE GHOST DRIVE

A container sees the drives that were mounted **when it started**. Replug wd1 and Jellyfin keeps looking at an empty folder, and a scan then marks everything missing.

> **FIX:** `podman pod restart stream` first, scan after.

! EVERYONE IS THE GATE

Without Known proxies, every remote viewer looks like 10.6.9.10 on the LAN. Remote bitrate limits never kick in and the logs are useless.

> **FIX:** Dashboard → Networking → Known proxies: `10.6.9.10`. Check: play from the-box, Activity shows the-box's address.

! THE FIRST-VISITOR CROWN

Until the setup wizard is finished, whoever opens the page first becomes admin.

> **FIX:** Finish the wizard on the-box right after the container first starts (admin `eels`).

! THE "SHOWS" MIX-UP

The TV folder on wd1 and wd2 is called **series**. Point a library at `/media/video/shows` and it stays empty forever.

> **FIX:** Shows library = `/media/video/series`.

! THE POWER-SAVER AMBUSH

Laptop guides say "install TLP" or "run powertop --auto-tune". On the SGIN both switch on USB autosuspend, which drops the hub, every drive **and the network adapter**, its only link.

> **FIX:** Never install TLP or auto-tune here. Looking with powertop is fine.

! THE :Z RELABEL

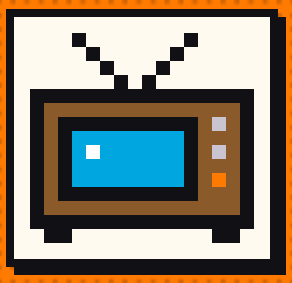
`:Z` on `/srv/storage/disks` would relabel the whole library for one container, and Samba shares those folders too.

> **FIX:** App data gets `:Z`; the library never does.

WORLD 2-2 · CHARACTER SELECT

ERSATZTV

The Broadcast Tower: real TV channels made from your own library



Remember flipping channels and landing on a rerun you hadn't picked? ErsatzTV brings that back. It builds always-on TV channels out of your movies and shows, complete with a schedule, a program guide and commercial breaks from your filler folder.

It doesn't have viewers of its own. ErsatzTV hands Jellyfin two links, a **channel list** (M3U) and a **program guide** (XMLTV), and Jellyfin shows the channels under **Live TV** on every app: Roku, Apple TV, phones, browsers. Tune in and ErsatzTV starts playing that channel's schedule from wherever "now" is, transcoding it live on the Quick Sync GPU.

Both live in the `stream` pod, so they talk to each other over `127.0.0.1`. ErsatzTV answers on port 8409 for the LAN and WireGuard, has **no login page at all**, and the gate never publishes it. People away from home watch the channels through Jellyfin at tv.songofeo.com instead.



CLAUDE-VIPER

Picture it, Commander: "Cobra Channel 1, all monster movies, all night", with your old bumpers between them. The Joes will never find the remote.

STATS

HOST	cobra-stream 10.6.9.20, pod `stream`
IMAGE	`ghcr.io/ersatztv/legacy:latest`
PORT	8409 (LAN + WireGuard only)
URL	`http://10.6.9.20:8409`
DATA	`/srv/stream/ersatztv`
SCRATCH	`/transcode`, 2 GB in RAM
GPU	`/dev/dri`, VAAPI
SIGN-IN	None. Never publish it
TOUGHNESS	■■■□□
TRAP LEVEL	■■■□□
FUN FACT	■■■■■

NOW SHOWING: YOUR LIBRARY

CHANNEL IDEAS

COBRA 1, "Late Show": a smart collection of horror and sci-fi movies, shuffled, with a Pad preset so each film starts on the half hour.

COBRA 2, "Saturday Morning": a playlist of cartoon series in order, a pre-roll of old commercials before every episode.

COBRA 3, "Marathon": one show, chronological, forever. Pick it up where the channel is, like the old days.

THREE KINDS OF BREAK

Live TV channels: ErsatzTV's own filler presets, from `wd1/filler`. This chapter.

Episodes on demand: cobra-breaks' "with breaks" playlists in Jellyfin (manual part 6).

Movies: a pre-roll from the Local Intros plugin, fed from `wd1/pre-rolls`.

HOW IT WORKS LEVEL MAP

CHANNEL 3 GOES ON THE AIR

1 JELLYFIN library synced in ▶ 2 SCHEDULE collections + filler ▶ 3 PLAYOUT built timeline ▶ 4 ERSATZTV :8409 M3U + XMLTV ▶ 5 LIVE TV in every Jellyfin app

WHERE THE SHOWS COME FROM

Media Sources → Jellyfin: ErsatzTV logs into Jellyfin at `http://127.0.0.1:8096` with an API key and syncs the Movies and TV libraries, posters and all.

Media Sources → Local: a library of type *Other Videos* on `/media/wd1/filler` for the break videos.

Both containers mount `/srv/storage/disks` at the same `/media` path, read-only. So a file path Jellyfin reports is one ErsatzTV can open too. That shared path is the whole trick.

THE GPU, AGAIN

Every channel someone is watching is a live transcode, so ErsatzTV gets the same `/dev/dri` and `keep-groups` wiring as Jellyfin. Its **FFmpeg profile** says: hardware acceleration **VAAPI**, driver **iHD**, device `/dev/dri/renderD128`, **H.264** (plays everywhere), 1080p.

THE BUILDING BLOCKS

BLOCK	WHAT IT IS
Collection	"All of this show" or "these 40 movies"
Smart collection	A saved search, e.g. `genre:comedy`
Playlist	A fixed order you choose, mixing shows and movies
Filler preset	Breaks: Pre-roll before each item, or Pad to the half hour
Schedule	Items + order (chronological, shuffle, random) + filler
Channel	Number, name, logo, FFmpeg profile, MPEG-TS
Playout	Channel + schedule, built into a timeline

WHY RAM?

While someone watches, ErsatzTV writes the channel's live video segments to `/transcode`. That's a 2 GB **tmpfs**, a folder in RAM, so the constant little writes never wear the SGIN's SSD.

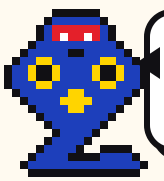
THE TWO MAGIC LINKS

`http://127.0.0.1:8409/iptv/channels.m3u`

`http://127.0.0.1:8409/iptv/xmltv.xml`

WHAT THE GUIDE LOOKS LIKE

CH	8:00 PM	8:30 PM	9:00 PM	9:30 PM
1	Movie from the "Late Show" smart collection, padded to 9:30			Next movie
2	Episode 4	Episode 5	Episode 6	Episode 7
3	Two-part special		Episode 12	Episode 13



CLAUDE-VIPER

This is the grid Jellyfin draws from the XMLTV guide. And ErsatzTV never copies a single movie: channels are just a timeline of pointers into the same read-only library Jellyfin uses, so a whole lineup costs almost no disk space.

TIPS & TRICKS POWER-UPS

BUILD A CHANNEL IN 6 MOVES

1. **Jellyfin** (the-box browser) → Dashboard → API Keys → **+** → name `ersatztv` → copy the key.
2. **ErsatzTV** `http://10.6.9.20:8409` → Media Sources → Jellyfin: `http://127.0.0.1:8096` + the key → Save → tick Movies and TV → Sync. Then Local → Other Videos → `/media/wd1/filler` → Scan.
3. Settings → FFmpeg Profiles → VAAPI, iHD, `/dev/dri/renderD128`, H.264, 1080p.
4. Lists → Collections (or Smart Collections, Playlists) → Filler Presets → Schedules.
5. Channels → Add → Playouts → Add → **Build**.
6. Jellyfin → Dashboard → Live TV: **M3U Tuner** with the channels link, **XMLTV** with the guide link → Refresh Guide Data.

★ PRO TIP: REAL TV TIMING

A **Pre-roll** filler preset (Mode: Count 1) drops one break before each episode. Add a **Pad** preset too and ErsatzTV fills each episode out to the next 30-minute mark, so shows start on the hour and half hour like real TV.

★ PRO TIP: THE FILLER STASH

Old commercials, bumpers, station IDs, trailers: drop them in `\\cobra-stream\media\wd1\filler` from the-box, subfolders allowed. ErsatzTV, cobra-breaks and Jellyfin's Filler library all use that one folder. Aim for 50+.

★ PRO TIP: THEME NIGHTS

A **smart collection** is a saved search that updates itself. `genre:comedy` today includes the comedy you add next month, with no rebuild of the list.

CODES

SGIN · podsvc

```
❯ podman logs --tail 50 ersatztv
❯ podman restart ersatztv
```

Channels, schedules and playouts live in /srv/stream/ersatztv, so a restart loses nothing.

THE-BOX · browser

```
❯ http://10.6.9.20:8409/iptv/channels.m3u
```

Opens the channel list as text: proof it's alive.

★ PRO TIP: REBUILD

Changed a schedule a lot? Playouts → **Build** again. The guide in Jellyfin catches up on its next refresh, or press Refresh Guide Data yourself.

★ PRO TIP: CHANNEL LOGOS

Each channel takes a number, a name and a **logo**. A little pixel logo per channel makes the Jellyfin guide look like a real cable box.

★ PRO TIP: WATCH FROM AWAY

On cellular, open Jellyfin at `https://tv.songofeo.com` → Live TV. The channel comes through Jellyfin and the gate; ErsatzTV never needs a public name.



CLAUDE-VIPER

Start with one channel and one schedule. Get it playing in Jellyfin end to end, then build the rest of the lineup.



JOE SABOTAGE REPORT

Footlocker findings from manual 3.8a and the research notes.

! THE OPEN TOWER

ErsatzTV has no login screen. Anyone who reaches port 8409 can rewrite every channel.

> **FIX:** Never give it a gate hostname. nftables keeps 8409 LAN and WireGuard only; remote viewers use Jellyfin's Live TV.

! THE WRONG HOST

The M3U and XMLTV buttons in ErsatzTV copy links with whatever address your browser used. Paste `10.6.9.20` or a hostname and Jellyfin goes the long way, or not at all.

> **FIX:** Replace the host part with `127.0.0.1:8409` before pasting into Jellyfin.

! THE VAAPI HICCUP

The manual marks the VAAPI profile "verify": some builds error on it.

> **FIX:** If channels won't start, try **QSV** with the same device in the FFmpeg profile.

! THE IMPOSTOR IMAGE

Old guides say `ghcr.io/ersatztv/ersatztv`. The maintained line is `ghcr.io/ersatztv/legacy` (v26.x). A rewrite, ErsatzTV Next, is still in early testing.

> **FIX:** Stay on `legacy` until Next calls itself stable.

! THE DEAD KEY

Delete or recreate the `ersatztv` API key in Jellyfin and the sync silently stops finding new episodes.

> **FIX:** New key → ErsatzTV → Media Sources → Jellyfin → paste → Sync.

! THE EMPTY FILLER

A channel with a filler preset but an empty or unscanned filler library plays straight through with no breaks, or gaps.

> **FIX:** Fill `wd1/filler`, then Media Sources → Local → Scan.

! THE GHOST DRIVE, PART 2

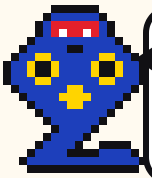
Just like Jellyfin, ErsatzTV only sees drives mounted when it started. Replug wd1 and the channels can't open their files.

> **FIX:** `podman pod restart stream`, then rescan.

! THE EMPTY GUIDE

Channels added in Jellyfin, but the guide shows "no information" for hours.

> **FIX:** Dashboard → Live TV → **Refresh Guide Data** after each new channel or playout build.



CLAUDE-VIPER

Channels transcode only while someone watches. Nobody tuned in, nobody working: the GPU stays free for Jellyfin.

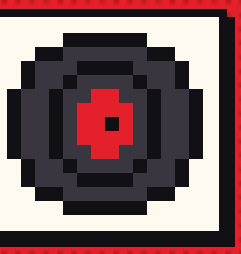
ON-AIR CHECKLIST

- [ ] Jellyfin source synced, filler library scanned
- [ ] FFmpeg profile on VAAPI (or QSV), H.264, 1080p
- [ ] Playout built for every channel
- [ ] M3U and XMLTV links use 127.0.0.1:8409
- [ ] Guide data refreshed in Jellyfin
- [ ] A channel plays on a phone on cellular via tv.

WORLD 2-3 · CHARACTER SELECT

NAVIDROME

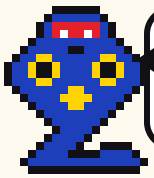
The Music Floor: one jukebox, a shelf for each person



Navidrome is a small, fast music server. Point it at a folder of songs and it reads their tags, builds an album and artist library, and plays it in a web player or any Subsonic app on your phone. Cobra Island runs one of it.

Since version 0.58 one Navidrome can hold several **libraries**, and the admin ticks which libraries each user may see. The Commander's music is library **Eels**, Rachel's is library **Rachel**. Rachel's account only sees hers; the admin sees both. Her music never lands in your shuffle unless you pick her shelf too.

Two names lead to the same jukebox: `music.` and `mustang.`. Stand-up and The Ward moved to Audiobookshelf, which remembers where you stopped in a long track (the last two pages of this chapter).



CLAUDE-VIPER

One jukebox, two shelves, one set of keys per person. Less to feed, less to break, Commander.

STATS

HOST	cobra-stream 10.6.9.20, pod `stream`
IMAGE	`deluan/navidrome:latest`
PORT	4533
LIBRARIES	`music/eels`, `music/rachel`, read-only
DATA	`/srv/stream/navidrome`
SESSION	8 hours
SIGN-IN	Gate sign-in, then trusted
TOUGHNESS	
TRAP LEVEL	
FUN FACTOR	

1 JUKEBOX 2 SHELVES

LIBRARY	FOLDER	SEEN BY	NAMES
Eels (Library 1)	`dock1a/music/eels`	eels	music.
Rachel	`dock1a/music/rachel`	rachel, eels	mustang. (or music.)

★ PRO TIP: QUICK CHECK

From the-box, `http://10.6.9.20:4533` → Settings → Libraries lists both shelves, and Settings → Users shows who sees which.

WHERE IT CAME FROM

Four separate Navidromes (PR #46) and the N.A.V. clones (PR #52) were folded into one on 8 November 2026, the Commander's call. The old databases are kept in a backup.

CDS IN, MUSIC OUT

Navidrome **plays** ripped music; it doesn't rip CDs. The path from a disc to your phone: rip it on the-box, fix the tags with MusicBrainz Picard, copy the album folder into `music\eels` or `music\rachel` on `\\cobra-stream\media\dock1a`, and Navidrome finds it on its next scan.

HOW IT WORKS LEVEL MAP

RACHEL OPENS MUSTANG. ON CELLULAR

1 BROWSER
mustang.songofeo.com

▶

2 AUTHELIA
signs in, says "rachel"; the music wall lets her by

▶

3 HEADER
Remote-User: rachel

▶

4 NAVIDROME
10.6.9.20:4533, her shelf only

ONE SIGN-IN, NOT TWO

Navidrome runs with `ND_REVERSEPROXYWHITELIST=10.6.9.10/32`: "if a request comes from the gate and carries a `Remote-User` header, that person is already signed in." Authelia fills in the header, so Navidrome lets you straight in. It only believes that header from 10.6.9.10, and nginx always **wipes** any Remote-User header a visitor sends.

THE MUSIC WALL

Navidrome makes an account for anyone the gate signs in. So Authelia has a wall in front of `music.` and `mustang.`: only the people listed (eels and rachel) get through, everyone else gets *forbidden*. `gate-music-one.sh` writes it; add a name with `MUSIC_USERS`.

SHELVES DON'T OVERLAP

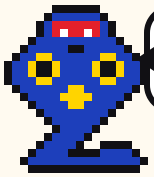
Navidrome won't let one library sit inside another. That's why the main shelf is `music/eels` and not all of `music/`: then `music/rachel` can be a shelf of its own. A person sees whole shelves, never single folders inside one.

WHY ONE AND NOT FOUR?

Four Navidromes don't serve four people any faster. One already streams to many at once; playing a song is just reading a file, and a transcode costs the same CPU in any container. Extra instances only add RAM, another scan of the same disk, and another password per person.

WHAT EACH PERSON KEEPS

Stars, ratings, playlists and play counts belong to each user. Phone apps only get the shelves that user may see, and search, albums and artists are filtered the same way.



CLAUDE-VIPER
Make Rachel's account yourself before she opens it, with only her shelf ticked. Then her first look is already the right one.

WHAT'S IN THE DATA FOLDER

`/srv/stream/navidrome` holds the database: users, shelves, playlists, stars and play counts, plus a cache of covers. Back that up; the music itself lives on dock1a.

WHAT THE GATE SEES

nginx's `$station` map sends music. and mustang. to :4533, and comedy. and sessions. to Audiobookshelf on :13378, all on 10.6.9.20. One server block for all of them.

TIPS & TRICKS POWER-UPS

★ PRO TIP: PHONE APPS

Navidrome speaks **Subsonic**, so any Subsonic app works: Symfonium or Substreamer on the phone, Feishin on the desktop. Server `https://music.songofeo.com`, your Navidrome username and password.

★ PRO TIP: THE APP PASSWORD

An account the gate made has a random password. Set a real one once: Settings → Users → the person → Change password. The app uses that on `/rest/`.

★ PRO TIP: CLEAN TAGS

Fix tags on the-box before copying music in. **MusicBrainz Picard** matches albums and writes artist, album, track numbers and cover art.

★ PRO TIP: PICK YOUR SHELVES

The library picker in the sidebar switches between shelves, or shows them together. The admin sees every shelf.

★ PRO TIP: CELLULAR DATA

Many Subsonic apps let you pick a lower bitrate on mobile data; Navidrome converts on the fly. Or download albums at home for offline play.

CODES

SGIN · podsvc

```
❯ podman ps --filter name=navidrome
```

One line, Up.

```
❯ podman logs --tail 30 navidrome
❯ systemctl --user restart navidrome
```

★ PRO TIP: A THIRD PLAYER

Someone new needs no new container, port or name:

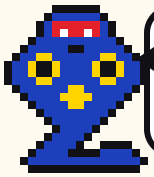
1. Add them to Authelia's users.yml on the gate.
2. Navidrome → Settings → Users → **+**, same username, tick their shelves (or make them a shelf: `music/<name>` and Settings → Libraries → **+**).
3. Run `gate-music-one.sh` again with them in `MUSIC_USERS`.

CODES

gate · podsvc

```
❯ bash ~/cobra-server/scripts/oneshot/gate-sso
-check.sh
```

Read-only. Shows the music wall and who is on it.



CLAUDE-VIPER

Want Rachel to hear your pop shelf too? Split `music/eels` into one shelf per genre and tick the ones she likes. Ask me when you want it.



JOE SABOTAGE REPORT

From manual 3.8, part 5.4 and the merge of November 2026 (docs/music.md).

! THE FIRST-VISITOR CROWN

A brand-new Navidrome makes whoever opens it first the admin.

> **FIX:** Open it from the-box right after it starts and create `eels`, the same username as in Authelia.

! THE NAME MISMATCH

Authelia says "rachel", Navidrome only knows "Rachel2". The trusted header points at nobody.

> **FIX:** One username per person, spelled the same in users.yml and in Navidrome.

! THE OVERLAPPING SHELF

Library 1 still reads all of `music/`, so Navidrome refuses `music/rachel` as a second shelf.

> **FIX:** Library 1's path must be `/media/dock1a/music/eels` (`sgin-music-one.sh` sets it).

! THE OPEN SHELF

A shelf marked *default for new users* lands in every new account, Rachel's included.

> **FIX:** Untick it on both shelves and tick shelves per person on the Users page.

! THE MISSING WALL

Without the music wall, any friend with a gate login who opens music. gets a Navidrome account of their own.

> **FIX:** Run `gate-music-one.sh`; `gate-sso-check.sh` says who's on the wall.

! THE READ-ONLY SURPRISE

Navidrome can't fix a wrong tag or delete a duplicate: its library is mounted read-only.

> **FIX:** Fix files on the-box over the share (Picard); Navidrome picks the change up on its next scan.

! THE STAND-UP SHUFFLE

A comedy album dropped into `music/` turns up in the Commander's shuffle.

> **FIX:** Stand-up goes in `dock1a/comedy`, podcasts in `dock1a/sessions`: Audiobookshelf's shelves.

! THE WATCHER LIMIT

Navidrome, Jellyfin, Kavita and ABS all watch the library, and Linux's default limit on file watchers is low.

> **FIX:** Part 3.3 raises it in `/etc/sysctl.d/90-inotify.conf`. Keep that file.

! THE WIDE-OPEN WHITELIST

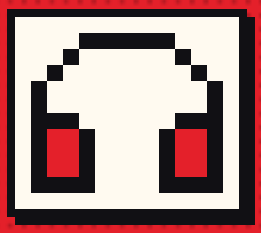
The trusted-header setting is the master key. Widen `ND_REVERSEPROXYWHITELIST` to the whole LAN and any device at home could claim to be any user.

> **FIX:** Keep it at exactly `10.6.9.10/32`, the gate and nothing else.

PLAYER 1 · NAVIDROME

MUSIC.

The Commander's door: every shelf in the jukebox



STATS

NAME	`navidrome`
PORT	4533
LIBRARY 1	`/media/dock1a/music/eels`
DATA	`/srv/stream/navidrome`
URL	`music.songofeo.com`
ADMIN	eels
RANGE	

The flagship. As the admin, the Commander sees every shelf: Eels, Rachel and any shelf added later. It's the Music floor of the station.

`ND_MUSICFOLDER` becomes Library 1, renamed **Eels**. Every other shelf is added by hand in Settings → Libraries, with its path as the container sees it (`/media/dock1a/...`).

★ PRO TIP: ALL-ACCESS

Admins see every shelf on their own, so a new person's shelf shows up here the moment it's added.

★ PRO TIP: THE RADIO

The Song of Eo radio shuffles `dock1a/music`, both shelves. Anything you add can turn up on the air (chapter 2-9).

! THE OLD ROOT

Library 1 still says `/media/dock1a/music`?

> FIX: Settings → Libraries → Library 1 → path `/media/dock1a/music/eels`, Save, let it rescan.

CODES

SGIN · podsvc

```
❯ podman logs --tail 30 navidrome
❯ podman exec navidrome env
```

Shows ND_MUSICFOLDER and the other settings.

THE ROUTE

AT HOME music.songofeo.com ▶ PI-HOLE answers 10.6.9.10 ▶ GATE one sign-in, music wall ▶ NAVIDROME 10.6.9.20:4533

★ PRO TIP: WEB PLAYER

No app needed on the-box: `https://music.songofeo.com` is the full web player, with queue, playlists and album art.

WHAT'S INSIDE

Shelf Eels (`music/eels`, by genre) and shelf Rachel (`music/rachel`).

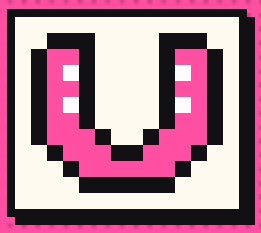
LOADOUT

- [ ] Admin `eels` made from the-box at `http://10.6.9.20:4533`
- [ ] Shelves Eels and Rachel, neither *default for new users*
- [ ] Symfonium or Substreamer on the S21

PLAYER 2 · NAVIDROME

MUSTANG.

Rachel's own door: only her shelf, dock1a/music/rachel



STATS

NAME	`navidrome` (the same one)
PORT	4533
SHELF	`/media/dock1a/music/rachel`
USER	rachel
URL	`mustang.songofeo.com`
HORSEPOWER	■■■■■

Mustang. is Rachel's name for the jukebox. It opens the same Navidrome as music., and her account only has her shelf ticked, so her library, playlists, stars and play counts are hers alone.

Her username is `rachel`, the same as in Authelia, so signing in at the gate carries straight in. `music.` works for her too and shows the same thing.

★ PRO TIP: HER OWN APP

On her phone, a Subsonic app with server `https://mustang.songofeo.com`, username `rachel` and the password set on the Users page.

★ PRO TIP: ADDING SONGS

New music for Rachel goes in `\\cobra-stream\media\dock1a\music\rachel`. It shows on her shelf, and in the Commander's view too.

! THE PEEK

If Rachel's account is made by her first visit while shelf Eels is *default for new users*, she sees your whole collection.

> **FIX:** Make her account yourself first, with only Rachel ticked.

CODES

the-box · browser

```
› http://10.6.9.20:4533
```

Settings → Users → rachel → Libraries: only Rachel.

THE ROUTE

ANYWHERE mustang.songofeo.com ▶ GATE one sign-in, music wall ▶ NAVIDROME 10.6.9.20:4533 ▶ HER SHELF library Rachel

★ PRO TIP: OLD PLAYLISTS

Her old Navidrome (`navidrome-rachel`) is retired, data kept. Playlists made there don't move over on their own.

WHO SEES HER MUSIC

Rachel, and the Commander as admin. Nobody else gets past the music wall.

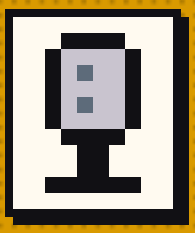
LOADOUT

- [ ] Navidrome user `rachel`, only shelf Rachel ticked
- [ ] Authelia user `rachel`, on the music wall

TALK FLOOR 1 · AUDIOBOOKSHELF

COMEDY.

The stand-up club: picks up right where the laugh stopped



STATS

APP	`audiobookshelf` (chapter 2-4)
PORT	13378
LIBRARY	Comedy, type Book
READS	`/media/dock1a/comedy`
URL	`comedy.songofeo.com`
LAUGHS	■■■■■

A forty-minute set is talk, not a song. Audiobookshelf remembers where each person stopped, has playback speed and a sleep timer, and never drops a bit into a music shuffle.

Comedy is a **Book** library in the Audiobookshelf already on the SGIN: comedian = author, special = book, tracks play in order. `comedy.songofeo.com` opens it, behind the same gate sign-in as the other floors.

★ PRO TIP: FOLDERS

`comedy/Comedian/Special (Year)/` with the tracks inside. Audiobookshelf reads the folder names first, then the tags.

★ PRO TIP: RESUME

Stop on the-box, carry on in the phone app: the position follows you.

! THE WRONG SHELF

Comedy copied into `music/` shows up in Navidrome and on the radio.

> **FIX:** Move it to `dock1a/comedy`; both rescan on their own.

CODES

SGIN · podsvc

```
❯ podman logs --tail 30 audiobookshelf
❯ systemctl --user restart audiobookshelf
```

THE ROUTE

ANYWHERE comedy.songofeo.com ▶ GATE one sign-in ▶ AUDIOBOOKSHELF 10.6.9.20:13378 ▶ LOGIN its own, once per device

★ PRO TIP: THE APP

The Audiobookshelf phone app, server `https://books.songofeo.com` (comedy. works too), then pick Comedy at the top.

WHO GETS IN

Whoever you tick Comedy for in Audiobookshelf's Settings → Users. A friend can have comedy without your music.

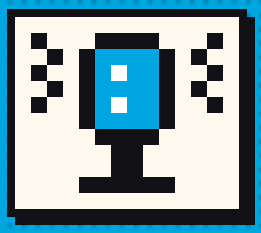
LOADOUT

- [ ] Library Comedy, type Book, folder `/media/dock1a/comedy`
- [ ] Stand-up only in `\\cobra-stream\media\dock1a\comedy`

TALK FLOOR 2 · AUDIOBOOKSHELF

SESSIONS.

The Ward: your own recorded podcasts



STATS

APP	audiobookshelf
PORT	13378
LIBRARY	Sessions, type Podcast
READS	/media/dock1a/sessions
URL	sessions.songofeo.com
ON AIR	

The Ward is where your own recordings live: one folder per show, one file per episode. As a Podcast library, each episode is marked played or not, per listener, and resumes where you stopped.

Audiobookshelf can also **Open RSS feed** on a show, so The Ward can be followed in any podcast app with the feed's address.

★ PRO TIP: DATE FIRST

Start each file name with the date, `2026-11-08 Episode title.mp3`, so episodes sort in order.

★ PRO TIP: COVER ART

A square `cover.jpg` in each show folder gives every show its own tile.

! THE LOOSE EPISODE

A file straight in `sessions/`, outside a show folder, isn't picked up as an episode.

FIX: One folder per show, episodes inside.

CODES

the-box · browser

```
› http://10.6.9.20:13378
```

Settings → Libraries → Sessions → Scan.

THE ROUTE

ANYWHERE sessions.songofeo.com ▶ GATE one sign-in ▶ AUDIOBOOKSHELF 10.6.9.20:13378 ▶ LOGIN its own, once per device

★ PRO TIP: FILE FORMAT

Export episodes as MP3 or M4A: every phone app plays them.

A FEED, IF YOU WANT

The RSS feed is off until you open it, per show, and you can close it again.

LOADOUT

- [ ] Library Sessions, type Podcast, folder `/media/dock1a/sessions`
- [ ] Shows in `\\cobra-stream\media\dock1a\sessions`, one folder each

WORLD 2-4 · CHARACTER SELECT

AUDIOBOOKSHELF

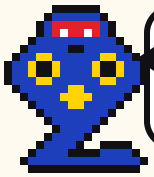
The Library: audiobooks and e-books at books.songofeo.com



Audiobookshelf (ABS to its friends) is the Library floor. It turns folders of audiobook files into a proper bookshelf with covers, chapters, series and narrators, and remembers exactly where every listener stopped, down to the second, across every device.

It also shelves e-books in a second library, with a reader in the browser. Audiobooks are where it shines: nothing else self-hosted comes close for audio, and its phone app can download books for offline listening on a long drive.

ABS lives in the `stream` pod on port **13378**. Its own settings and its database sit in `/srv/stream/abs/config`, and the covers and caches it makes in `/srv/stream/abs/metadata`, both on the SGIN's SSD. The books come in read-only from dock1a.



CLAUDE-VIPER

Start a book on the S21 on the drive home, keep going on the-box at your desk. ABS hands over your place in the story like a relay baton.

STATS

HOST	cobra-stream 10.6.9.20, pod `stream`
IMAGE	`ghcr.io/advplyr/audiobookshelf`
PORT	13378
URL	`books.songofeo.com`
DATA	`/srv/stream/abs/config`, `/metadata`
LIBRARY	`dock1a/audiobooks`, `dock1a/books`
SIGN-IN	Gate sign-in from outside + ABS login
TOUGHNESS	■■■■□
TRAP LEVEL	■■□□□
FUN FACT	■■■■□

NEVER LOSE YOUR PAGE

LIBRARY	TYPE	FOLDER IN ABS	ON THE SHARE
Audiobooks	Book	`/media/dock1a/audiobooks`	`\\cobra-stream\media\dock1a\audiobooks`
Books	Book (e-books)	`/media/dock1a/books`	`\\cobra-stream\media\dock1a\books`

★ PRO TIP: ONE ADDRESS

`https://books.songofeo.com` works at home, on WireGuard and on cellular. At home the Pi-holes send it to the gate inside the house, and the gate skips its sign-in.

FRESH SHELVES

dock1a was wiped and reloaded in the build, from the Commander's own backups of the audiobooks and e-books. ABS only had to scan them.

HOW IT WORKS LEVEL MAP

CHAPTER 7 RESUMES ON THE PHONE

1 ABS APP
books.songofeo.com
▶
2 GATE
nginx, sign-in from outside
▶
3 ABS
10.6.9.20:13378
▶
4 PROGRESS
last position, from the DB
▶
5 STREAM
audio from dock1a

WHY PORT 13378?

On its own, ABS listens on port 80. Inside a pod every container shares **one** network, so two apps can't both have the same port, and a rootless container can't have 80 anyway. The container is started with `PORT=13378`, and the pod publishes 13378 to the LAN.

HOW IT READS A BOOK

ABS treats **each folder as one book**. A tidy layout is `Author / Book Title / files`, with a series folder in between if you like: `Author / Series / Book 1 Title / files`. One file or fifty MP3 parts, it's all one book to ABS, and chapters come from the file or the parts.

It then looks up covers, descriptions and narrators from online sources and saves what it finds in its own metadata folder, because the library is read-only.

WHAT LIVES WHERE

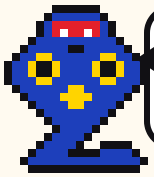
FOLDER	HOLDS
`/srv/stream/abs/config`	Database: users, libraries, listening progress, bookmarks
`/srv/stream/abs/metadata`	Covers, cached items, ABS's own backups
`/media` (read-only)	Every drive, so libraries can point anywhere on them

THE TWO LOCKS

From outside, books. is a station floor: the gate's sign-in first, then ABS's own login. At home and over WireGuard the gate skips its sign-in, and you meet ABS's login directly. The ABS login is what keeps each listener's progress their own.

WHO'S THE ROOT?

The very first visit creates the **root** user, the top admin. Everyone else gets a normal user under Settings → Users.



CLAUDE-VIPER

The books are never touched. ABS only reads them and writes its notes in its own folders on the SSD. Wipe ABS and the library is still sitting there, safe on dock1a.

A TIDY SHELF

```
audiobooks/
  Author Name/
    Standalone Title/
      book.m4b
    Series Name/
      Book 1 - First Title/
        01.mp3  02.mp3  03.mp3
```

One folder per book, under its author, with an optional series folder. A single `.m4b` with built-in chapters or a pile of numbered MP3s: both are one book to ABS.

TIPS & TRICKS POWER-UPS

FIRST VISIT, 4 MOVES

1. **the-box browser:** `http://10.6.9.20:13378` → create the **root** user and a strong password.
2. Settings → Libraries → Add: `Audiobooks`, type **Book**, folder `/media/dock1a/audiobooks`.
3. Again: `Books`, folder `/media/dock1a/books`, for e-books.
4. Settings → Users: one user per listener.

★ PRO TIP: THE PHONE APP

Install the Audiobookshelf app, server `https://books.songofeo.com`, your ABS user. Away from home, turn WireGuard on first: phone apps can't show the gate's sign-in page. Downloaded books play offline anywhere.

★ PRO TIP: ADDING BOOKS

From the-box, copy the book's folder into `\\cobra-stream\media\dock1a\audiobooks\Author\`. ABS spots it and shelves it. If not, the library's menu has a scan button.

★ PRO TIP: FIX A COVER

Wrong cover or blurb? Open the book → edit (pencil) → Match, and pick the right edition. The fix is stored in ABS's metadata, not the files.

★ PRO TIP: SERIES ORDER

Put the series name and number in the book's details (or in the folder name, like `Book 1 - Title`) and ABS lines the series up in reading order on its own shelf.

★ PRO TIP: SLEEP TIMER

The player has a playback-speed control and a sleep timer. Bedtime stories, Cobra style.

CODES

SGIN · podsvc

```
❯ podman logs --tail 50 audiobookshelf
❯ podman restart audiobookshelf
```

Safe: progress and users live in /srv/stream/abs.

THE-BOX · browser

```
❯ http://10.6.9.20:13378
```

★ PRO TIP: E-BOOK READERS

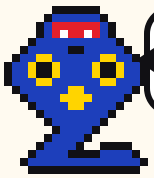
ABS reads EPUBs in the browser, but its e-reader sync is basic. Kavita (next chapter) also reads EPUB well if you prefer its reader.

WHICH SCREEN FOR WHAT

the-box browser: setting up libraries, users, fixing covers and matches. Always do admin here.

Phone app: listening, downloads for offline, the sleep timer in bed.

Any browser: listening or reading an e-book anywhere you're signed in.



CLAUDE-VIPER

Admin from the-box, listening from anywhere. That's the Cobra way of doing books.



JOE SABOTAGE REPORT

From manual 3.8 and the stream pod's wiring.

! THE ROOT GRAB

The first person to open ABS becomes **root**. Leave it unclaimed and that could be anyone on the LAN.

> **FIX:** Create root from the-box as soon as the container first starts.

! THE PORT 80 HABIT

Guides on the internet run ABS on port 80. In the shared pod, and rootless, it can't have 80.

> **FIX:** Keep `PORT=13378` in the container and 13378 in pod.sh and nftables.

! THE UPLOAD WALL

The library is mounted **read-only**, so anything that writes into the book folders, like the web Upload page or saving covers next to the files, can't.

> **FIX:** Add books by copying them over the share from the-box. ABS keeps its own covers in its metadata folder.

! THE LOOSE FILES

Fifty MP3s dumped straight into `audiobooks/` become fifty "books", or one mess.

> **FIX:** One folder per book: `Author / Title / files`.

! THE GHOST DRIVE

dock1a unplugged and plugged back: ABS still sees the empty folder from before, and its library looks gone.

> **FIX:** `podman pod restart stream`, then scan the library.

! THE WRONG TYPE

ABS libraries are either **Book** or **Podcast**. Pick Podcast for audiobooks and nothing lines up.

> **FIX:** Both libraries here are type Book.

! THE SHARED ACCOUNT

Two people on one ABS user overwrite each other's progress: you'll resume at chapter 20 of a book you were on chapter 3 of.

> **FIX:** One ABS user per listener.

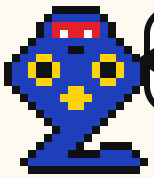
! THE DOUBLE LOGIN SURPRISE

From outside you sign in at the gate, then ABS asks again. That's normal: ABS doesn't read the gate's sign-in the way Navidrome does.

> **FIX:** Nothing to fix in the browser. For the phone app away from home, use WireGuard, which skips the gate's sign-in.

LIBRARIAN'S CHECKLIST

- [ ] root user made from the-box
- [ ] Audiobooks library on `/media/dock1a/audiobooks`
- [ ] Books library on `/media/dock1a/books`
- [ ] One user per listener, app on each phone



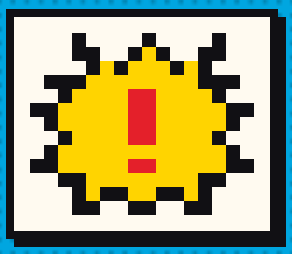
CLAUDE-VIPER

Most ABS trouble is folders, not software. Get the shelf tidy and the rest takes care of itself.

WORLD 2-5 · CHARACTER SELECT

KAVITA

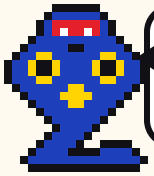
The Comics Shop: comics, manga and graphic novels at comics.songofeo.com



Kavita is the island's comic shop. It shelves your CBZ and CBR comic archives, manga and graphic novels by series, draws the covers, and opens a fast page-turning reader right in the browser, phone or tablet included.

It remembers where each reader stopped in every issue, shows what's next in a series, and speaks **OPDS**, the common catalog language that many reading apps understand. It reads EPUB and PDF too, so it can double as an e-book reader.

Kavita lives in the `stream` pod on port **5000**. Its database, covers and settings stay in `/srv/stream/kavita` on the SGIN's SSD; the comics come in read-only from `dock1a/comics`.



CLAUDE-VIPER

KA-POW! Every issue from the footlockers, filed, sorted and readable from the couch. The Joes can keep their crumpled paper copies.

STATS

HOST	cobra-stream 10.6.9.20, pod `stream`
IMAGE	`jvmilazz0/kavita:latest`
PORT	5000
URL	`comics.songofeo.com`
DATA	`/srv/stream/kavita`
LIBRARY	`/media/dock1a/comics`
SIGN-IN	Gate sign-in from outside + Kavita login
TOUGHNESS	■■■■□
TRAP LEVEL	■■□□□
FUN FACTOR	■■■■■

TURN EVERY PAGE!

WHAT IT READS

Comic archives: `.cbz` (a ZIP of page images) and `.cbr` (a RAR). CBZ is the friendliest.

Books: EPUB and PDF, with a reader for each.

Metadata: a `ComicInfo.xml` inside the archive fills in series, number, writer and summary.

★ PRO TIP: OPDS LINK

OPDS is a catalog feed that reading apps understand. Each Kavita user finds their own OPDS link in their user settings.

ONE ADDRESS

`https://comics.songofeo.com` everywhere. At home it opens straight to Kavita's login; outside, the gate's sign-in comes first.

HOW IT WORKS LEVEL MAP

ISSUE #12 OPENS ON THE TABLET

1 TABLET
comics.songofeo.com

▶

2 GATE
nginx, sign-in from outside

▶

3 KAVITA
10.6.9.20:5000 login

▶

4 SERIES
covers, progress, next issue

▶

5 PAGES
from the archive on dock1a

SERIES FOLDERS

Kavita groups by **series**, and it expects the files inside folders under the library root, not loose in it. The safe layout is one folder per series:

`comics / Series Name / Series Name 001.cbz`

The folder and file names give Kavita its first guess; a `ComicInfo.xml` inside the archive, when there is one, gives it the real details.

THE LIBRARY TYPE MATTERS

When you add a library, its **type** (Comic, Manga, Book and so on) tells Kavita how to read names and how to show the pages. The island's library is `Comics`, type **Comic**, on `/media/dock1a/comics`. Manga could get its own library of type Manga later.

WHAT LIVES WHERE

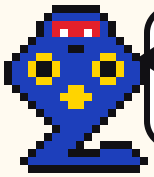
FOLDER	HOLDS
`/srv/stream/kavita` (as `/kavita/config`)	Database, users, reading progress, cover images, logs
`/media` (read-only)	Every drive; the library points at `dock1a/comics`

WATCHING FOR NEW ISSUES

Kavita watches the library folder and notices new files on its own, and it also rescans on a schedule. Linux limits how many folders one user may watch; part 3.3 raised that limit for podsvc, because Kavita, Jellyfin, ABS and the Navidromes all watch at once.

THE ADMIN

The first visit registers the admin account (the email can be anything local). After that, the admin adds or invites one user per reader.



CLAUDE-VIPER

Comics, Library, Music, Arcade: each floor of the station is its own app on the SGIN, with its own door on the gate. Kavita's door is comics.

A TIDY RACK

```
comics/
  Series Name/
    Series Name 001.cbz
    Series Name 002.cbz
  Another Series (1990)/
    Another Series (1990) 001.cbz
```

One folder per series, and the issue number in each file name. Kavita then stacks them in order, shows the next unread issue, and tracks each reader's progress through the run.

TIPS & TRICKS *POWER-UPS*

FIRST VISIT, 3 MOVES

1. **the-box browser:** `http://10.6.9.20:5000` → register the admin account.
2. ⚙ → Libraries → Add: name `Comics`, type **Comic**, folder `/media/dock1a/comics` → Save → **Scan**.
3. Users → invite or add one per reader.

★ PRO TIP: CBZ IS KING

A CBZ is just a ZIP of page images renamed. Scans of your own comics can be zipped up on the-box and renamed `.cbz`, and Kavita reads them like any other issue.

★ PRO TIP: READING APPS

Kavita's web reader works well on phones and tablets. For a dedicated app, look for one that supports **OPDS**; Kavita gives each user an OPDS link in their settings.

★ PRO TIP: READING LISTS

Crossovers jump between series. A **reading list** puts issues from several series in the order you want to read them.

★ PRO TIP: READER SETTINGS

In the reader, the settings menu sets reading direction, page fit and single or double pages. Manga reads right to left; set it once per library type.

★ PRO TIP: WANT TO READ

Mark series as "want to read" and use **collections** to group them ("Cobra classics", "Rachel's picks"). They're shelves on top of the folders, nothing moves on disk.

★ PRO TIP: ADDING ISSUES

From the-box, copy into `\\cobra-stream\media\dock1a\comics\Series Name\`. Kavita picks it up; if you're impatient, the library's Scan button does it now.

CODES

SGIN · podsvc

```
❯ podman logs --tail 50 kavita
❯ podman restart kavita
```

Safe: everything Kavita knows lives in /srv/stream/kavita.

SGIN · eels

```
❯ ls /srv/storage/disks/dock1a/comics
```

One folder per series is what you want to see.

★ PRO TIP: E-BOOKS TOO

Kavita reads EPUB nicely. If you like its reader better than ABS's, add a second library of type **Book** pointed at `/media/dock1a/books`.



CLAUDE-VIPER

Read on the couch, pick up on the tablet in bed. Kavita remembers the page.



JOE SABOTAGE REPORT

From manual 3.8 and the footlockers.

! THE ADMIN SNATCH

Until someone registers, the first visitor gets the admin account.

FIX: Register from the-box the moment Kavita first starts.

! THE LOOSE ISSUES

Archives sitting directly in `comics/` with no series folder confuse Kavita's grouping, or get skipped.

FIX: One folder per series under the library root.

! THE SPLIT SERIES

"Batcave 01.cbz", "Bat-Cave #2.cbz" and "batcave_003.cbz" can turn one series into three.

FIX: Name files the same way, or add a ComicInfo.xml with the series name.

! THE WRONG TYPE

A Comic library pointed at manga reads names wrong and turns pages the wrong way round.

FIX: Manga in its own library of type Manga.

! THE GHOST DRIVE

Replug dock1a while Kavita runs and it scans an empty folder: series vanish from the shelves.

FIX: `podman pod restart stream` first, scan after.

! THE READ-ONLY SHELF

Kavita can't rename, delete or fix files: the library is mounted read-only.

FIX: Tidy files on the-box over the share, then scan.

! THE WATCHER CAP

Too many apps watching too many folders, and new issues stop showing up until a scheduled scan.

FIX: Keep part 3.3's `/etc/sysctl.d/90-inotify.conf`.

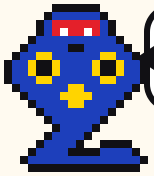
! THE PHONE-APP WALL

An OPDS app on cellular hits the gate's sign-in page, which it can't show.

FIX: Turn WireGuard on first; the gate then lets it through to Kavita's own login.

SHOPKEEPER'S CHECKLIST

- [ ] Admin registered from the-box
- [ ] Comics library, type Comic, scanned
- [ ] One folder per series on dock1a
- [ ] One Kavita user per reader



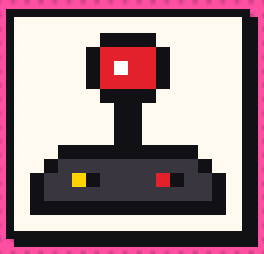
CLAUDE-VIPER

Name files the same way every time and Kavita rewards you with perfect series. Sloppy names are the Joes' favourite trick.

WORLD 2-6 · CHARACTER SELECT

ROMM

The Arcade: your ROM library, playable in a browser, at arcade.songofeo.com



RomM is the browser arcade. It catalogs every ROM by platform, fetches box art and titles, and with EmulatorJS built in, lets anyone you've signed in play right in their browser. Press start on a laptop, a tablet, even a phone.

The trick: the game runs **in the player's browser**, not on the server. The SGIN only hands out files, so ten people playing ten games barely tickle its CPU. RomM keeps its catalog in a MariaDB database (next chapter) and its covers and settings on the SGIN's SSD.

The ROMs live on **dock1b**, in `dock1b/roms`. Their main copy is on the cabinet, `cobra-arcade`'s `D:\Roms`, and the cabinet sends them up every night, so the web arcade keeps working while the cabinet sleeps.



CLAUDE-VIPER

Status report, Commander: RomM is staged and ready in the arcade runbook (`docs/arcade-runbook.md`), waiting for the live arcade session. This chapter is your pre-mission briefing.

STATS

HOST	cobra-stream 10.6.9.20, pod `stream`
IMAGE	`rommapp/romm:latest`
PORT	8080
URL	`arcade.songofeo.com`
DATA	`/srv/stream/romm/` resources, assets, config, redis
ROMS	`dock1b/roms`, writable
DATABASE	`romm-db` (MariaDB 11)
SIGN-IN	Gate sign-in from outside + RomM login

TOUGHNESS
TRAP LEVEL
FUN FACTOR

INSERT COIN? NO NEED!

TWO ARCADES, ONE ROM PILE

The cabinet (OptiPlex, LaunchBox and RetroArch) plays from `D:\Roms` with the Tankstick. Its saves live on the cabinet.

The web arcade (RomM and EmulatorJS) plays the nightly copy on dock1b in any browser. Its saves live in RomM, per user.

PLATFORM FOLDERS

Examples from the manual: `roms/snes`, `roms/n64`, `roms/arcade`. RomM needs its own names for platforms; the cabinet's folder map translates where the cabinet's names differ.

★ PRO TIP: GAMEPADS

EmulatorJS runs in the browser, and browsers can read most USB and Bluetooth gamepads. Plug one into the-box, press a button on the game page, play.

HOW IT WORKS LEVEL MAP

FROM CABINET TO BROWSER

1 CABINET
D:\Roms\<platform>
▶
2 NIGHTLY
robocopy to \\cobra-stream\roms
▶
3 DOCK1B
roms/<platform>/
▶
4 ROMM
scan, art, :8080
▶
5 BROWSER
EmulatorJS plays it

ONE FOLDER PER PLATFORM

RomM expects `roms/<platform>/`: `roms/snes`, `roms/n64`, `roms/arcade` and so on. Each folder shows up as a platform after a scan. The cabinet's own folder names don't always match RomM's, so the cabinet script `03-roms-map.ps1` writes a folder map, and `romm-platforms.sh` adds it to RomM's `config.yml` under `system:`.

ONE-WAY TRAFFIC

The nightly copy runs **cabinet to SGIN only** (`robocopy /E /XO`: adds and updates, never deletes). New ROMs go onto the cabinet's `D:\Roms`; the next night they appear on dock1b, then a RomM scan picks them up.

WHY RomM MAY WRITE

Unlike the read-only media apps, RomM's roms folder is mounted writable, with `keep-groups` so it keeps podsvc's `media` group. No `:Z` on it, because Samba shares the same folder for the cabinet's copy.

WHAT LIVES WHERE

FOLDER	HOLDS
`romm/resources`	Fetched covers and screenshots
`romm/assets`	Uploaded saves, states, screenshots
`romm/config`	`config.yml`: netplay, folder map
`romm/redis`	Its built-in task queue's data
`romm/db`	MariaDB's files (romm-db)

SECRETS, NOT SCRIPTS

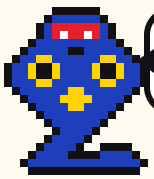
The database password and RomM's auth key are made once and kept as **podman secrets** (`romm-db-password`, `romm-auth-secret`) when built by the scripts. They never appear in the script or in `podman inspect`. In Quadlet mode they're in the 600-protected `romm*.container` files instead.

THE SCOREBOARD IS ELSEWHERE

The hiscore board at scores.songofeo.com runs on the gate, not here. RomM is just the arcade floor.

THE NIGHTLY COPY

On the cabinet, `02-nightly.ps1` sets up the nightly job (run it once with `-RunNow` for the first copy). It logs to `C:\Cobra\nightly-copy.log` and writes `=== end` when done. The copy logs in to the SGIN's Samba as user `arcade`, saved once with `cmdkey`.



CLAUDE-VIPER
Cabinet by day, browser by night, one ROM pile for both. The Joes don't even know which arcade you're in.

TIPS & TRICKS POWER-UPS

★ PRO TIP: COVERS + NAMES

Free metadata keys (IGDB through Twitch, ScreenScraper, SteamGridDB) give RomM box art and proper titles. With the scripts, put them in `~/.config/cobra/romm.env` as `IGDB_CLIENT_ID=...` lines and rerun `./romm.sh`; with Quadlets, as `Environment=` lines in `romm.container`.

★ PRO TIP: NETPLAY

Netplay lets two browsers play one game together (WebRTC). `romm.sh` writes it switched on in `config.yml`. Friends outside the house usually also need a **TURN relay**; RomM's EmulatorJS docs list free ones under `ice_servers`.

★ PRO TIP: BIOS FILES

Some systems won't boot without BIOS files. RomM → the platform → **Firmware** → upload. Several files for one system? Zip them together first.

★ PRO TIP: ONE USER PER PLAYER

Administration → Users. Each player's saves and save states then stay their own.

CODES

SGIN · podsvc

```
❯ bash /tmp/cobra-arcade/romm-up.sh
```

The runbook's starter: picks Quadlet or scripts mode, never overwrites.

```
❯ bash /tmp/cobra-arcade/romm-platforms.sh
```

Adds the cabinet's folder map, with a backup of config.yml.

```
❯ podman logs --tail 50 romm
❯ podman restart romm
```

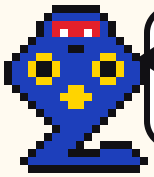
SGIN · eels

```
❯ bash /tmp/cobra-arcade/check.sh
```

Read-only check. Ends with ALL GOOD, or prints each fix.

FIRST VISIT

1. **the-box browser:** `http://10.6.9.20:8080` → setup wizard → admin `eels`, password into Vaultwarden.
2. After the cabinet's first nightly copy: Library → **Scan**.
3. Phone on cellular: `https://arcade.songofeo.com` → gate sign-in → one game plays.



CLAUDE-VIPER

The gate needs nothing new: arcade. is already in nginx's map, pointing at 10.6.9.20:8080. Start RomM and the door is already open.

RUNBOOK ORDER

1 SGIN check.sh ▶ 2 CABINET scripts + base ▶ 3 MAP 03-roms-map.ps1 ▶ 4 ROMM romm-up.sh ▶ 5 COPY 02-nightly - RunNow ▶ 6 SCAN in RomM



JOE SABOTAGE REPORT

From manual 3.8b, the arcade runbook and the scripts' own warnings.

! THE WRONG-WAY ROM

ROMs dropped straight into `\\cobra-stream\roms` never reach the cabinet: the copy only runs cabinet to SGIN.

> **FIX:** Add ROMs on the cabinet in `D:\Roms\<platform>`.

! THE TOO-BIG PILE

dock1b is a 1 TB drive; `D:\Roms` on the 4 TB drive could outgrow it.

> **FIX:** The nightly script stops by itself if `D:\Roms` is bigger than dock1b's free space. Check its log.

! THE FORGOTTEN PASSWORD

RomM's database already exists with an old password, and a fresh run makes a new one: RomM can't log in to its own database. `romm-up.sh` stops rather than guess.

> **FIX:** Give the old values once: `ROMM_DB_PASSWORD=... ROMM_AUTH_SECRET=...` in front of the command.

! THE PUBLISHED-PORT GAP

No `PublishPort=8080:8080` in `stream.pod`, and RomM answers nobody.

> **FIX:** Add the line, `daemon-reload`, restart stream-pod. romm-up.sh checks it for you.

! THE SAVE SPLIT

Beat a level on the cabinet, open the same game in the browser: no save. They share ROMs, never saves, and browser netplay can't join a RetroArch netplay game.

> **FIX:** Pick one place per playthrough.

! THE LONELY NETPLAY

Netplay works at home but a friend across town never connects.

> **FIX:** Add a TURN relay under `ice_servers` in config.yml.

! THE DOUBLE CONFIG

`romm-platforms.sh` refuses if `config.yml` already has a `platforms:` or `system:` section, so it never writes a duplicate.

> **FIX:** Paste the map in by hand with nano, under the existing section.

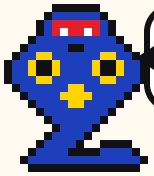
! THE EMPTY SCAN

Scan before the first nightly copy and RomM finds nothing.

> **FIX:** Watch `C:\Cobra\nightly-copy.log` on the cabinet for `=== end`, then Scan.

ARCADE OPERATOR'S CHECKLIST

- [ ] check.sh says ALL GOOD
- [ ] romm-up.sh ends with "RomM is up"
- [ ] Folder map added, Library scanned
- [ ] One game plays from a phone on cellular



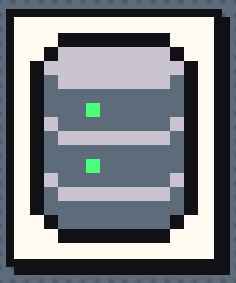
CLAUDE-VIPER

Every one of these traps prints its own fix in the runbook scripts. Read the red line, Commander, it's talking to you.

WORLD 2-7 · CHARACTER SELECT

ROMM-DB

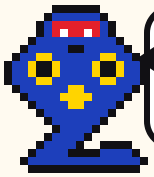
The Arcade's memory: a MariaDB database only RomM talks to



Every arcade needs someone keeping the books. romm-db is a MariaDB database server, and it remembers everything RomM knows: which games exist on which platform, their titles and metadata, the users, and who played what.

It has no web page and no public name. It sits in the `stream` pod, listening on MariaDB's port 3306 on `127.0.0.1`, which only containers inside the pod can reach. That port is **not** in the pod's published list, so nothing on the LAN can talk to it, let alone the internet.

Its files live in `/srv/stream/romm/db` on the SGIN's SSD. The ROMs themselves aren't in it: the database holds the catalog, the drive holds the games.



CLAUDE-VIPER

The quiet operative. No page, no port to the LAN, no fame. Just a perfect memory. Every good Cobra unit needs one.

STATS

HOST	cobra-stream 10.6.9.20, pod `stream`
IMAGE	`library/mariadb:11`
PORT	3306, inside the pod only
DATABASE	`romm`, user `romm`
DATA	`/srv/stream/romm/db`
SECRETS	`romm-db-password`, `romm-db-root-password`
SIGN-IN	Password, RomM only
TOUGHNESS	■■■■■
TRAP LEVEL	■■■□□
FUN FACT	□□□□□

NO PAGE
NO PORT
NO FAME

HOW IT WORKS *LEVEL MAP*

A RomM SCAN WRITES TO THE BOOKS

1 ROMM
finds roms/snes/*.sfc

▶

2 127.0.0.1
:3306, inside the pod

▶

3 ROMM-DB
user romm, db romm

▶

4 DISK
/srv/stream/romm/db

HOW IT GETS ITS PASSWORDS

On the very first start, with an empty `db` folder, MariaDB creates the `romm` database and user with the password it's given (`MARIADB_PASSWORD`) and sets the root password. After that, the password is **stored inside the database files**. Changing the variable later doesn't change it.

With the scripts, the passwords come from podman secrets, handed in as environment variables only inside the container. With the manual's Quadlets, they're written into `romm-db.container`, which is `chmod 600`.

THE HEALTH CHECK

The script gives the container a health check (`healthcheck.sh --connect --innodb_initialized`) and waits for **healthy** before starting RomM. In Quadlet mode, `romm.container` says `Requires=romm-db.service` and `After=` it instead, so systemd starts the database first.

WHY NOT THE GATE'S DB?

The gate's Postgres belongs to Paperless and lives on another machine. RomM wants MariaDB or MySQL, and keeping its database next to it in the same pod means the arcade never depends on the gate being up.

TIPS & TRICKS

CODES

SGIN · podsvc

```
❯ podman ps --filter name=romm
```

romm-db and romm, both Up.

```
❯ podman logs --tail 30 romm-db
```

```
❯ podman secret ls
```

Scripts mode: lists the secret names, never the values.

```
❯ podman secret inspect --showsecret --format '{{.SecretData}}' romm-db-password
```

Prints the password. Only when you need it, and not on a shared screen.

★ PRO TIP: A SPARE COPY

A dump is a plain-text copy of the whole database you can restore from. Inside the container the password is already in the environment, so the dump command doesn't need you to type it:

CODES

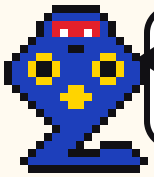
SGIN · podsvc

```
❯ podman exec romm-db sh -c 'mariadb-dump -u romm -p"$MARIADB_PASSWORD" romm' > ~/romm-db.sql
```

Writes ~/romm-db.sql in podsvc's home. Move it to dock1a/backups if you want it kept.

★ PRO TIP: IT'S IN THE BACKUP

`/srv/stream` is the SGIN's app data. Back it up with RomM stopped (or use a dump), so the database files are consistent.



CLAUDE-VIPER

Lose the database and RomM forgets its users and saves list, but not a single ROM: those sit safe on dock1b, and a rescan rebuilds the catalog.



JOE SABOTAGE REPORT

Includes a trap that hit the gate's Postgres for real on 6 November 2026.

! THE CHOWN DISASTER

On the gate, re-running a `chown -R podsvc: /srv/gate` after Postgres had made its files broke them ("pg_filenode.map: Permission denied"). Rootless database files belong to a mapped user ID inside the container, not podsvc. The same move on `/srv/stream` would break romm-db.

> **FIX:** Never `chown -R` a database folder. If it happened, repair from podsvc with `podman unshare chown -R` to the database's own user (999:999 for these official images).

! THE NEW-PASSWORD TRAP

Rebuild RomM with fresh random passwords while `romm/db` still holds the old database: the database keeps the old password, RomM sends the new one, and gets "access denied".

> **FIX:** Reuse the old values (`ROMM_DB_PASSWORD=... ./romm.sh`), read from the old `romm.container` or the saved secret.

! THE PERMISSION LOOP

A container that keeps restarting with `Permission denied` in its log: the image switched to a non-root user inside, and its folder belongs to someone else.

> **FIX:** Manual 3.8's last bullet: find the user with `podman exec <name> id` and add `UserNS=keep-id:uid=...,gid=...`.

! THE BIG-JUMP UPGRADE

The image is pinned to `mariadb:11`, so updates stay within MariaDB 11. Changing the tag to a new major version blindly can leave the files needing an upgrade step.

> **FIX:** Dump first, read the release notes, then change the tag.

! THE PUBLISHED DATABASE

Adding 3306 to the pod's published ports "to look at it from the-box" would put a database on the LAN.

> **FIX:** Leave it out. Use `podman exec -it romm-db mariadb -u romm -p romm` on the SGIN instead.

! THE IMPATIENT START

RomM started before the database was ready fails its first connection. Usually it just retries.

> **FIX:** The script waits for healthy; in Quadlet mode systemd's `After=` orders them. Still stuck: `podman logs romm-db`.

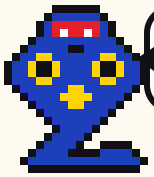
! THE PASSWORD ON SCREEN

`--showsecret` prints the real password in the terminal and in scrollback.

> **FIX:** Run it only when needed, then clear the screen. Keep a copy in Vaultwarden.

DATABASE KEEPER'S CHECKLIST

- [ ] romm-db Up, before romm
- [ ] Password saved in Vaultwarden
- [ ] 3306 never published
- [ ] No `chown -R` on /srv/stream/romm/db, ever



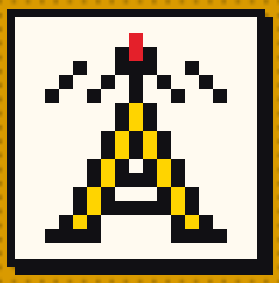
CLAUDE-VIPER

Treat the database folder like the vault: look, don't touch. podman exec is the only key you need.

WORLD 2-8 · CHARACTER SELECT

ICECAST

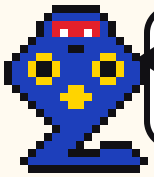
The Transmitter: the tower every radio listener tunes in to



Icecast is the radio tower of Song of Eo Radio. It doesn't pick songs and it doesn't play anything by itself. It takes one live audio stream in from the DJ and hands copies out to every listener who tunes in.

The DJ is the next chapter's container, **radio** (Liquidsoap), which connects to Icecast with a **source password** and pushes an endless MP3 stream to the mount point `/stream`. Listeners open `http://10.6.9.20:8000/stream` at home, or `https://radio.songofeo.com/stream` through the gate, in a browser, VLC or a radio app.

Icecast is part of manual part 8, the landing site and radio. It joins the `stream` pod on port **8000**, which needs adding to the pod's published ports and to the SGIN's nftables list. It keeps no data folder at all: everything it knows comes from its settings.



CLAUDE-VIPER

This is Cobra Island broadcasting on all frequencies! Well, on one mount point, to at most 50 listeners, behind a login. Still counts.

STATS

HOST	cobra-stream 10.6.9.20, pod `stream`
IMAGE	`libretime/icecast:latest`
PORT	8000
MOUNT	`/stream` (MP3, 192 kb/s)
URL	`radio.songofeo.com`
LISTENERS	50 max
DATA	None, settings only
SIGN-IN	Gate sign-in; `/admin` LAN + WG only

TOUGHNESS ■■■■□
TRAP LEVEL ■■■□□
FUN FACTOR ■■■■□

ON THE AIR!

HOW IT WORKS LEVEL MAP

A SONG REACHES A LISTENER

1 RADIO — Liquidsoap picks a song ▶ 2 SOURCE — 127.0.0.1:8000 + password ▶ 3 ICECAST — mount /stream ▶ 4 GATE — radio. + sign-in ▶ 5 LISTENER — browser or VLC

THREE PASSWORDS

- **Source:** what the DJ uses to push audio in. The radio container gets the same one.
- **Admin:** for Icecast's admin pages (listeners, kicking a source).
- **Relay:** for other Icecast servers relaying this one. Unused here, but set.

With the scripts they're generated once as podman secrets (`icecast-source-password` and friends). With the manual's Quadlets they're random values written into `icecast.container`, `chmod 600`.

THROUGH THE GATE

Manual 8.3 gives radio. its own nginx server block. `location /` asks Authelia (password only, `one_factor`), turns buffering off so audio flows straight through, and allows a listener to stay tuned for 24 hours. `location /admin` allows only 10.6.9.0/24 and 10.6.99.0/24.

WHY BEHIND A LOGIN?

Streaming music you don't hold the rights to, to anyone on the internet, needs public-performance licences, even for a tiny station. So radio. stays behind the gate's sign-in for you and the people you add. Fully public only with music you have rights to: your own, Creative Commons or royalty-free (manual 8.4).

TIPS & TRICKS

★ PRO TIP: LAN TEST

On the-box, open `http://10.6.9.20:8000/stream` in VLC (Media → Open Network Stream) or the browser. Music = Icecast and the DJ both work. Silence or 404 = the DJ isn't connected.

★ PRO TIP: THE STATUS PAGE

`http://10.6.9.20:8000/` shows Icecast's status page: the mount, what's playing and how many are listening.

CODES

SGIN · podsvc

```
❯ podman logs --tail 30 icecast
❯ podman restart icecast
```

The DJ reconnects by itself.

```
❯ podman secret inspect --showsecret --format
'{{.SecretData}}' icecast-admin-password
```

Scripts mode: the admin password, for /admin.

SGIN · eels

```
❯ sudo nft list ruleset | grep 8000
```

8000 must be in the LAN/VPN port list, or the gate can't reach Icecast.

★ PRO TIP: UPLOAD MATH

Each listener at 192 kb/s uses about 0.2 Mb/s of Xfinity upload. Ten listeners, about 2 Mb/s.

WAYS TO TUNE IN

Browser: `https://radio.songofeo.com/stream`, signed in at the gate. Plays right in the tab.

VLC on the-box: `http://10.6.9.20:8000/stream` on the LAN, no sign-in needed.

Phone player apps: they can't show the gate's sign-in, so use the LAN address with WireGuard on.



JOE SABOTAGE REPORT

From manual part 8 and the stream pod's port rules.

! THE PIRATE STATION

Removing the sign-in from radio. turns a private jukebox into a public broadcast of music you may not have the rights to stream.

> **FIX:** Keep the Authelia rule unless every track is yours, Creative Commons or royalty-free.

! THE CLOSED FIREWALL

Icecast plays fine on the SGIN itself, but radio. through the gate times out. Port 8000 isn't in nftables' list.

> **FIX:** Part 8.2's sed line adds 8000, then `sudo systemctl reload nftables`.

! THE UNPUBLISHED PORT

Quadlet mode: forget `PublishPort=8000:8000` in `stream.pod` and the LAN can't reach Icecast at all. (pod.sh already has 8000.)

> **FIX:** Add it under `[Pod]`, daemon-reload, restart stream-pod.

! THE OPEN CONTROL ROOM

Icecast's `/admin` pages can kick the DJ off the air.

> **FIX:** nginx allows `/admin` only from the LAN and WireGuard. Test: from the S21 on cellular, `radio.songofeo.com/admin/` must be refused.

! THE PASSWORD MISMATCH

Change Icecast's source password but not the radio container's, and the DJ is locked out: dead air.

> **FIX:** Both read the same secret (scripts) or the same value (Quadlets). Change both, restart both.

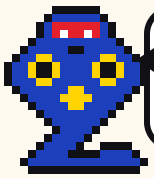
! THE FULL HOUSE

Listener 51 gets turned away: `ICECAST_MAX_CLIENTS=50`. Your upload speed runs out long before that matters.

> **FIX:** Raise it only if the upload can carry it.

TRANSMITTER CHECKLIST

- [ ] 8000 published in the pod and open in nftables
- [ ] `http://10.6.9.20:8000/stream` plays on the-box
- [ ] `radio.songofeo.com/stream` asks for the sign-in once
- [ ] `/admin` refused from cellular



CLAUDE-VIPER

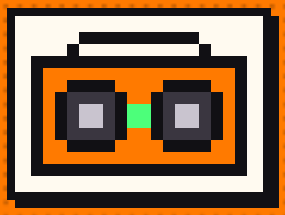
check.sh on the gate shows radio. as a NOTE until part 8.3's server block exists. A NOTE there isn't a failure, just a floor still under construction.

SAME RADIO, TWO WAYS	SCRIPTS (radio.sh)	MANUAL QUADLETS
Passwords	podman secrets	random values in the 600 files
Port 8000	already in pod.sh	sed adds it to stream.pod
Start order	waits for :8000	Requires= and After=

WORLD 2-9 · CHARACTER SELECT

RADIO

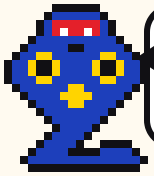
The AutoDJ: Liquidsoap shuffles the music, all day, every day



Meet the DJ who never sleeps. The radio container runs Liquidsoap, a programming language made for radio. Its script, `apps/radio/radio.liq`, is fifteen lines long: shuffle the music folder, crossfade between songs, never go silent, send it to Icecast.

The container reads the music library read-only, at `/media/music`, which is `dock1a/music` on the SGIN, the same folder the Commander's Navidrome reads. It runs the script from `/srv/stream/radio/radio.liq` (mounted read-only too), encodes MP3 at 192 kb/s, and pushes it to Icecast at `127.0.0.1:8000`, mount `/stream`.

The station's name comes from the environment: `RADIO_NAME="Song of Eo Radio"`. The music folder can be changed the same way with `RADIO_MUSIC`.



CLAUDE-VIPER

No playlists to build, no requests to take, no coffee breaks. Drop a song into the library and it joins the rotation by itself.

STATS

HOST	cobra-stream 10.6.9.20, pod `stream`
IMAGE	`savonet/liquidsoap:v2.4.x-latest`
SCRIPT	`/srv/stream/radio/radio.liq`
MUSIC	`dock1a/music` as `/media/music`
OUTPUT	MP3 192 kb/s to Icecast `/stream`
PORT	None of its own
NAME	Song of Eo Radio

TOUGHNESS ■■■■□
TRAP LEVEL ■■□□□
FUN FACTOR ■■■■■

24/7 SHUFFLE

THE WHOLE SCRIPT, DECODED

LINE	WHAT IT DOES
`playlist(mode="randomize", reload_mode="watch", music_dir)`	Shuffles every track under the folder and notices new files without a restart
`crossfade(duration=4., music)`	4-second smooth hand-off between songs
`mksafe(radio)`	Silence instead of a crash if a file is bad
`output.icecast(%mp3(bitrate=192), ...)`	Sends it to 127.0.0.1:8000, mount /stream, with the source password

★ PRO TIP: STATION NAME

`RADIO_NAME` is what listeners' players show as the station. Change it in the container's settings, not in the script.

PINNED TO 2.4

The image tag `v2.4.x-latest` follows Liquidsoap's 2.4 line, so updates bring fixes, not a new language version that could break radio.liq.

HOW IT WORKS LEVEL MAP

A NEW SONG JOINS THE ROTATION

1 THE-BOX copy to dock1a\music ▶ 2 WATCH playlist sees the file ▶ 3 LIQUIDSOAP shuffle + crossfade ▶ 4 ICECAST /stream ▶ 5 LISTENERS through radio.

STARTING ORDER

The DJ needs the tower. With the scripts, `radio.sh` starts Icecast, waits for port 8000, then starts radio. With Quadlets, `radio.container` has `Requires=icecast.service` and `After=` it, plus `Restart=on-failure` every 10 seconds, so a radio that started too early just tries again.

THE ENTRYPOINT

The Liquidsoap image needs to be told which program to run. The script mode passes `--entrypoint liquidsoap` and then `/radio/radio.liq`. In Quadlet mode, if the log says it can't run `/radio/radio.liq` as a command, the line must be `Exec=liquidsoap /radio/radio.liq`.

WHERE ITS SETTINGS COME FROM

- `RADIO_MUSIC`: folder to shuffle (default `/media/music`)
- `RADIO_NAME`: the station name listeners see
- `ICECAST_SOURCE_PASSWORD`: shared with Icecast

WHAT IT PLAYS

All of `dock1a/music`: the Commander's and Rachel's music both. Comedy and your recorded sessions live in their own folders, so they never get on the air.

WHAT IT DOESN'T DO (YET)

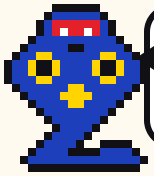
No station IDs, no requests, no schedule. That's the whole point of the fifteen-line script: easy to read, easy to change. Liquidsoap can do all of those later if the Commander wants them.

UP-ALL SKIPS IT

`sgin/up-all.sh` runs `radio.sh` only if `/srv/stream/radio/radio.liq` exists. No script in place, no radio, and the rest of the pod still starts: "skipping the radio (no ... yet, part 8)".

NO PORT, NO DATA

radio has no port of its own and writes nothing. Its only output goes over 127.0.0.1 to Icecast, inside the pod.



CLAUDE-VIPER

Tower and DJ, two containers, one job each. If the music stops, ask the DJ (podman logs radio). If nobody can hear it, ask the tower.

THE HOUSE BAND

radio. streams the shuffle to anyone signed in at the gate.

music. lets you pick songs yourself, from the same folder.

The website's players on songofeo.com use the floors behind the gate's sign-in.

TIPS & TRICKS POWER-UPS

★ PRO TIP: A STATION CRATE

Want a playlist just for the radio? Make `\\cobra-stream\media\dock1a\radio`, copy songs in, then point the music mount and `RADIO_MUSIC` at that folder instead (manual 8.2).

★ PRO TIP: COPY, THEN EDIT

The repo's `apps/radio/radio.liq` is the original. The live copy is `/srv/stream/radio/radio.liq`. Edit the live copy, and keep a backup next to it before big changes.

★ PRO TIP: CHANGE THE SOUND

Longer fades? Change `duration=4.` in radio.liq. Better quality? Raise `bitrate=192`, and remember each listener uses more upload. Then restart radio.

CODES

SGIN · podsvc

```
> podman logs --tail 30 radio
```

Want: connected to Icecast, and a track starting.

```
> podman restart radio
```

SGIN · eels

```
> sudo cp /srv/stream/radio/radio.liq /srv/stream/radio/radio.liq.bak
> sudo nano /srv/stream/radio/radio.liq
```



JOE SABOTAGE REPORT

! THE MISSING SCRIPT

`radio.sh` stops with "radio.liq is missing" if the script isn't in `/srv/stream/radio`.

> FIX: Copy `apps/radio/radio.liq` there first, owned by podsvc (part 8.2).

! THE NAMELESS PROGRAM

Quadlet with `Exec=/radio/radio.liq`: the log says it can't run that as a command.

> FIX: `Exec=liquidsoap /radio/radio.liq`, daemon-reload, restart radio.

! THE WRONG PASSWORD

radio and Icecast disagree on the source password: the DJ is refused and the station is dead air.

> FIX: Same secret or value for both, then restart both.

! THE BAD FILE

A broken MP3 in the library could crash a less careful DJ.

> FIX: `mksafe` plays silence instead and the shuffle moves on. Find the file in the log and fix or remove it on the-box.

! THE PUNCHLINE ON AIR

A stand-up album copied into `music/` ends up in the rotation.

> FIX: Comedy belongs in `dock1a/comedy`, which the radio never reads.

! THE GHOST DRIVE

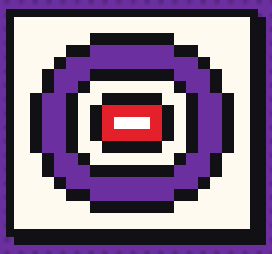
dock1a replugged: the DJ shuffles an empty folder and goes quiet.

> FIX: `podman pod restart stream`.

BONUS STAGE · NOT IN ANY POD

PI-HOLE ×2

The island's phone book, and the ad blocker in front of it



Before your browser can knock on the gate, it has to ask "where is music.songofeo.com?" On Cobra Island, two Pi-holes answer that question for every machine, block the ad and tracker names, and send the rest up to OPNsense.

Neither Pi-hole is a container in this guide's pods. **Pi-hole #1** is a small Proxmox container, CT 101 `cobra-pihole-1` at 10.6.9.3 on the Latitude. **Pi-hole #2** is installed natively on the SGIN at 10.6.9.20, right beside the stream pod. OPNsense's DHCP hands every machine both addresses, so if one goes down for an update, the other keeps answering.

Their best trick is the **wildcard**: any name under songofeo.com gets answered with **10.6.9.10**, the gate. At home your phone never goes out to the internet and back to reach your own server; it walks straight down the hall. That's why the floors open without the sign-in at home: the gate sees a 10.6.9.x address, and Authelia lets home addresses through.

STATS

#1	CT 101 on cobra-command, `10.6.9.3`
#2	native on cobra-stream, `10.6.9.20`
ADMIN	`http://10.6.9.3/admin` and `http://10.6.9.20/admin`
UPSTREAM	OPNsense `10.6.9.1` (Unbound, DNS over TLS)
LISTEN	`ALL`, so WireGuard clients get answers
TOUGHNESS	■■■■□
TRAP LEVEL	■■■■■
FUN FACTOR	■■■□□



CLAUDE-VIPER

DNS trouble looks like every other kind of trouble. When "nothing works", ask the phone book first.

LEVEL MAP: WHO ANSWERS WHAT

ASK tv.songofeo.com ▶ WILDCARD 10.6.9.10, the gate | ASK songofeo.com, www. ▶ PASS UP Porkbun's site | ASK vpn.songofeo.com ▶ PASS UP your public IP

HOW IT WORKS *THREE MAGIC LINES*

THE DNSMASQ LINES (SAME ON BOTH)

- `address=/*.songofeo.com/10.6.9.10` : every name *under* songofeo.com goes to the gate. The `*.` leaves the bare `songofeo.com` alone, so the station on Porkbun's hosting still loads at home.
- `server=/www.songofeo.com/#` : www is the station too, so it's passed up to the real DNS.
- `server=/vpn.songofeo.com/#` : the WireGuard endpoint. It must be your **public** address even at home, or a laptop that sets up its tunnel at home would point it at the gate.

SHORT NAMES

Both Pi-holes carry the same 8 host entries, each with a full and a short name: `cobra-firewall` .1, `cobra-command` .2, `cobra-pihole-1` .3, `cobra-switch` .5, `cobra-ap` .6, `cobra-gate` .10, `cobra-stream` .20, `cobra-arcade` .30.

The short ones matter over WireGuard: a VPN client doesn't get the `home.arpa` search suffix, so `\\cobra-stream` would otherwise not resolve.

WHY "ALL" IS SAFE HERE

Pi-hole's default **LOCAL** mode ignores clients on subnets it has no interface on, and WireGuard's 10.6.99.0/24 is one of those. So both run `listeningMode ALL`. That's safe because nothing outside reaches them: CT 101 sits only on the LAN, and the SGIN's nftables lets port 53 in from the LAN and WireGuard only.

HOW MACHINES FIND THEM

OPNsense runs **Dnsmasq DNS & DHCP**. Its DHCP option 6 (DNS servers) hands out `10.6.9.3, 10.6.9.20`, with the domain `home.arpa`. A machine that renews its lease gets both.

WHERE THE REST GOES

Names that aren't blocked and aren't local go up to OPNsense at 10.6.9.1, whose Unbound resolver sends them out over DNS over TLS. Your ISP sees an encrypted stream, not your lookups.

CODES: ASK THE PHONE BOOK

THE-BOX · PowerShell

```
❯ nslookup tv.songofeo.com 10.6.9.3
Must answer 10.6.9.10.
❯ nslookup tv.songofeo.com 10.6.9.20
Same answer from Pi-hole #2.
❯ nslookup vpn.songofeo.com 10.6.9.3
Must be your public IP, never 10.6.9.10.
❯ nslookup songofeo.com 10.6.9.3
Porkbun's address: the station has to load at home too.
```

TIPS & TRICKS

★ PRO TIP: SYNC THE TWINS

Change blocklists on Pi-hole #1, then copy them across: #1 → Settings → **Teleporter** → Export, and #2 → Teleporter → Import. Do it on the-box.

★ PRO TIP: SITE BROKEN?

Query Log on either admin page, filter by your machine, reload the site. Blocked names show in red.

CODES

THE-BOX · PowerShell

```
❯ ipconfig /renew
❯ ipconfig /all
```

DNS Servers must list 10.6.9.3 and 10.6.9.20, suffix home.arpa.

CODES

SGIN · eels

```
❯ pihole status
❯ sudo pihole -up
```

Update #2, check it answers, then do #1 (pct enter 101 on cobra-command). While one restarts, machines ask the other.

```
❯ sudo pihole setpassword
```

Forgot the admin password? Set a new one.

★ PRO TIP: XFINITY WI-FI

Xfinity's router has no NAT loopback, so songofeo names don't load from the Xfinity Wi-Fi. Use the WireGuard **home** profile (endpoint `10.0.0.2:51820`) until the cobra access point arrives.



JOE SABOTAGE REPORT

! THE GREEDY WILDCARD

The wildcard also grabbed `vpn.`, which must stay public for WireGuard.

> **FIX:** The `server=/vpn.songofeo.com/#` line on both Pi-holes passes it up.

! THE STATIC DESKTOP

the-box had a hand-typed IP with one DNS server and no suffix, so short names never resolved.

> **FIX:** Switched to DHCP with a reservation at .50.

! THE SILENT DHCP

OPNsense's Dnsmasq had no DNS option at all, so every machine got 10.6.9.1 and skipped the Pi-holes.

> **FIX:** DHCP option 6 = 10.6.9.3, 10.6.9.20.

! THE VPN THAT EATS DNS

PIA VPN on the-box takes over DNS and reconnects by itself, so lookups skip the Pi-holes.

> **FIX:** Disconnect PIA before testing anything DNS.

! THE MISSING TWIN ENTRIES

Pi-hole #1 was missing `cobra-switch` and `cobra-ap` while #2 had all 8, so a name worked or failed depending on which twin your machine asked.

> **FIX:** Both now carry the same 8 hosts. Change one, change the other the same day (Settings → Local DNS Records, or Teleporter).

TOP SECRET · TEAR OUT AND TAPE INSIDE YOUR FOOTLOCKER

SECRET CODES

WORLD 1 · THE GATE

EYES ONLY

GET IN

THE-BOX · Windows Terminal

```
❯ ssh eels@10.6.9.10
```

GATE · eels

```
❯ sudo machinectl shell podsvc@
```

Every command below runs as podsvc unless the tag says eels.

LOOK AROUND

GATE · podsvc

```
❯ podman ps --pod
❯ podman logs --tail 50 authelia
❯ podman logs -f nginx
❯ podman stats --no-stream
```

CPU and memory per container, one snapshot.

RESTART (QUADLETS)

GATE · podsvc

```
❯ systemctl --user restart authelia nginx
❯ systemctl --user status paperless
❯ systemctl --user restart gate-pod
```

The whole pod, every container in it.

```
❯ systemctl --user daemon-reload
```

After editing any .container file.

★ PRO TIP: WHERE IT LIVES

Every gate app keeps its own data under `/srv/gate/<app>`, and the Quadlet files sit in `~/.config/containers/systemd/` as podsvc. Both ride along in the nightly VM backup. The shared files (scans, youtube, art) live on the SGIN's drives and come over the CIFS mount.

CHECK EVERYTHING

GATE · eels or podsvc

```
❯ ~podsvc/cobra-server/scripts/podman/gate/check.sh
❯ podman exec nginx nginx -t
❯ ss -tlnH
```

Only 22, 80, 443 and 9090 may listen on the gate.

UPDATES

GATE · podsvc

```
❯ podman auto-update --dry-run
```

Which Quadlet images have a newer version (AutoUpdate=registry).

```
❯ podman auto-update
```

Read Authelia's and Paperless's release notes first.

THE SHARE FROM THE SGIN

GATE · eels

```
❯ findmnt /srv/storage/disks
❯ grep -c cifs /etc/fstab
```

Must print 1. More means the fstab line stacked up again.

```
❯ sudo mount -a
```

CERTIFICATE

GATE · eels

```
❯ systemctl list-timers lego-renew.timer
❯ journalctl -u lego-renew -n 20
```

TOP SECRET · PAGE 2 OF 2

SECRET CODES

WORLD 2 · THE STREAM

EYES ONLY

GET IN

THE-BOX · Windows Terminal

```
❯ ssh eels@10.6.9.20
```

SGIN · eels

```
❯ sudo machinectl shell podsvc@
```

LOOK AROUND

SGIN · podsvc

```
❯ podman ps --pod
❯ podman logs --tail 50 jellyfin
❯ podman restart navidrome
```

Works whether the SGIN runs Quadlets or the scripts.

READ A GENERATED PASSWORD

SGIN · podsvc

```
❯ podman secret ls
❯ podman secret inspect --showsecret --format
'{{.SecretData}}' icecast-admin-password
```

THE GPU

SGIN · eels

```
❯ ls -l /dev/dri
❯ sudo intel_gpu_top
```

Watch the Video bar move while a transcode runs. Needs intel-gpu-tools.

RUN WITH THE SCRIPTS

SGIN · podsvc

```
❯ cd ~/cobra-server/scripts/podman/sgin
❯ ./up-all.sh
❯ UPDATE=1 ./jellyfin.sh
❯ ./navidrome.sh
```

The scripts stop if a Quadlet already owns that app.

DRIVES

SGIN · eels

```
❯ findmnt -t ext4
❯ df -h /srv/storage/disks/*
❯ ls /srv/storage/disks/dock1a
```

FIREWALL AND SHARES

SGIN · eels

```
❯ sudo nft list ruleset
❯ sudo systemctl status smbd
❯ ss -tlnH
```

★ PRO TIP

Every app's own data lives under `/srv/stream/<app>` on the SGIN's SSD. The library drives are mounted **read-only** into the apps, so a confused container can't delete a movie.

LETTERS FROM THE BARRACKS

Q: Why do I sign in once on my phone, but my TV never asks?

A: tv. is the one floor nginx sends straight to Jellyfin without Authelia, because TV apps can't show a web sign-in page. Jellyfin's own login guards it instead.

Q: At home nothing asks me to sign in. Is that a hole?

A: No. The floors trust home (10.6.9.x) and WireGuard (10.6.99.x) addresses. The admin apps still ask for password and TOTP, even at home.

Q: Can I just open a port to Jellyfin on the SGIN?

A: Please don't. Everything on the SGIN answers without the gate. Exactly one forward lives on OPNsense: WAN 443 to 10.6.9.10.

Q: How do I give a friend Jellyfin?

A: On the-box, Jellyfin → Dashboard → Users → add them, with only the libraries you want. Then send them `https://tv.songofeo.com`. No gate account needed.

Q: The gate rebooted but the SGIN was off. Now what?

A: Paperless, MeTube and eo-art can't find their shared folder and fail. systemd retries them every minute, so they come back by themselves once the SGIN is up.

Q: Where do my passwords live?

A: Yours: in Vaultwarden at vault., LAN and WireGuard only. The apps' generated ones: as podman secrets, readable with the codes on the previous page. Never in the scripts.

Q: How do I know when something breaks?

A: Uptime Kuma at status. watches every app, and Homepage at command. shows a live dot on each tile.

Q: Who gets in through the gate sign-in?

A: Whoever is in Authelia's `users.yml` on the gate. Right now that's eels and rachel.



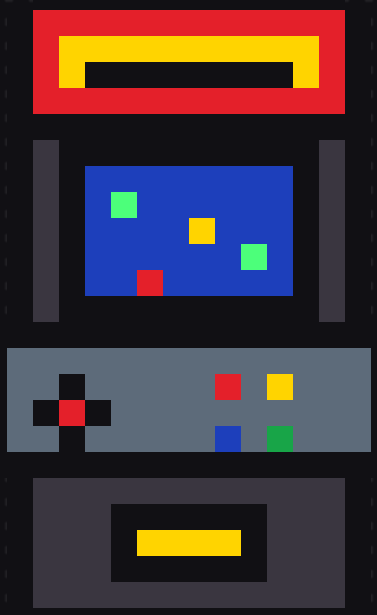
CLAUDE-VIPER

Got a question for the barracks? Ask in the project and the next issue might answer it. Cobra-ah... ahem. Carry on.

COMING NEXT ISSUE

COBRA ARCADE

TANKSTICK · TRACKBALL · LED-WIZ · HIGH SCORES



CABINET CLINIC

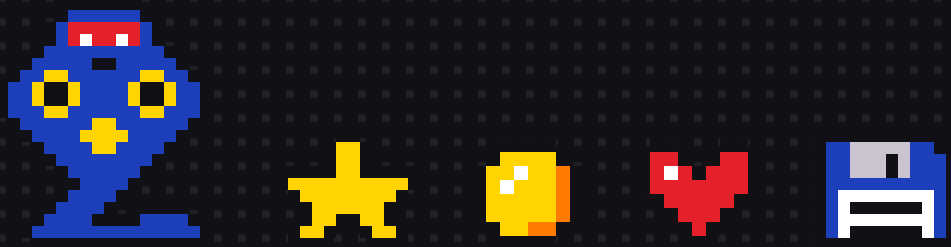
WIRE THE TANKSTICK AND LIGHT UP EVERY BUTTON

NIGHT OPS

THE NIGHTLY ROM COPY TO THE SGIN

HALL OF FAME

SCORES FROM THE CABINET TO SCORES.SONGOFEO.COM



HIGH SCORE TABLE

```
1ST EELS ..... 999990
2ND RACHEL ... 876540
3RD JOES ..... 00
```

CREDITS

Written and drawn by Claude-Viper and the Brush-Fang art den for Commander Eels. Built from the cobra-server manual and the logs of the actual build. All pixel art is original.

GAME OVER? NEVER. PRESS START TO CONTINUE.